
Design of UHPC based Reinforced Concrete Structural Sections

Limit states

ACI 239C Design Task Group
January 13th, 2023

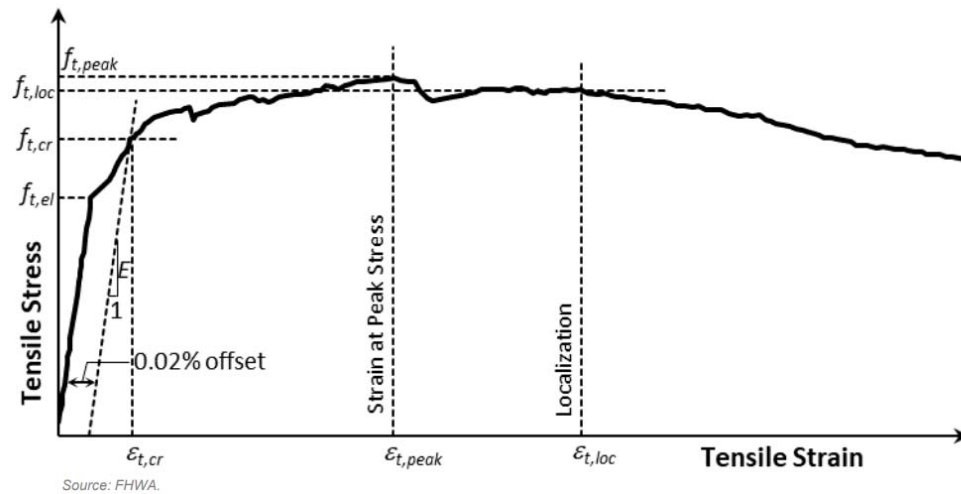
Barzin Mobasher
Arizona State University

Outline

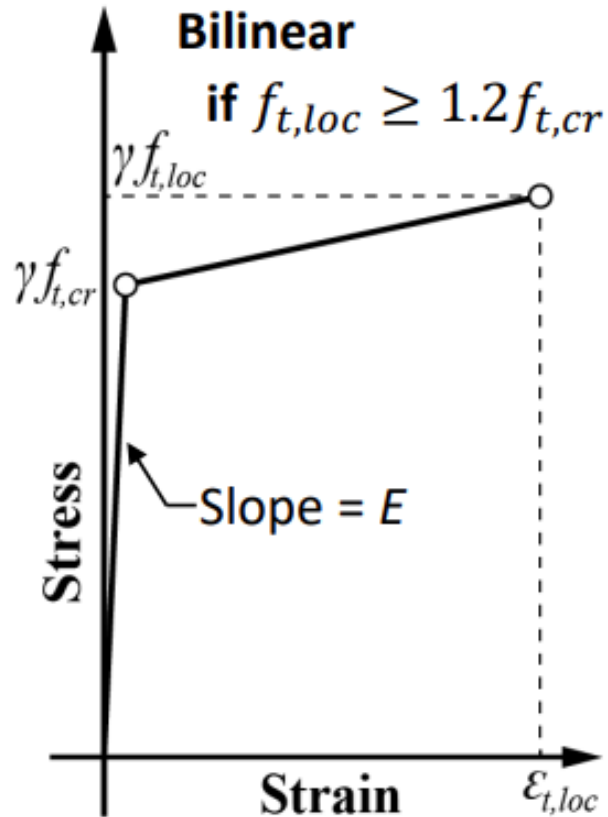
- Objectives
- Overview of UHPC material responses
- Parametric Material models
- Algorithms
- Closed form solutions for moment-curvature diagram
- Load deflection response
- Limit States-
- Simulations
- Conclusions

Limit States

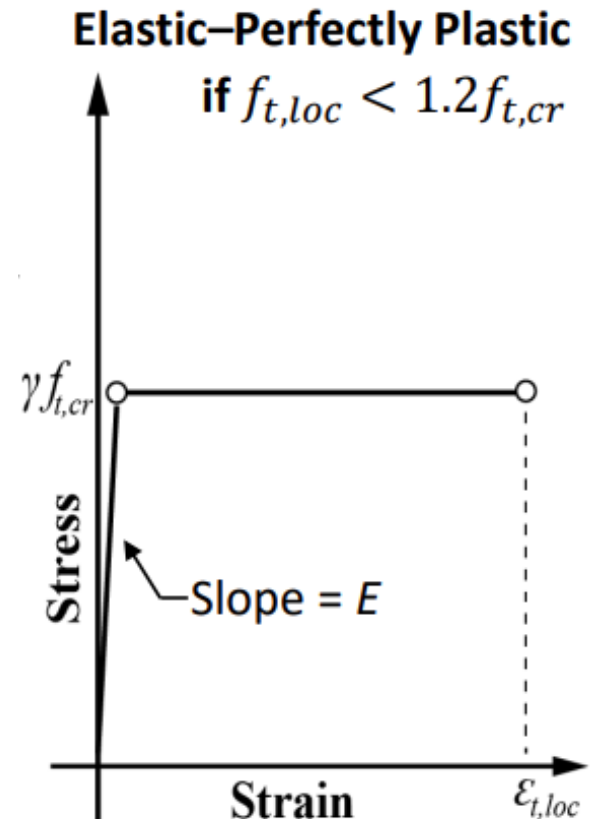
- Tension Modes,
 - Compression Modes,
 - Shear Modes, Bond and Pullout,
 - Section Ductility- Material Ductility, Structural Ductility
-
- End Goal, , Graybeal, B. 2022 ACI presentation



Simplified Material Models



Source: FHWA.



Tension and compression Dominates Limit states

- Tension Modes Beams
 - First crack strength
 - Elastic steel rebar, composite action, partially cracked
 - Yielding of the rebar ultimate limit state
 - Softening Contribution
- Compression Failure Beams
 - First crack
 - Elastic plastic and softening Behavior
 - Confinement and Compression reinforcement interaction
- Columns and compression –bending design limit states, Second Order Effects
 - Column Confinements

Limit States

ULS

- Bending
- Axial compression in linear members
- Bending in slabs
- Shear & Torsion
- Punching
- Anchorages and splices
- Structural ductility
- Orientation, settlement
- Lateral loads,
- Yield hinges,
- Upper bound Theory limit states
- Ductility, seismic loading

SLS

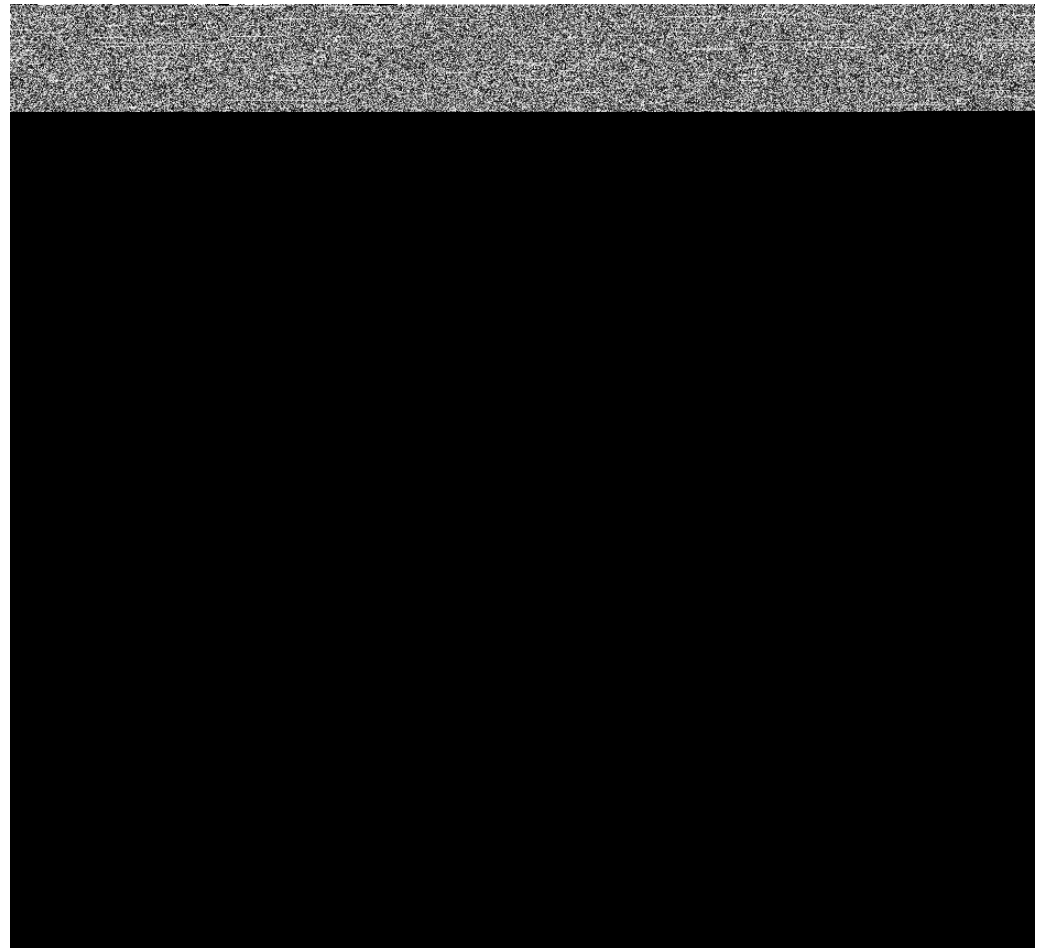
- Stresses
- Cracking
- Deformability
- Minimum reinforcement
- Long term behavior
- Fatigue

Serviceability Limit state, Deflection, stiffness reduction due to cracking

- Shear Failure Criteria
 - Diagonal Tension shear Cracking
 - Punching shear cracking in slabs
 - Pullout and development length
 - Alternative approaches
- Shear retention factor, Reduce the effective Shear modulus,
 - G to represent a value of 1 for full-aggregate bond and 0 no-aggregate bond, Popular with ABAQUS and Diana, Use typical 0.1. Has not improved in 20+ years

Strain Softening and Strain hardening

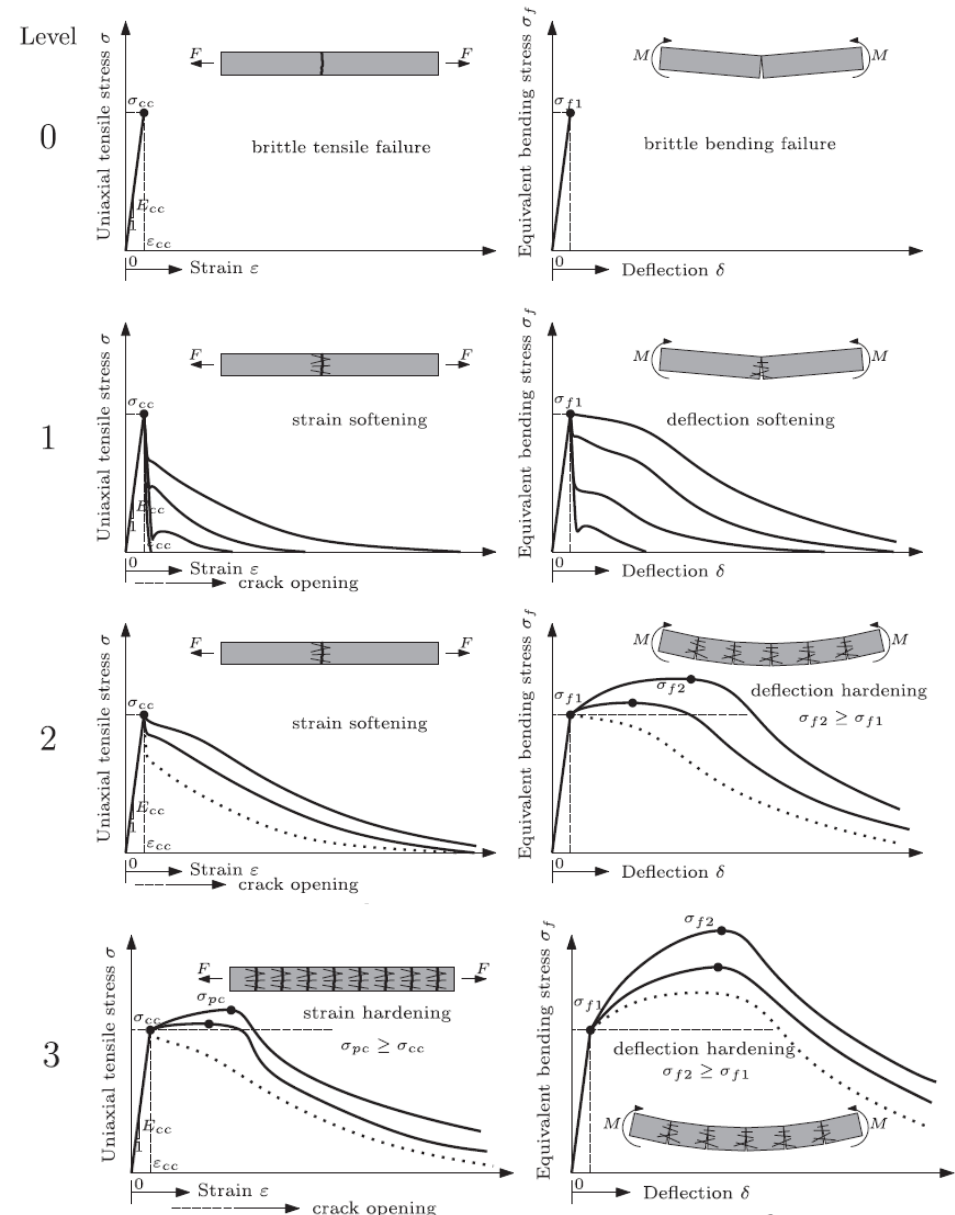
Tensile stress-strain curves of strain hardening FRCs



Typical stress-elongation curves in tension of fiber reinforced cement composites: (a) strain-softening. (b) strain-hardening.

Unified representation of FRC Composites

- Four Response Systems
 - Strain Softening/Hardening, Deflection Softening/Hardening
- Strain softening behavior
 - Discrete fiber systems
 - SFRC, PFRC
- Strain hardening behavior
 - Discrete & continuous fiber systems
 - Textile reinforced cement, TRC, UHPC
- Load-Deflection Calculations as well as reinforced sections are significantly affected by the tensile post cracking behavior



Tension Modes of response

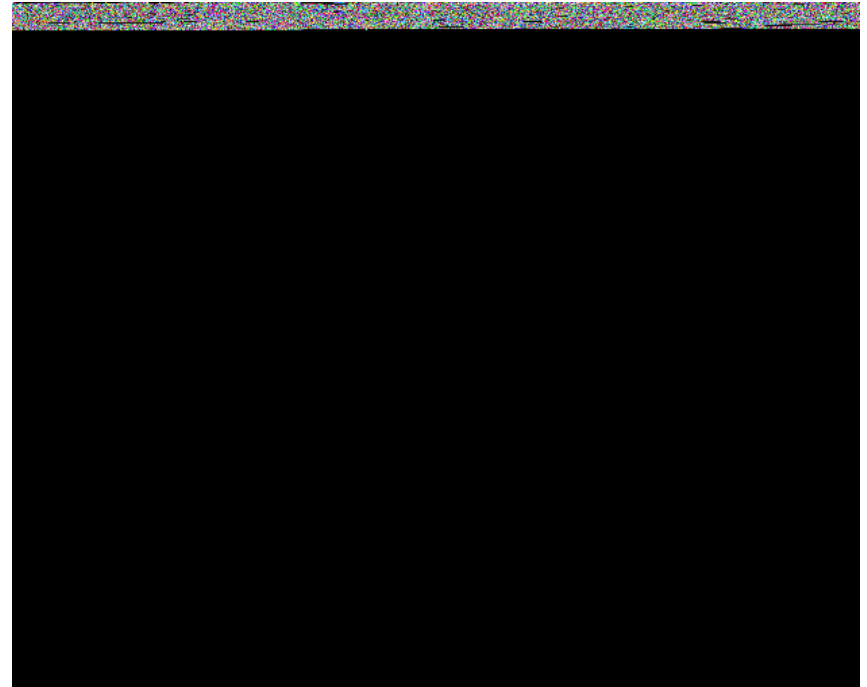
- Brittle
- Linear Tension Softening
- Bilinear Tension softening
- Multilinear Tension Softening
- Non-linear tension Softening
- Reinhardt, H. W., “Fracture mechanics of an elastic softening material like concrete. Heron 29, 2 (1984).

Shear Modes of Failure

$$V_n = V_{UHPC} + V_s$$

$$V_{UHPC} = \gamma f_{t,loc} b_v d_v \cot \theta$$

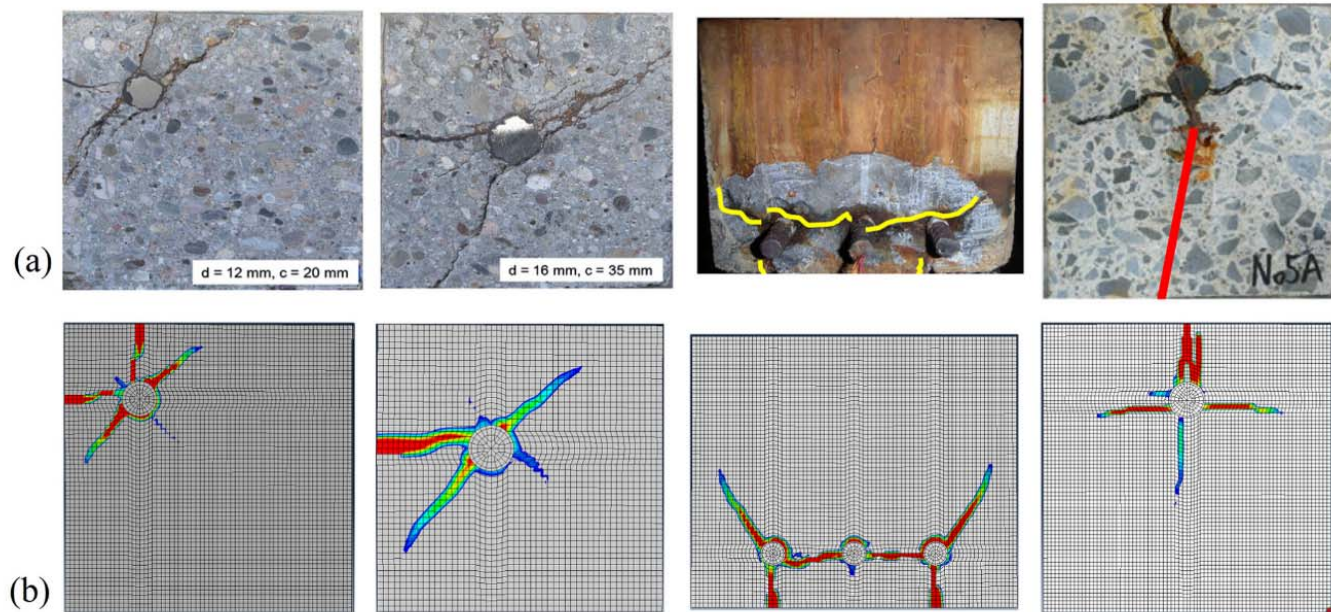
$$V_s = \frac{A_v f_v}{s} d_v \cot \theta$$



- Source, FHWA
- Scotta, R., Vitaliani, R., Saetta, A., Oñate, E., & Hanganu, A. (2001). A scalar damage model with a shear retention factor for the analysis of reinforced concrete structures: theory and validation. *Computers & Structures*, 79(7), 737–755. [https://doi.org/10.1016/S0045-7949\(00\)00178-4](https://doi.org/10.1016/S0045-7949(00)00178-4)

Durability Modes

- Durability limit state
- Chloride penetration
- Corrosion or rebar, delamination, debonding



Zhu, X., & Zi, G. (2017). A 2D mechano-chemical model for the simulation of reinforcement corrosion and concrete damage. *Construction and Building Materials*, 137, 330–344. <https://doi.org/10.1016/j.conbuildmat.2017.01.103>

Strain Hardening Cement Based Composites

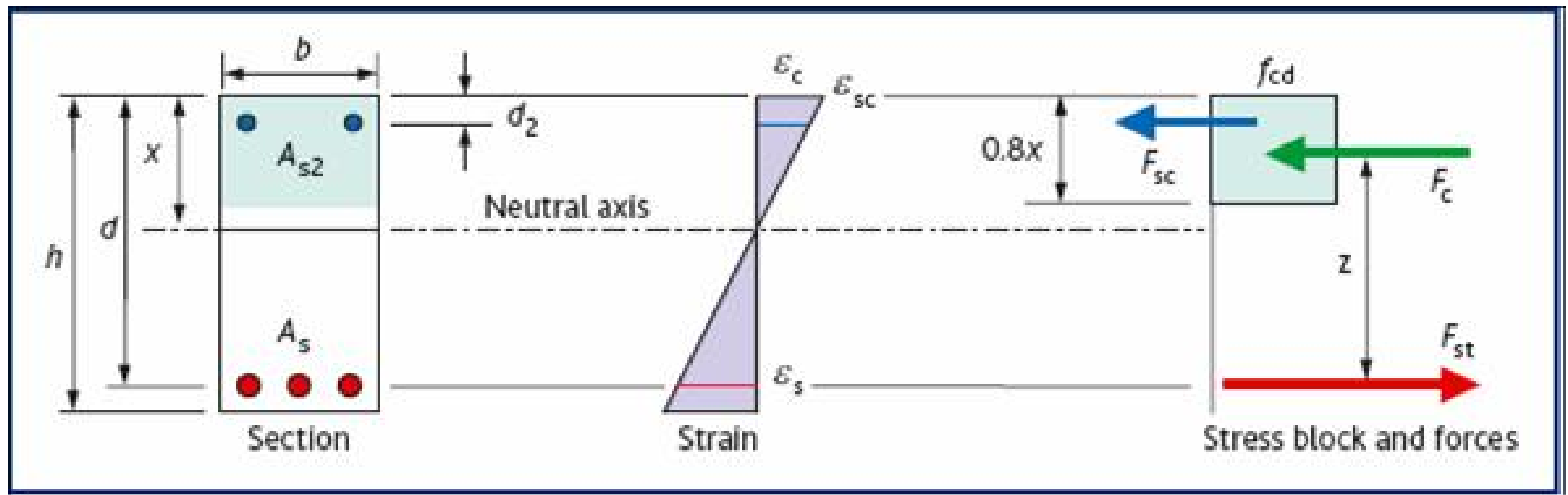
- Similarity in material constitutive models
 - Tensile response ~ various levels of ductility
 - Tensile strength $\ll$ Compressive strength
- Elastic response
 - Use standard composite materials models
 - Composite laminate theory
 - Finite difference approach for tension stiffening
- Allow damage-based models
 - Fracture mechanics
 - Damage based models
 - Plasticity based models
- Initial linear elastic response
- First cracking strength, a function of fiber content
- Distributed cracking start to saturation
- Saturation crack spacing
- Tension stiffening
- Ultimate strength



One can get multiple cracking from strain softening FRC assuming that the structural loading is statically indeterminate such as a fully supported round panel test

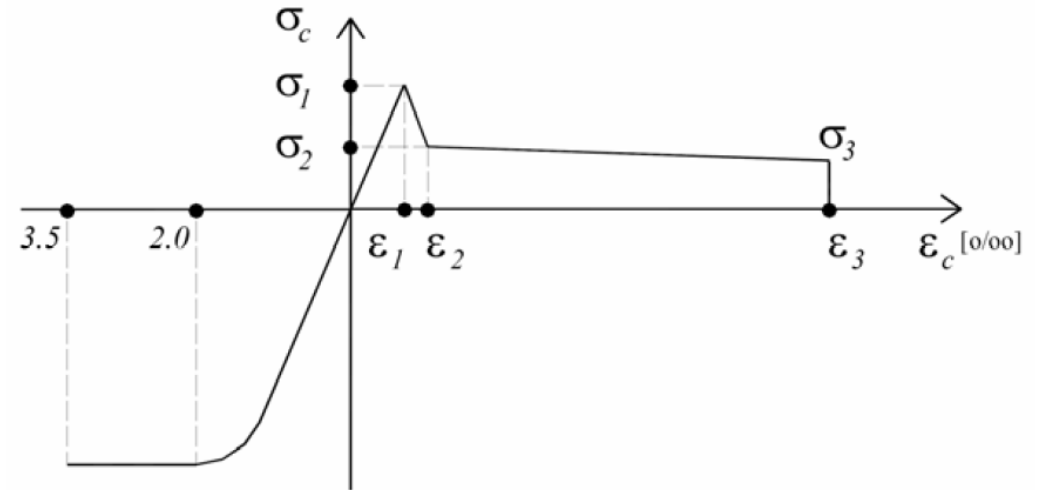
Flexure design of RC

- Cross sectional analysis
- Plane section remains plane
- Compression: stress block method
- Tension: ignored
- Ultimate limit state: moment capacity



RILEM Stress-Strain Model for FRC

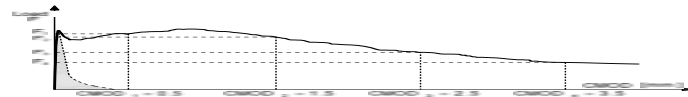
The model stress-strain diagram the same as proposed by RILEM and shown in the figure. The key points of the tension side of the diagram are found in a flexural test of a notched beam.



- σ_1 , and ϵ_1 – tensile strength and corresponding strain
- σ_2 , and ϵ_2 – stress and strain at the onset of the stable strain softening branch
- σ_3 , and ϵ_3 – stress and strain at the end of the softening branch

RILEM Method for Test and Design

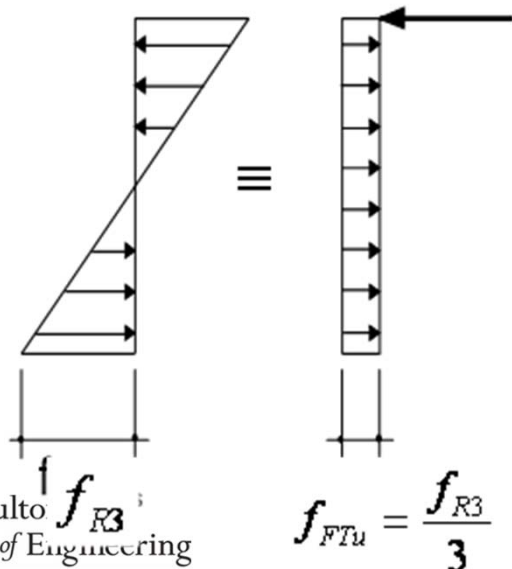
RILEM (EN 14651), a 3-point bending test on a notched beam and using the flexural load vs. crack opening (CMOD) response. Parameters of F_{r1} to F_{r4} correspond to apparent flexural load at crack opening of $CMOD_1$ (0.5 mm) to $CMOD_4$ (3.5 mm)



RILEM Method for Test and Design

Residual flexural-tensile strength parameters, f_{Rj} , are calculated from the Load-CMOD curves with regard to the beam dimensions. A simplified model is adopted to compute the equivalent ultimate tensile strength in uniaxial tension f_{FTu} by means of the residual nominal bending strength f_{R3} as shown.

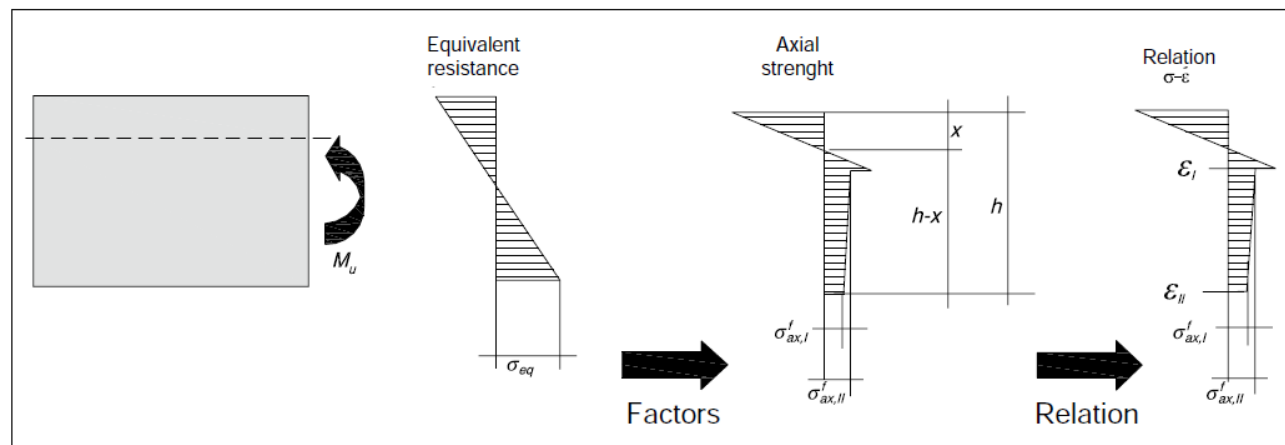
- F_j is the load corresponding with $CMOD_j$
- f_{Rj} is the residual flexural tensile strength corresponding with $CMOD_j$
- l is the beam span length [mm];
- b is the width of the specimen [mm];
- h_{sp} is the net height: distance between the notch tip and the top of the specimen [mm].



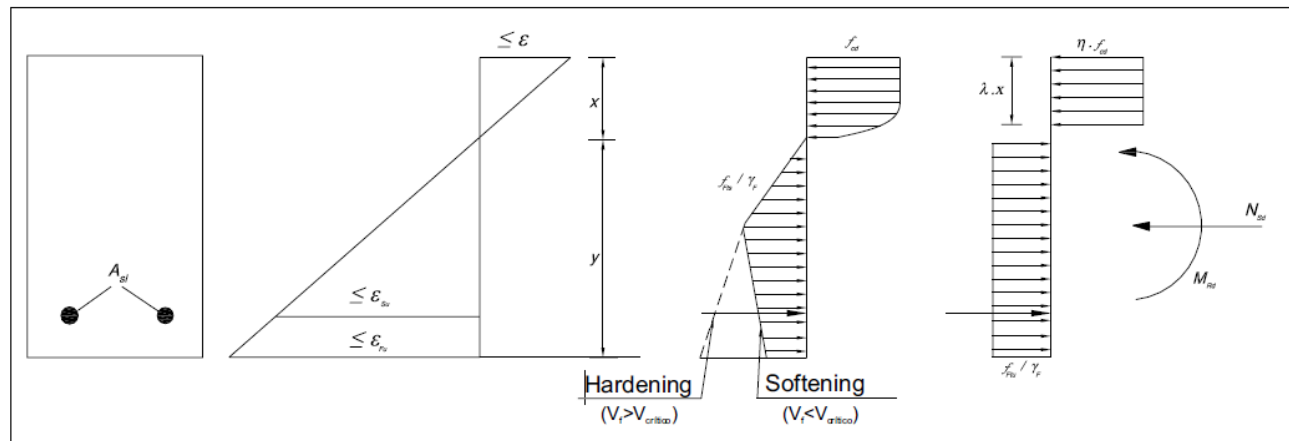
$$f_{R3} = \frac{F_3 \cdot l}{b \cdot h_{sp}^2} \quad \Rightarrow \quad f_{FTU} = \frac{f_{R3}}{3}$$

RILEM Method for Test and Design

Limit states for the combined compressive and bending stress: use the constituent simplified laws. Both softening and hardening responses can be modeled using a simplified stress block as shown in the figures.



Graphic 3.12 - Limit states for the combined compressive and bending stress: simplified method (stress-block as RILEM TC-162).

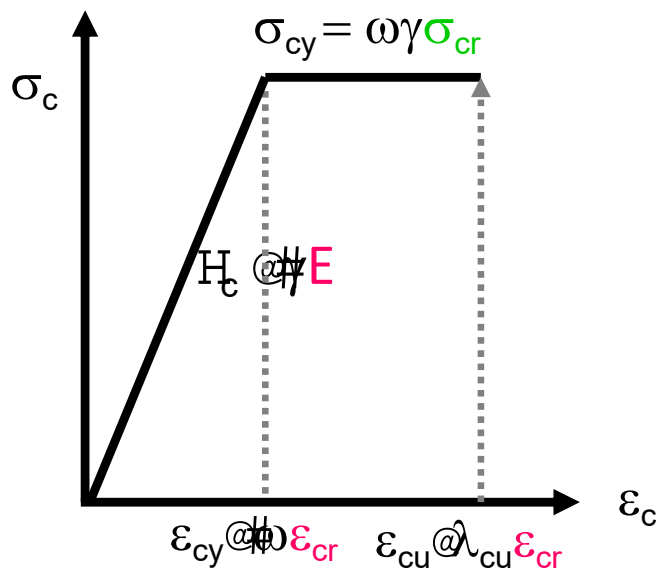


Graphic 3.11 - Limit states for the combined compressive and bending stress: use of the constituent simplified laws (stress-block with coefficients η and λ as EC2).

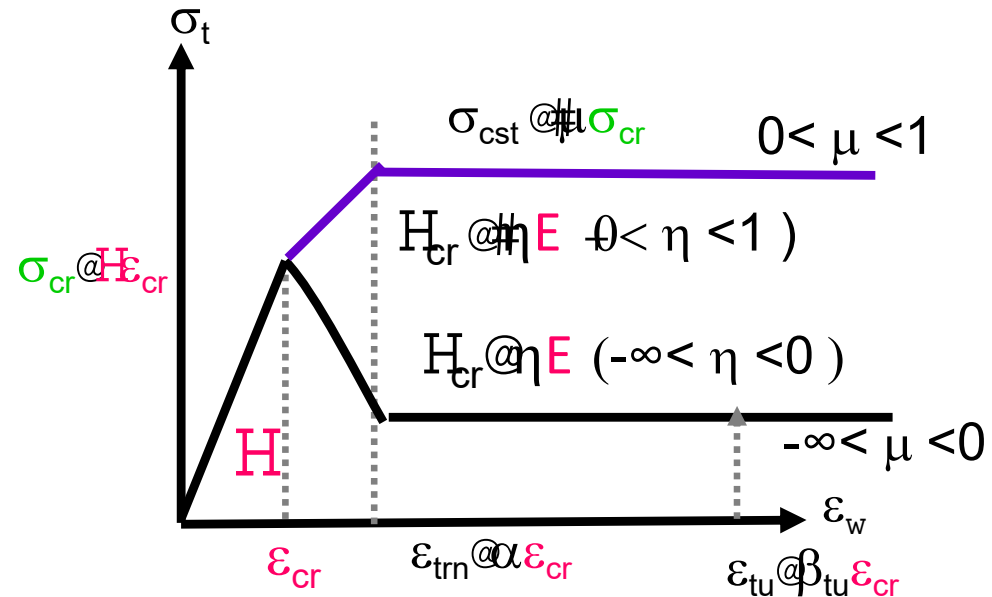
Material Model for Strain Hardening and Softening

- Two material parameters (ϵ_{cr} , E) and six normalized parameters (ω , γ , λ_{cu} , α , β_{tu} , η), $\mu = f(\eta)$

$$M = m(\omega, \gamma, \alpha, \eta, \beta_{tu}, \lambda_{cu}) M_{cr} \quad M_{cr} = \frac{1}{6} b d^2 E \epsilon_{cr} \quad \phi_{cr} = \frac{2\epsilon_{cr}}{d}$$



Compression model



Tension model

Parameters Simplification

- Tension and compression strains: $\beta = \frac{\varepsilon_{tbot}}{\varepsilon_{cr}}; \quad \lambda = \frac{\varepsilon_{ctop}}{\varepsilon_{cr}}$

- Stress and strain limiting parameters:

$$\omega = \frac{\varepsilon_{cy}}{\varepsilon_{cr}}; \quad \alpha = \frac{\varepsilon_{trn}}{\varepsilon_{cr}}; \quad \beta_{tu} = \frac{\varepsilon_{tu}}{\varepsilon_{cr}}; \quad \lambda_{cu} = \frac{\varepsilon_{cu}}{\varepsilon_{cr}}$$

$$\gamma = \frac{E_c}{E}; \quad \eta = \frac{E_{cr}}{E}$$

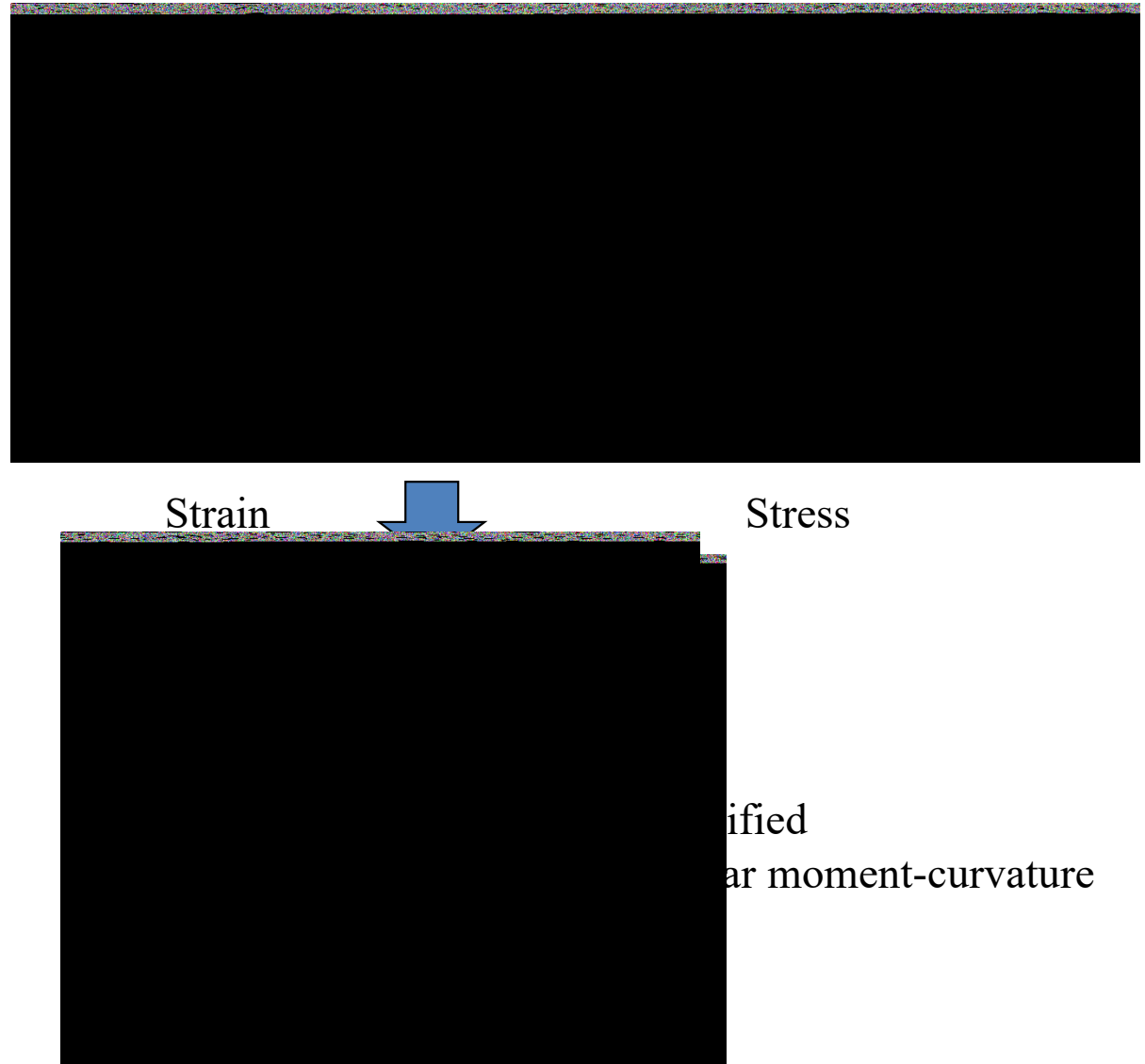
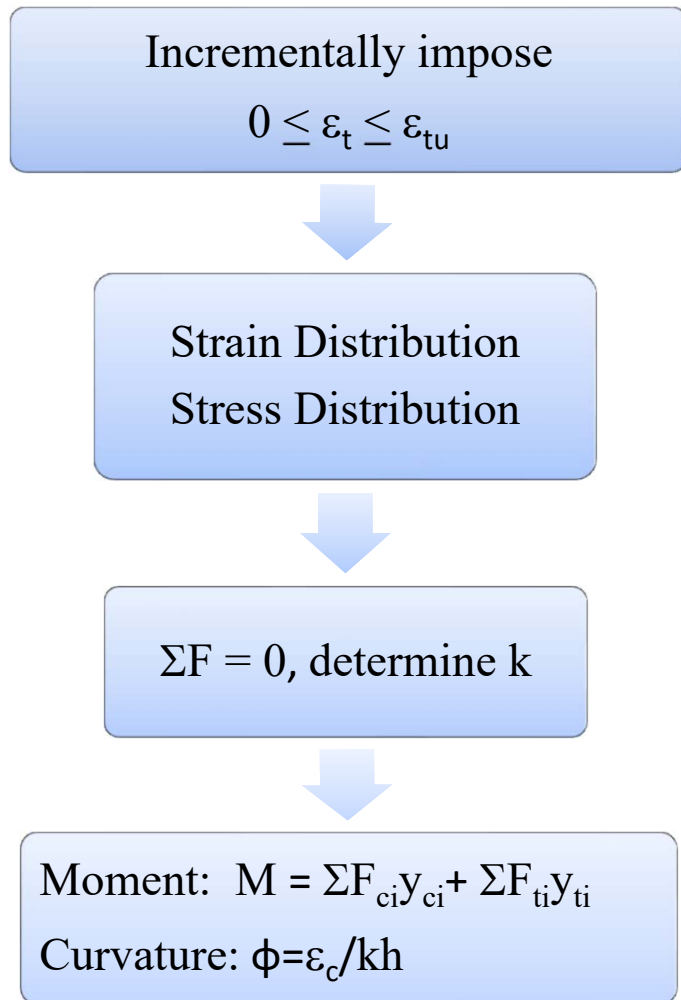
$$\mu = \frac{\sigma_{cst}}{E\varepsilon_{cr}}$$

- Moment and curvature

$$M_{cr} = \frac{1}{6} b d^2 E \varepsilon_{cr} \quad \phi_{cr} = \frac{2\varepsilon_{cr}}{d}$$

Derivation of Moment-Curvature Relationship

Stage : $\lambda > \omega$, $\beta > \alpha$



ified
ar moment-curvature

Normalized Force Terms and Moment Arms

Table 3

Normalized force component for each stage of normalized tensile strain at bottom fiber (β)

Normalized force component	Stage 1 ($0 \leq \beta \leq 1$) and ($0 \leq \lambda \leq \omega$)	Stage 2.1 ($1 < \beta \leq \alpha$) and ($0 \leq \lambda \leq \omega$)	Stage 2.2 ($1 < \beta \leq \alpha$) and ($\omega < \lambda \leq \lambda_{cu}$)	Stage 3.1 ($\beta > \alpha$) and ($0 \leq \lambda \leq \omega$)	Stage 3.2 ($\beta > \alpha$) and ($\omega < \lambda \leq \lambda_{cu}$)
$\frac{F_{c2}}{bdE_c e_{ct}}$	—	—	$\frac{\omega\gamma}{\beta}(\beta k + \omega k - \omega)$	—	$\frac{\omega\gamma}{\beta}(\beta k + \omega k - \omega)$
$\frac{F_{c1}}{bdE_c e_{ct}}$	$\frac{\beta\gamma k^2}{2(1-k)}$	$\frac{\beta\gamma k^2}{2(1-k)}$	$\frac{\omega^2\gamma}{2\beta}(1-k)$	$\frac{\beta\gamma k^2}{2(1-k)}$	$\frac{\omega^2\gamma}{2\beta}(1-k)$
$\frac{F_{t1}}{bdE_c e_{ct}}$	$\frac{\beta}{2}(1-k)$	$\frac{(1-k)}{2\beta}$	$\frac{(1-k)}{2\beta}$	$\frac{(1-k)}{2\beta}$	$\frac{(1-k)}{2\beta}$
$\frac{F_{t2}}{bdE_c e_{ct}}$	—	$\frac{(1-k)(\beta-1)(\eta\beta-\eta+2)}{2\beta}$	$\frac{(1-k)(\beta-1)(\eta\beta-\eta+2)}{2\beta}$	$\frac{(1-k)(\alpha-1)(\eta\alpha-\eta+2)}{2\beta}$	$\frac{(1-k)(\alpha-1)(\eta\alpha-\eta+2)}{2\beta}$
$\frac{F_{t3}}{bdE_c e_{ct}}$	—	—	—	$\frac{(1-k)(\beta-\alpha)\mu}{\beta}$	$\frac{(1-k)(\beta-\alpha)\mu}{\beta}$

Table 4

Normalized moment arm of force component for each stage of normalized tensile strain at bottom fiber (β)

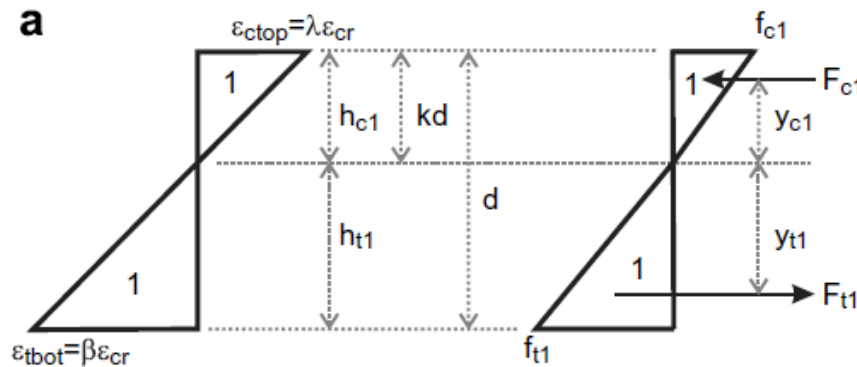
Normalized moment arm	Stage 1 ($0 \leq \beta \leq 1$) and ($0 \leq \lambda \leq \omega$)	Stage 2.1 ($1 < \beta \leq \alpha$) and ($0 \leq \lambda \leq \omega$)	Stage 2.2 ($1 < \beta \leq \alpha$) and ($\omega < \lambda \leq \lambda_{cu}$)	Stage 3.1 ($\beta > \alpha$) and ($0 \leq \lambda \leq \omega$)	Stage 3.2 ($\beta > \alpha$) and ($\omega < \lambda \leq \lambda_{cu}$)
$\frac{y_{c2}}{d}$	—	—	$\frac{\beta k + \omega(1-k)}{2\beta}$	—	$\frac{\beta k + \omega(1-k)}{2\beta}$
$\frac{y_{c1}}{d}$	$\frac{2}{3}k$	$\frac{2}{3}k$	$\frac{2}{3}\frac{\omega(1-k)}{\beta}$	$\frac{2}{3}k$	$\frac{2}{3}\frac{\omega(1-k)}{\beta}$
$\frac{y_{t1}}{d}$	$\frac{2}{3}(1-k)$	$\frac{2}{3}\frac{(1-k)}{\beta}$		$\frac{2}{3}\frac{(1-k)}{\beta}$	
$\frac{y_{t2}}{d}$	—	$\frac{2\eta\beta^2 - \eta\beta - \eta + 3\beta + 3}{3\beta(\eta\beta - \eta + 2)}(1-k)$		$\frac{2\eta\alpha^2 - \eta\alpha - \eta + 3\alpha + 3}{3\beta(\eta\alpha - \eta + 2)}(1-k)$	
$\frac{y_{t3}}{d}$	—	—	—	$\frac{(\alpha+\beta)}{2\beta}(1-k)$	

Soranakom C, Mobasher B. “Correlation of tensile and flexural responses of strain softening and strain hardening cement composites”, Cem Concr Comp 2008;30:465–477.

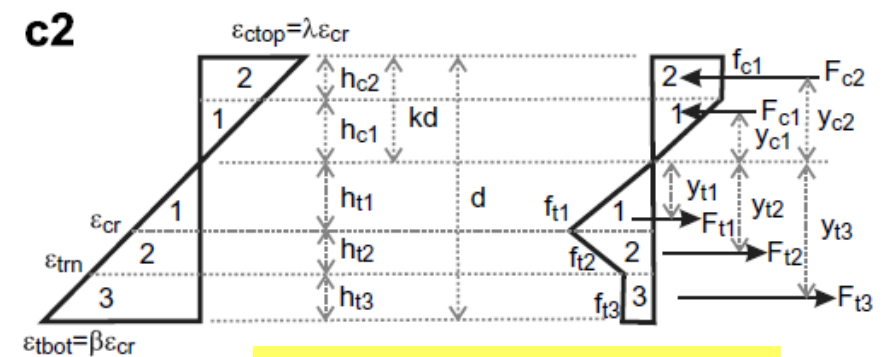
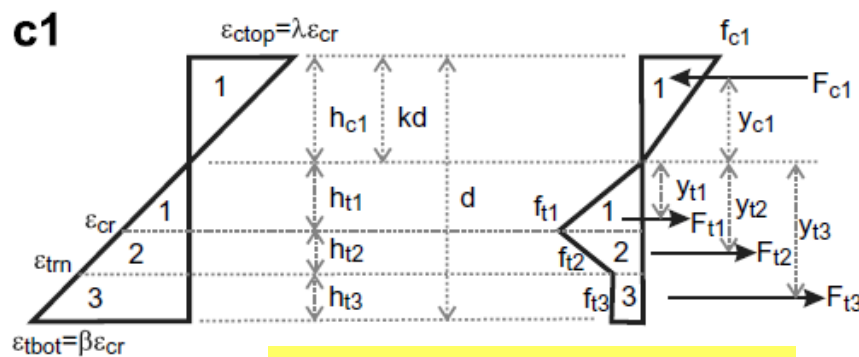
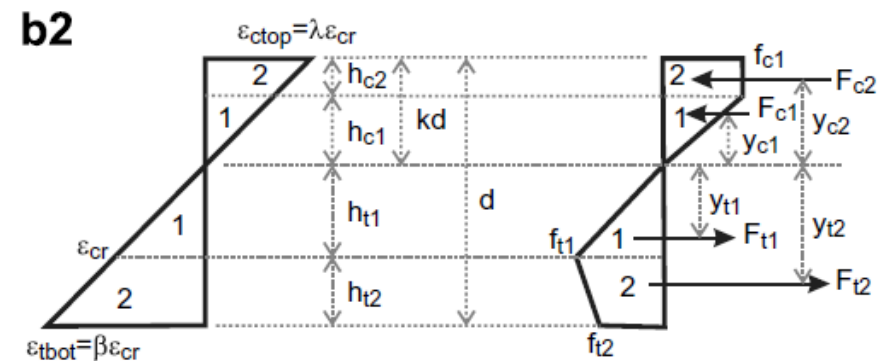
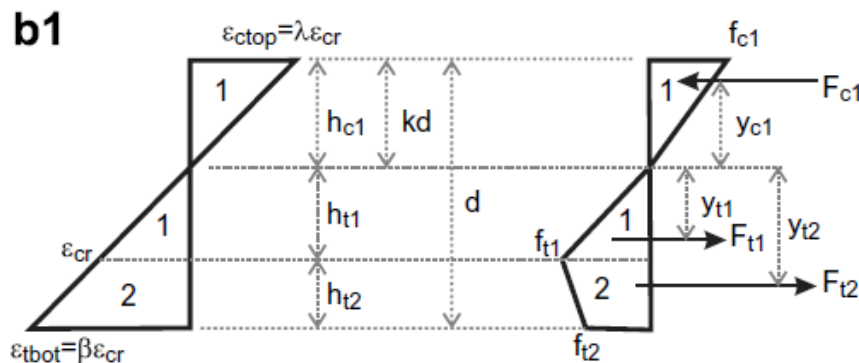
Stress and Strain Distributions in Different Stages

$$0 < \beta \leq 1 \text{ and } \lambda \leq \omega$$

$$1 < \beta \leq \alpha \text{ and } \lambda \leq \omega$$



$$1 < \beta \leq \alpha \text{ and } \lambda > \omega$$



$$\alpha < \beta \leq \beta_{tu} \text{ and } \lambda \leq \omega$$

$$\alpha < \beta \leq \beta_{tu} \text{ and } \lambda > \omega$$

Closed Form Solutions for Strain Hardening/Softening material

Table 6

Neutral axis depth ratio, normalized moment and curvature for each stage of normalized tensile strain at bottom fiber (β)

Stage	k	M'	ϕ'
1	$k_1 = \begin{cases} \frac{1}{2} & \text{for } \gamma = 1 \\ \frac{-1+\sqrt{\gamma}}{-1+\gamma} & \text{for } \gamma < 1 \text{ or } \gamma > 1 \end{cases}$	$M'_1 = \frac{2\beta[(\gamma-1)k_1^3 + 3k_1^2 - 3k_1 + 1]}{1-k_1}$	$\phi'_1 = \frac{\beta}{2(1-k_1)}$
2.1	$k_{21} = \frac{\beta^2\gamma + D_{21} - \sqrt{\gamma^2\beta^4 + D_{21}\gamma\beta^2}}{D_{21}}$ $D_{21} = \eta(\beta^2 - 2\beta + 1) + 2\beta - \beta^2\gamma - 1$	$M'_{21} = \frac{(2\beta\gamma + C_{21})k_{21}^3 - 3C_{21}k_{21}^2 + 3C_{21}k_{21} - C_{21}}{1-k_{21}}$ $C_{21} = \frac{-2\eta\beta^3 + 3\eta\beta^2 - 3\beta^2 - \eta + 1}{\beta^2}$	$\phi'_{21} = \frac{\beta}{2(1-k_{21})}$
2.2	$k_{22} = \frac{D_{22}}{D_{22} + 2\omega\gamma\beta}$ $D_{22} = \eta(\beta^2 - 2\beta + 1) + 2\beta + \omega^2\gamma - 1$	$M'_{22} = (3\omega\gamma + C_{22})k_{22}^2 - 2C_{22}k_{22} + C_{22}$ $C_{22} = \frac{2\eta\beta^3 - 3\eta\beta^2 + 3\beta^2 - \omega^3\gamma + \eta - 1}{\beta^2}$	$\phi'_{22} = \frac{\beta}{2(1-k_{22})}$
3.1	$k_{31} = \frac{D_{31} - \sqrt{\gamma\beta^2 D_{31}}}{D_{31} - \beta^2\gamma}$ $D_{31} = \eta(\alpha^2 - 2\alpha + 1) + 2\mu(\beta - \alpha) + 2\alpha - 1$	$M'_{31} = \frac{(C_{31} - 2\beta\gamma)k_{31}^3 - 3C_{31}k_{31}^2 + 3C_{31}k_{31} - C_{31}}{k_{31} - 1}$ $C_{31} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 + \eta - 1}{\beta^2}$	$\phi'_{31} = \frac{\beta}{2(1-k_{31})}$
3.2	$k_{32} = \frac{D_{32}}{D_{32} + 2\omega\gamma\beta}$ $D_{32} = \omega^2\gamma + \eta\alpha^2 + 2(\mu\beta - \eta\alpha - \mu\alpha + \alpha) + \eta - 1$	$M'_{32} = (C_{32} + 3\omega\gamma)k_{32}^2 - 2C_{32}k_{32} + C_{32}$ $C_{32} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 - \omega^3\gamma + \eta - 1}{\beta^2}$	$\phi'_{32} = \frac{\beta}{2(1-k_{32})}$

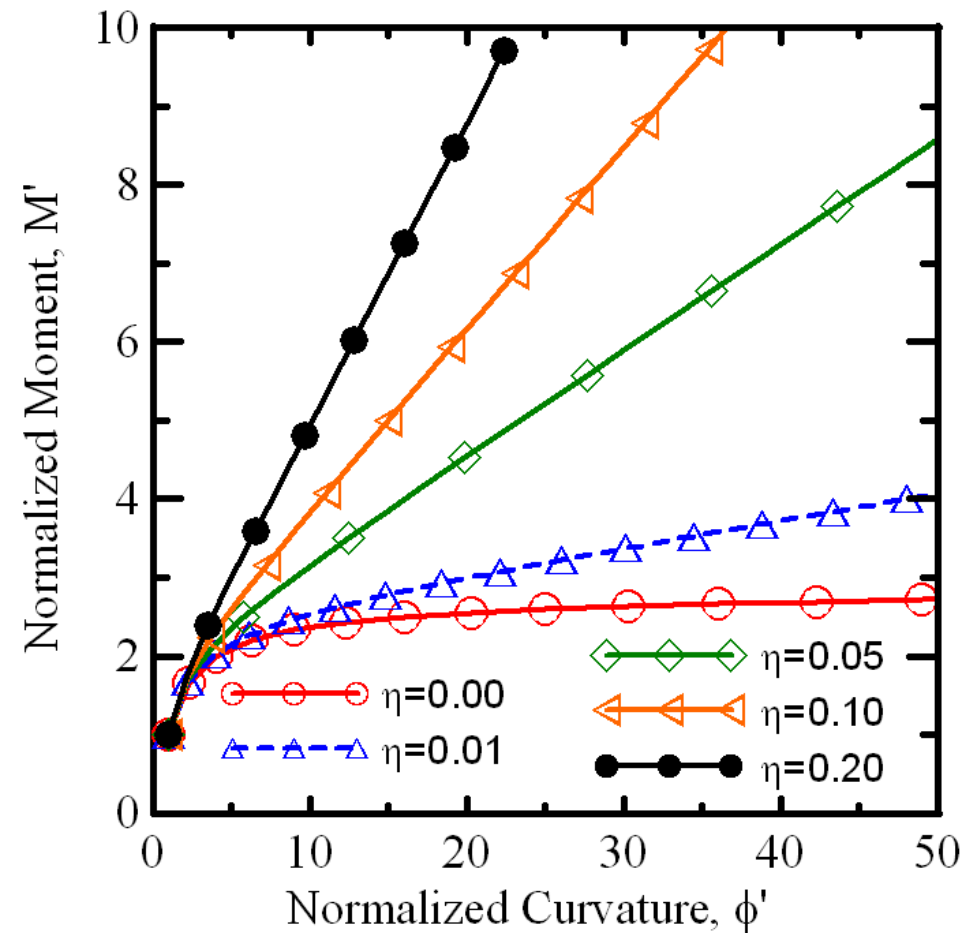
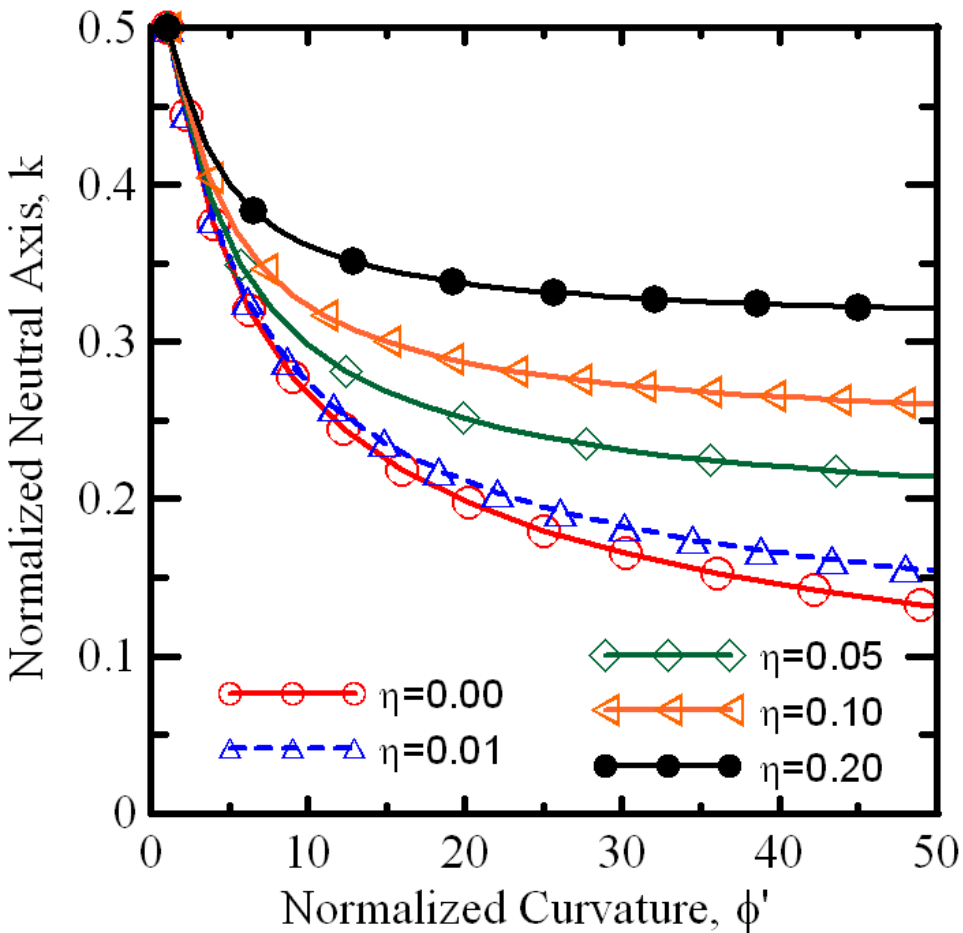
$$M = M' M_{cr}$$

$$\phi = \phi' \phi_{cr}$$

$$M_{cr} = \frac{1}{6} b d^2 E \varepsilon_{cr}$$

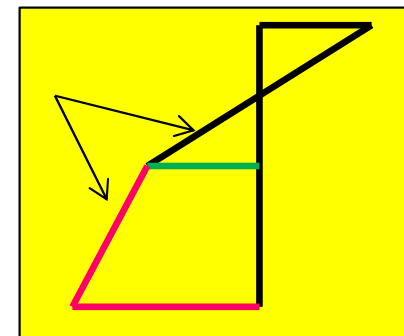
$$\phi_{cr} = \frac{2\varepsilon_{cr}}{d}$$

Range 2.1 - Compression Elastic - Tension Hardening



Elastic perfectly plastic tension ($\eta=0$)
Elastic ($\eta=1$)

η =ratio of Slopes



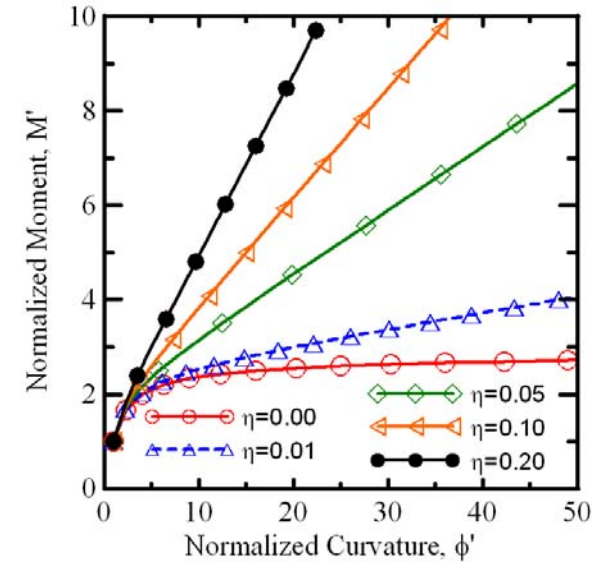
Range 2.1 -Compression Elastic - Tension Hardening

$$k = \frac{\sqrt{A}}{\sqrt{A} + \beta\sqrt{\gamma}}$$

$$A = \eta(\beta^2 + 1 - 2\beta) + 2\beta - 1$$

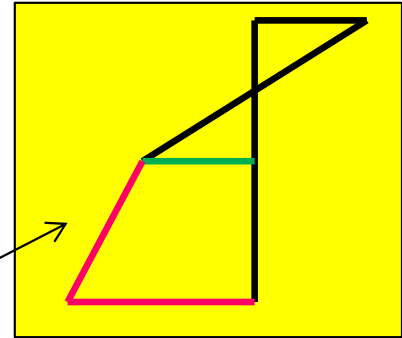
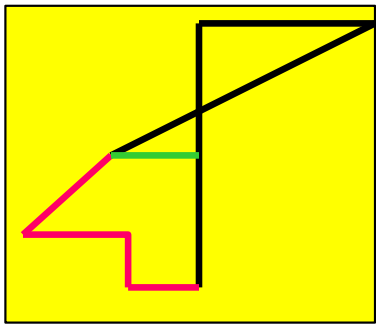
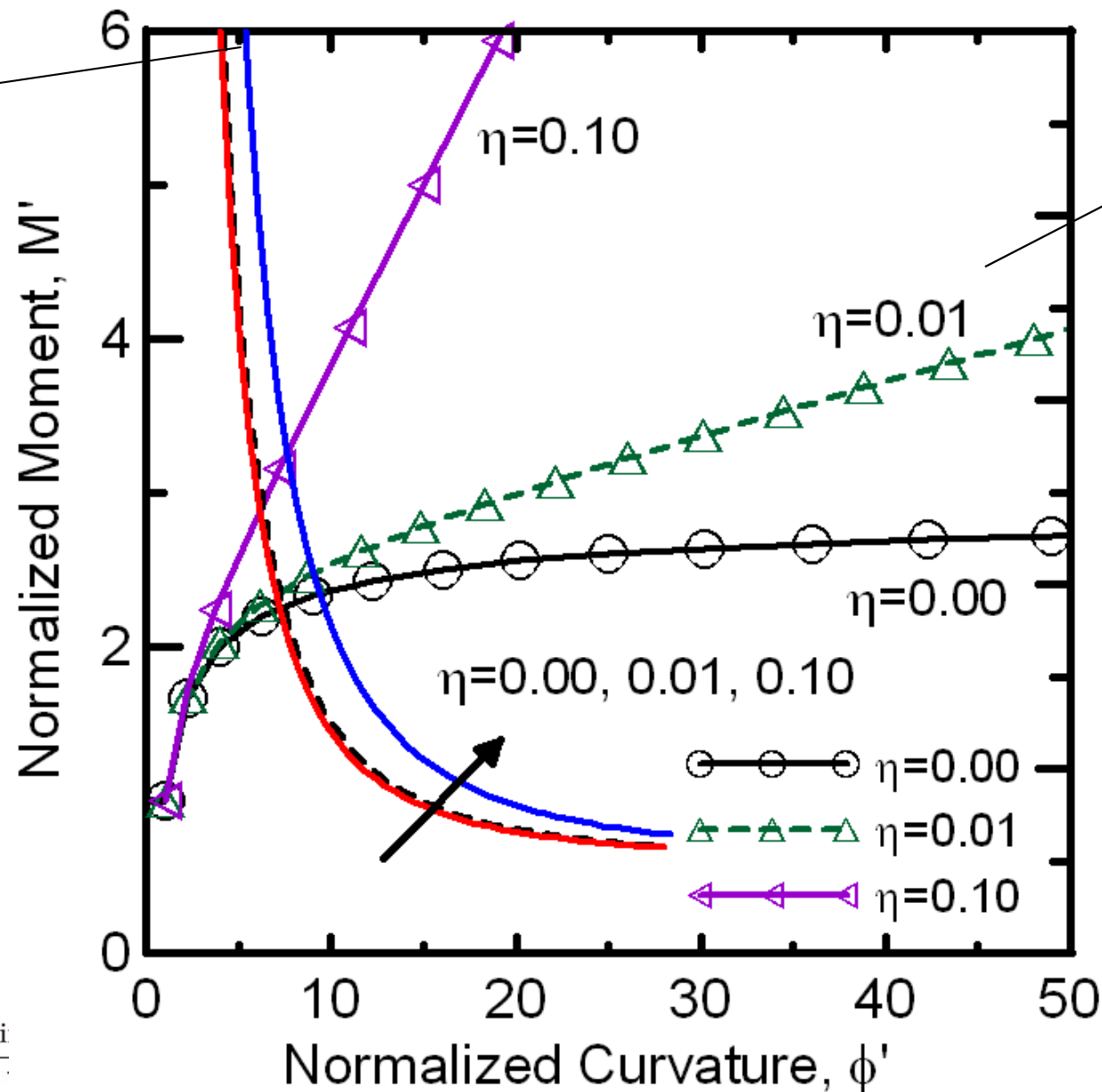
Special Case: $E_c = E_t$ ($\gamma=1$),
Elastic perfectly plastic tension ($\eta=0$)

$$k = \frac{\sqrt{2\beta-1}}{\sqrt{2\beta-1} + \beta}$$

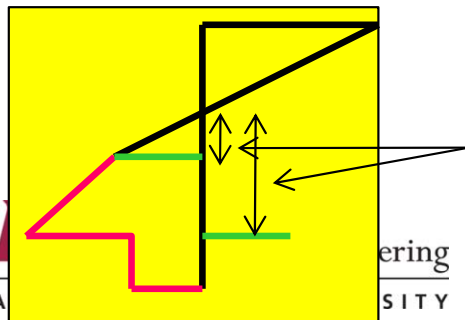
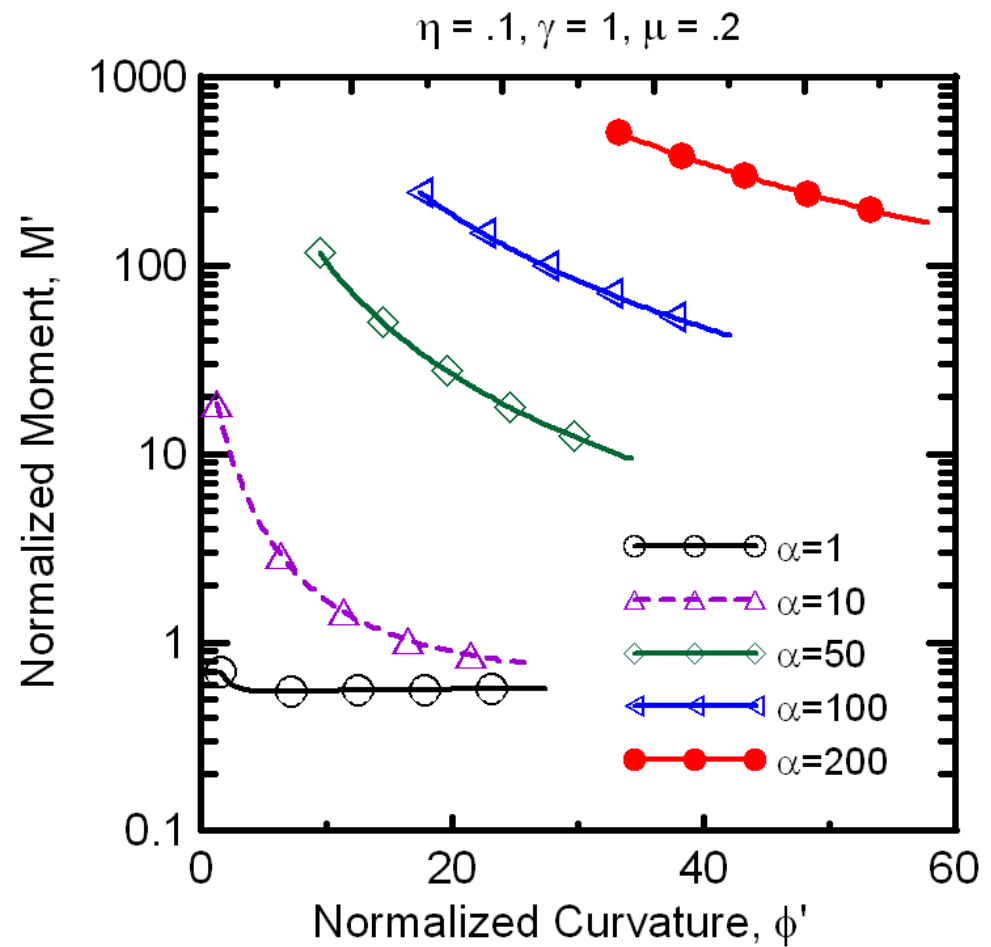
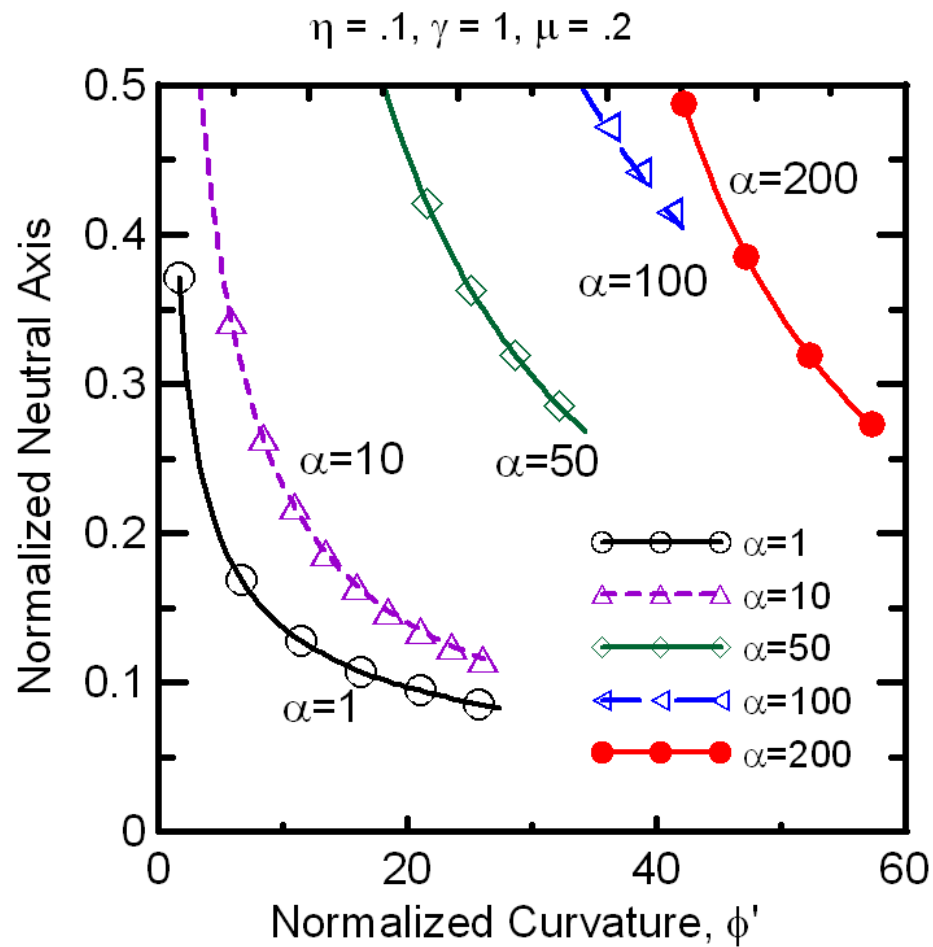


η	$A, (k = \frac{\sqrt{A}}{\sqrt{A} + \beta})$	$M'(k)$	$M'(\phi)$
0.5	$0.5(\beta^2 + 1 - 2\beta) + 2\beta - 1$	$-0.773 + 0.108 \times 10^{-1} k^{-6}$	$0.507 + 0.686 \phi$
0.2	$0.2(\beta^2 + 1 - 2\beta) + 2\beta - 1$	$0.654 + 0.516 \times 10^{-2} k^{-6}$	$1.105 + 0.383 \phi$
0.1	$0.1(\beta^2 + 1 - 2\beta) + 2\beta - 1$	$1.276 + 0.289 \times 10^{-2} k^{-6}$	$1.461 + .234 \phi$
0.05	$0.05(\beta^2 + 1 - 2\beta) + 2\beta - 1$	$1.645 + .1632 \times 10^{-2} k^{-6}$	$1.720 + .1401 \phi$
0.01	$0.01(\beta^2 + 1 - 2\beta) + 2\beta - 1$	$0.852 + 0.456 k^{-1}$	$1.342 + 0.371\sqrt{\phi}$

Interaction of Failure Zones (three tension modes, one compression)

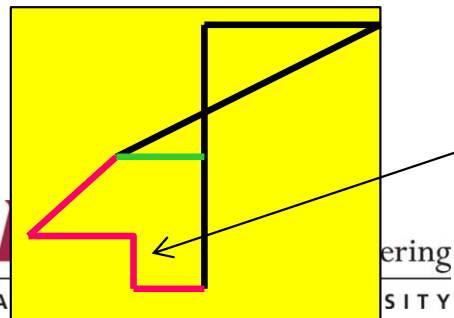
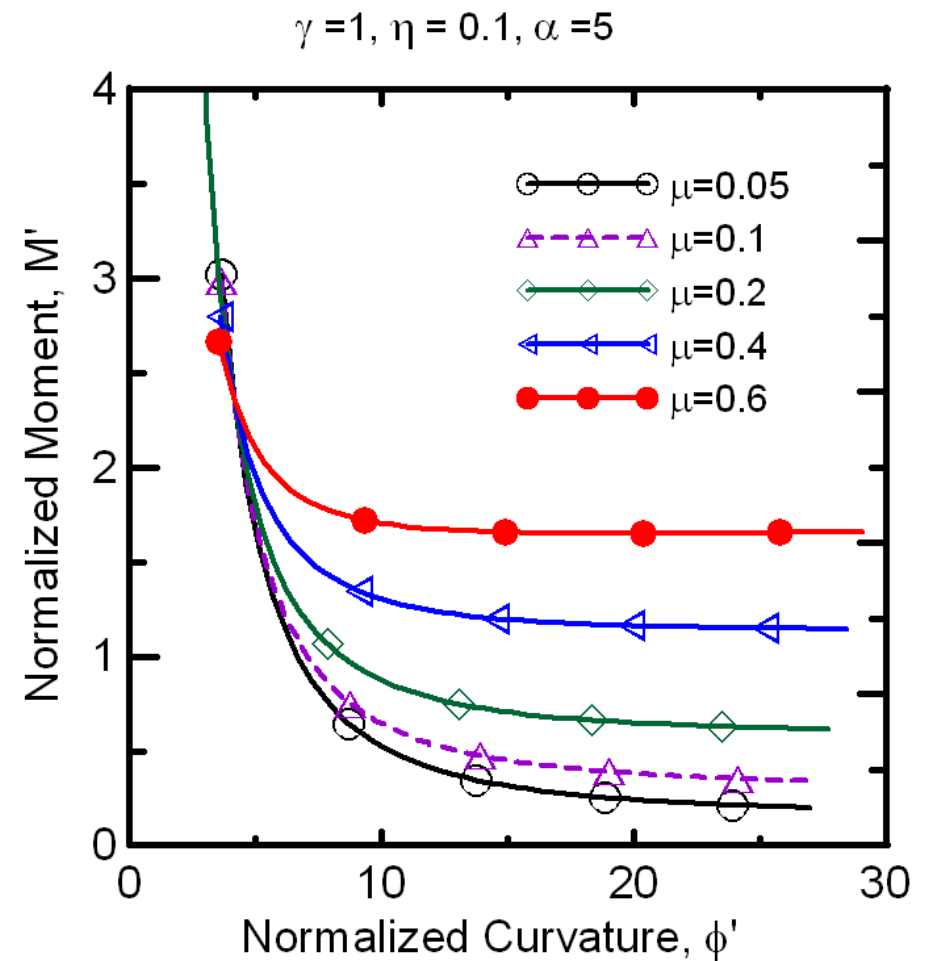
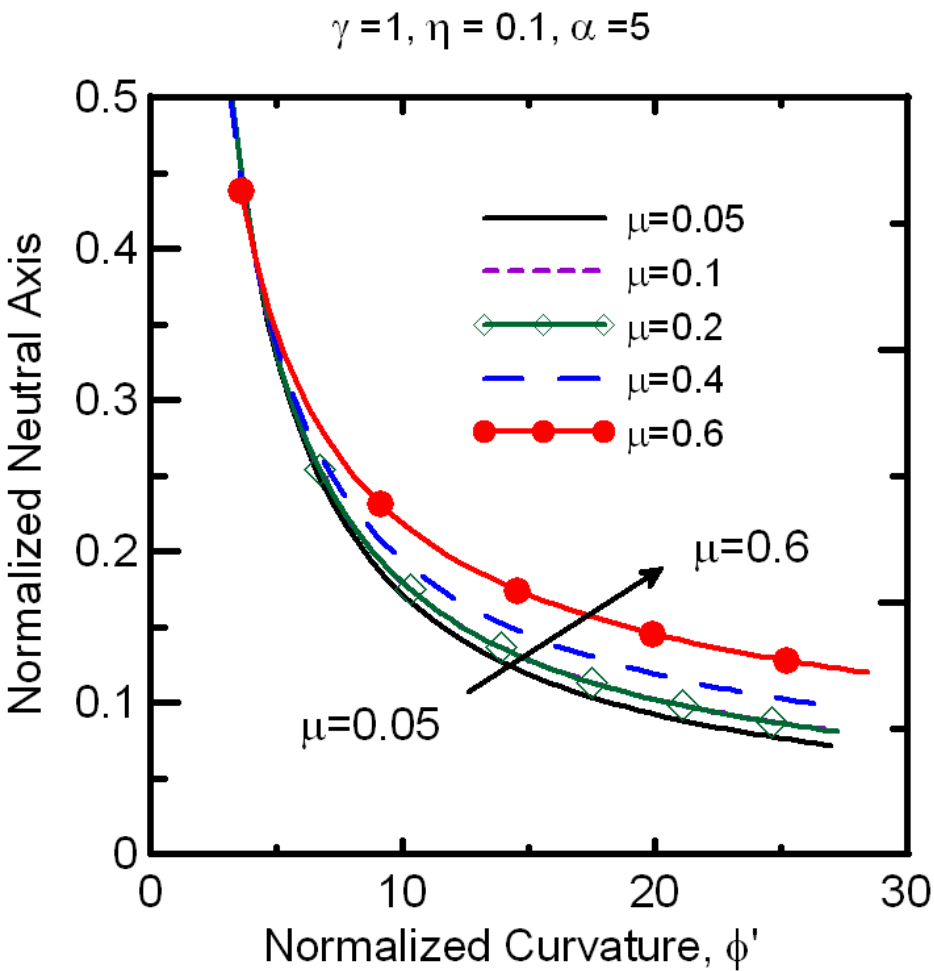


Range 3.1 Compression Elastic Tension-Softening Effect of Ultimate Strain Capacity, α



α =ratio of ultimate strain capacity to first crack strain

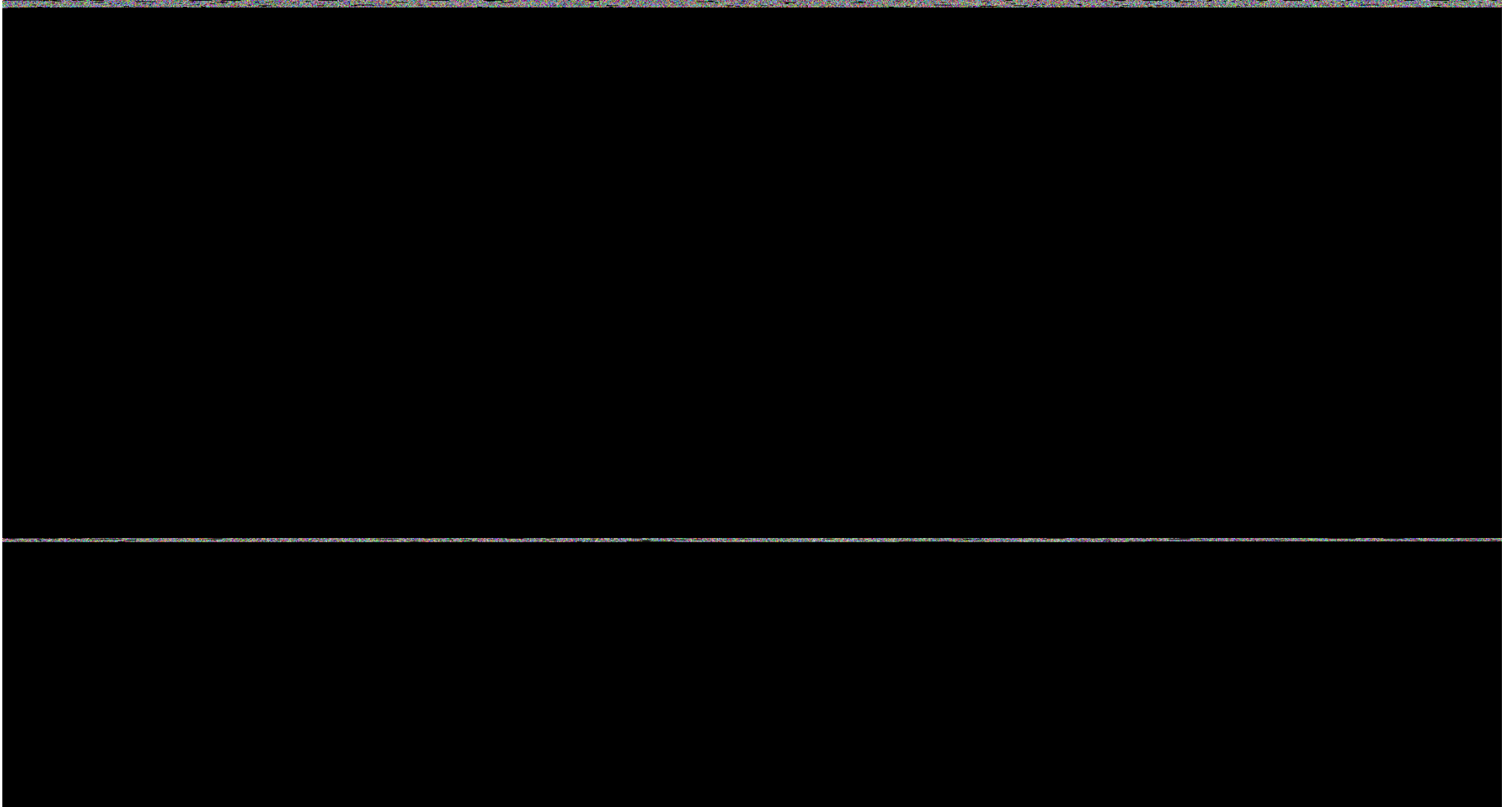
Range 3.1 Compression Elastic Tension-Softening Effect of Post peak Stress level, μ



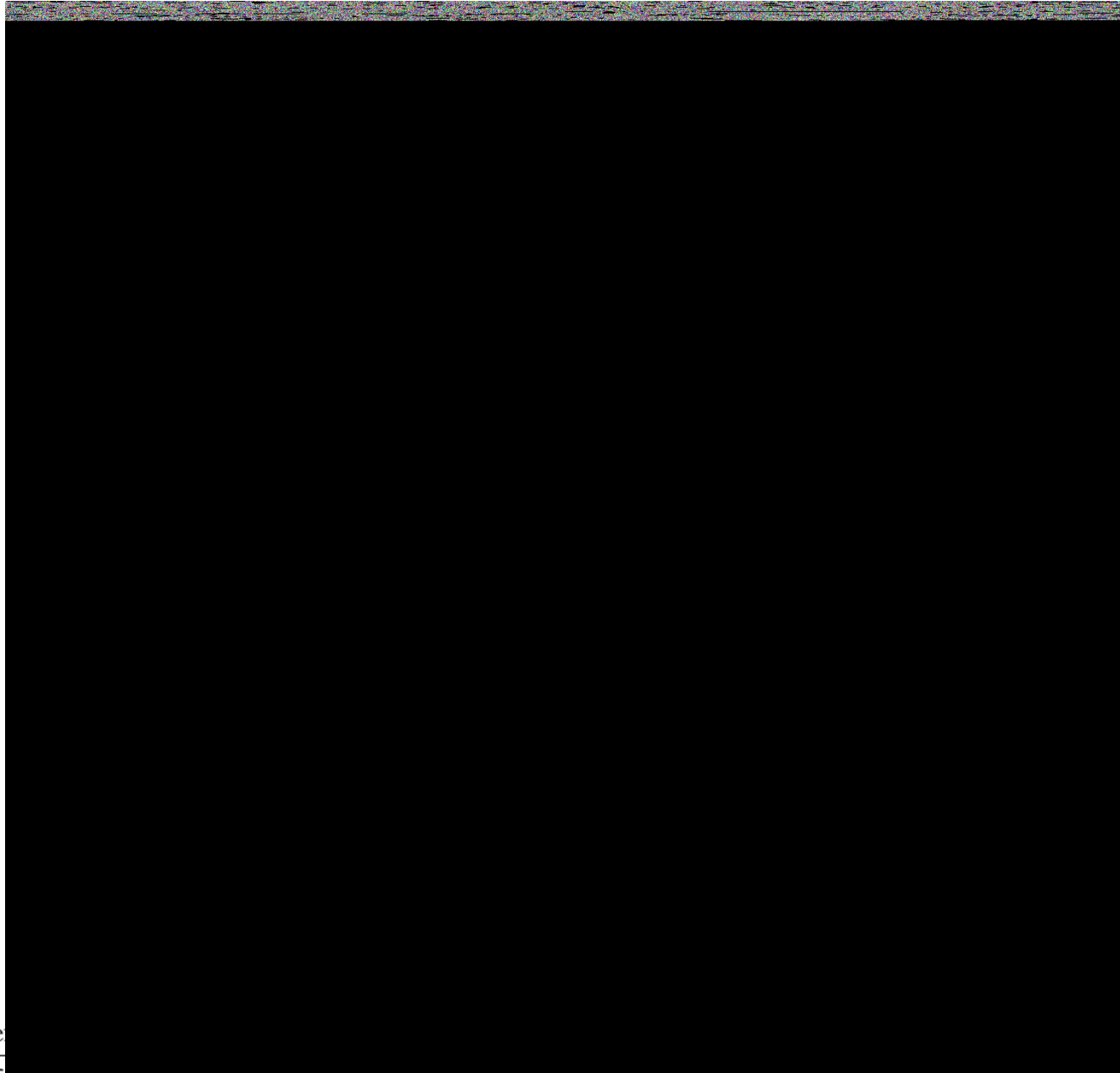
Post peak Stress level, μ

ering
SITY

Localized hinge Model for Reinforced Concrete

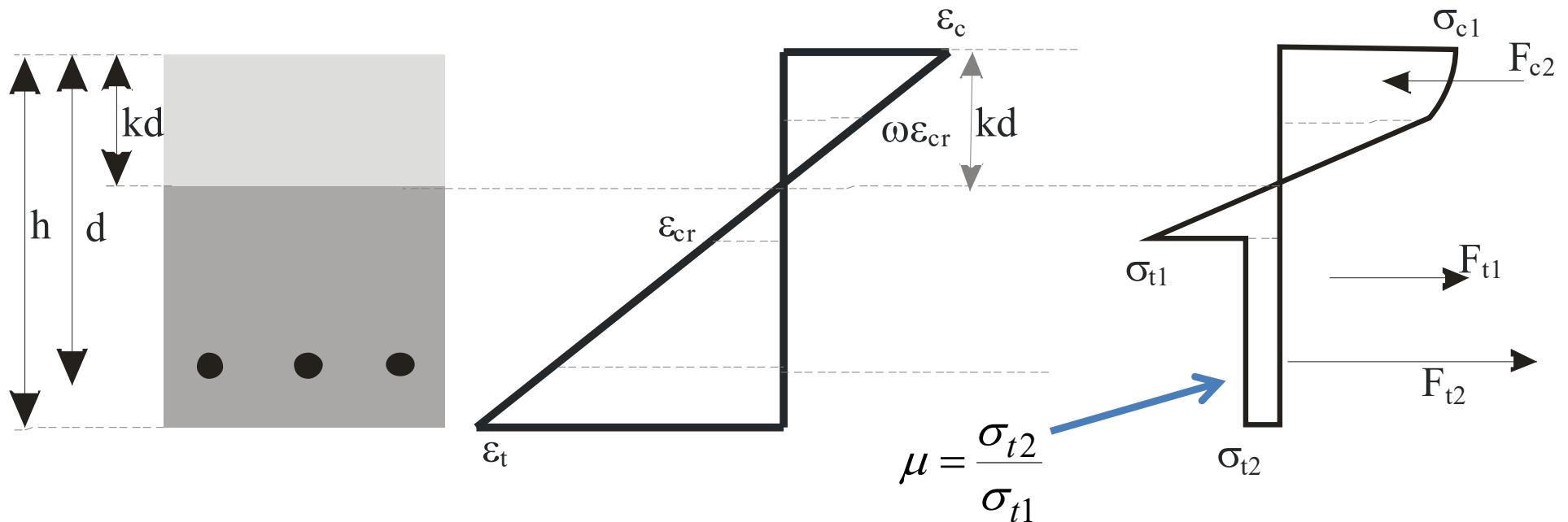


Parameterized Material Models



Simplified Design Approach for FRC

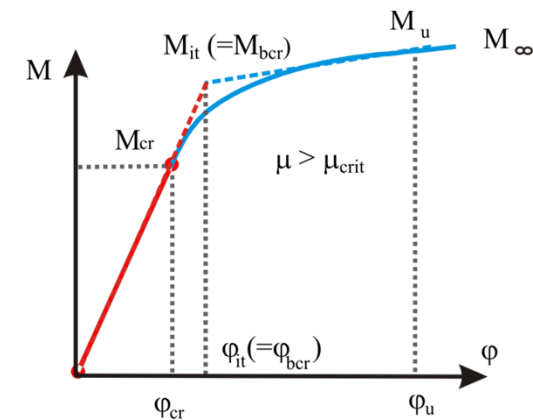
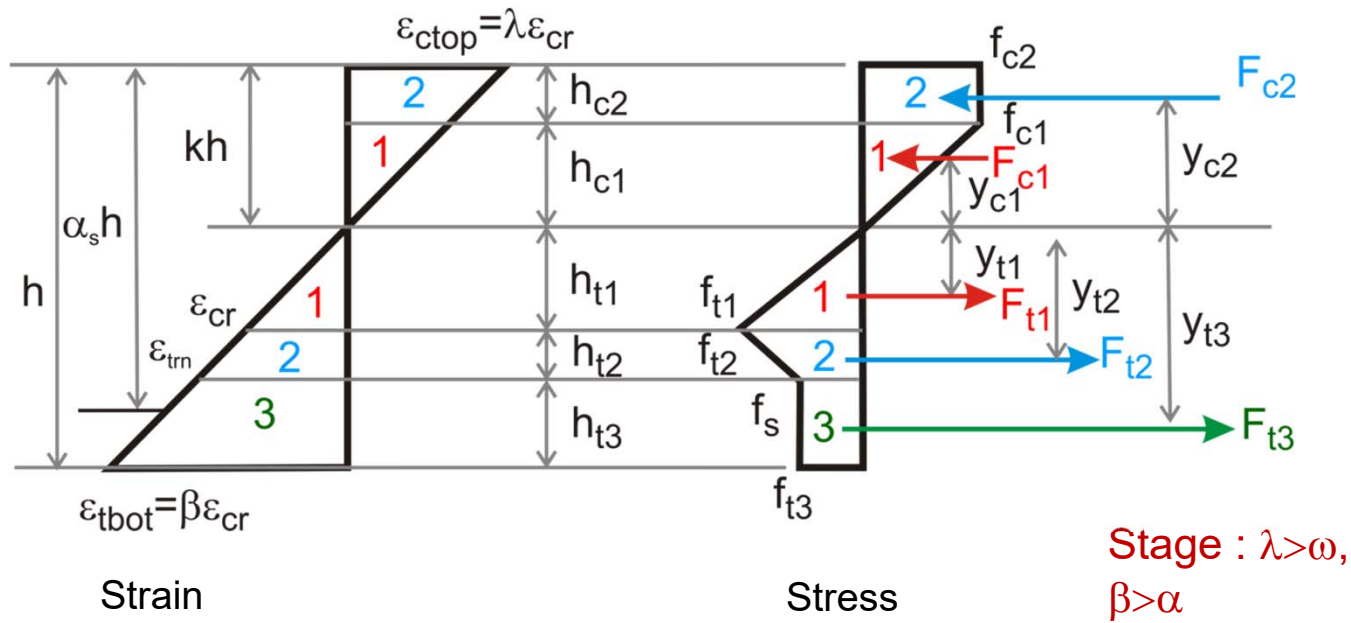
- materials model, testing techniques, and design codes
- Simplifies FRC Flexural design assumption for a beam with traditional and fiber reinforcement.
- Use a stress distribution analogous to Whitney's stress block diagram
- Tensile stresses in accordance with stress-strain response applied over the tensile domain
- Integrate to compute the moment capacity



tensile stress transmitted by the fibers across the flexural crack. A measure of residual tensile stress-strain response.

$$M_{ult} = \left[\frac{\mu \sqrt{f'_c}}{15.55\mu + 2\sqrt{f'_c}} \right] bd^2 E \epsilon_{cr}$$

Derivation of Moment-Curvature Relationship



Incrementally
impose
 $0 \leq \varepsilon_t \leq \varepsilon_{tu}$

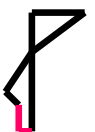
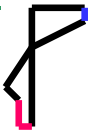
Strain Distribution
Stress Distribution

$\Sigma F = 0$,
determine k

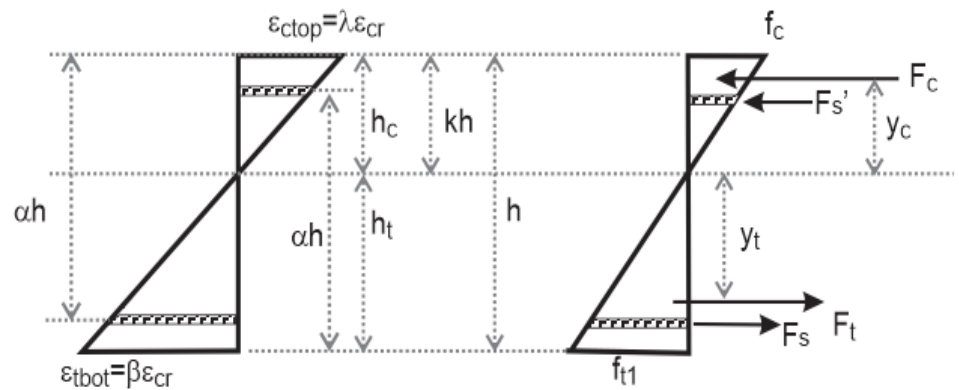
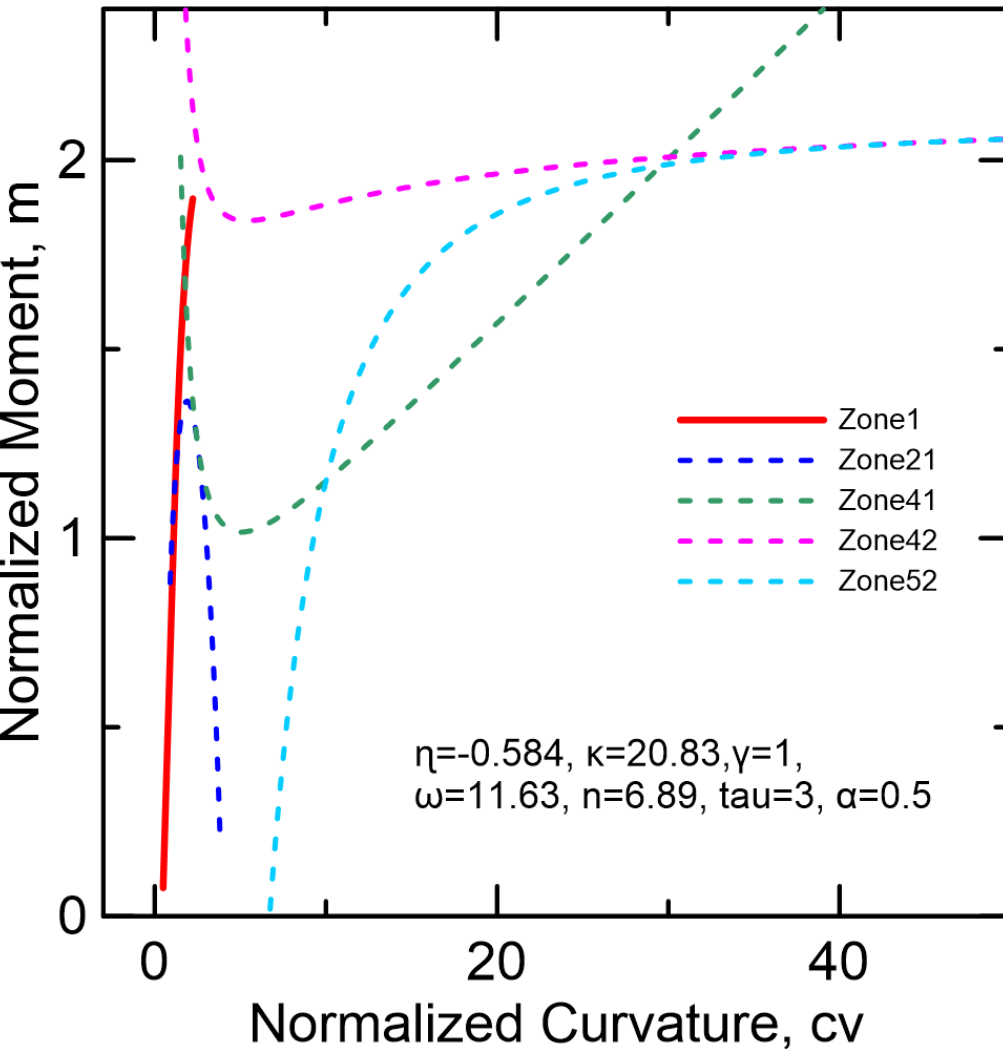
Moment: $M = \Sigma F_{ci} y_{ci} + \Sigma F_{ti} y_{ti}$
Curvature: $\phi = \varepsilon_c / kh$

Moment Curvature Diagram for hardening –softening materials

$$\phi' = \phi / \phi_{cr} \quad \phi'_i = \frac{\beta}{2(1 - k_i)}$$

Stage	Range	k	M' = M/M _{cr}	
1	(0 < β ≤ 1) and (λ ≤ ω)	$k_1 = \begin{cases} \frac{-1 + \sqrt{\gamma}}{-1 + \gamma} & \text{for } \gamma > 1 \\ \frac{1}{2} & \text{for } \gamma = 1 \end{cases}$	$M'_{11} = \frac{2\beta[(\gamma - 1)k_1^3 + 3k_1^2 - 3k_1 + 1]}{1 - k_1}$	
2.1	(1 < β ≤ α) and (λ ≤ ω)	$k_{21} = \frac{\beta^2\gamma + D_{21} - \sqrt{\gamma^2\beta^4 + D_{21}\gamma\beta^2}}{D_{21}}$ $D_{21} = \eta(\beta^2 - 2\beta + 1) + 2\beta - \beta^2\gamma - 1$	$M'_{21} = \frac{(2\beta\gamma + C_{21})k_1^3 - 3C_{21}k_1^2 + 3C_{21}k_1 - C_{21}}{1 - k_1}$ $C_{21} = \frac{-2\eta\beta^3 + 3\eta\beta^2 - 3\beta^2 - \eta + 1}{\beta^2}$	
2.2	(1 < β ≤ α) and (ω < λ ≤ λ _{cu})	$k_{22} = \frac{D_{22}}{D_{22} + 2\omega\gamma\beta}$ $D_{22} = \eta(\beta^2 - 2\beta + 1) + 2\beta + \omega^2\gamma - 1$	$M'_{22} = (3\omega\gamma + C_{22})k_{22}^2 - 2C_{22}k_{22} + C_{22}$ $C_{22} = \frac{2\eta\beta^3 - 3\eta\beta^2 + 3\beta^2 - \omega^3\gamma + \eta - 1}{\beta^2}$	
3.1	(β > α) and (λ ≤ ω)	$k_{31} = \frac{D_{31} - \sqrt{\gamma\beta^2 D_{31}}}{D_{31} - \beta^2\gamma}$ $D_{31} = \eta(\alpha^2 - 2\alpha + 1) + 2\mu(\beta - \alpha) + 2\alpha - 1$	$M'_{31} = \frac{(C_{31} - 2\beta\gamma)k_{31}^3 - 3C_{31}k_{31}^2 + 3C_{31}k_{31} - C_{31}}{k_{31} - 1}$ $C_{31} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 + \eta - 1}{\beta^2}$	
3.2	(β > α) and (ω < λ ≤ λ _{cu})	$k_{32} = \frac{D_{32}}{D_{32} + 2\omega\gamma\beta}$ $D_{32} = \omega^2\gamma + \eta\alpha^2 + 2(\mu\beta - \eta\alpha - \mu\alpha + \alpha) + \eta - 1$	$M'_{32} = (C_{32} + 3\omega\gamma)k_{32}^2 - 2C_{32}k_{32} + C_{32}$ $C_{32} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 - \omega^3\gamma + \eta - 1}{\beta^2}$	

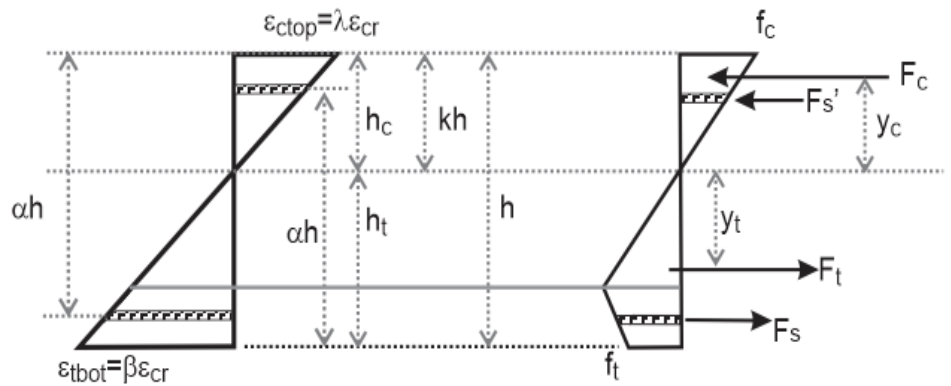
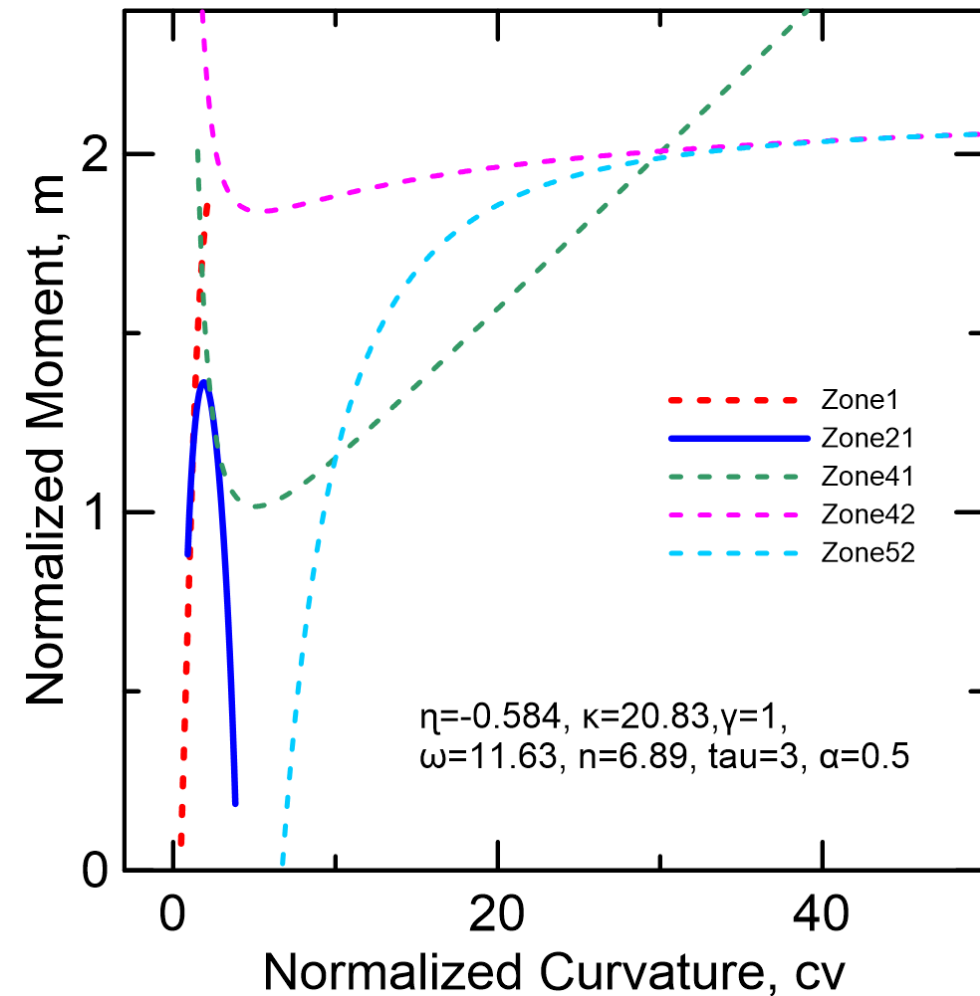
Interaction of modes of strain and stress distribution at all stages



Zone1 :

- compression and tension are in elastic zone
- Steel is elastic

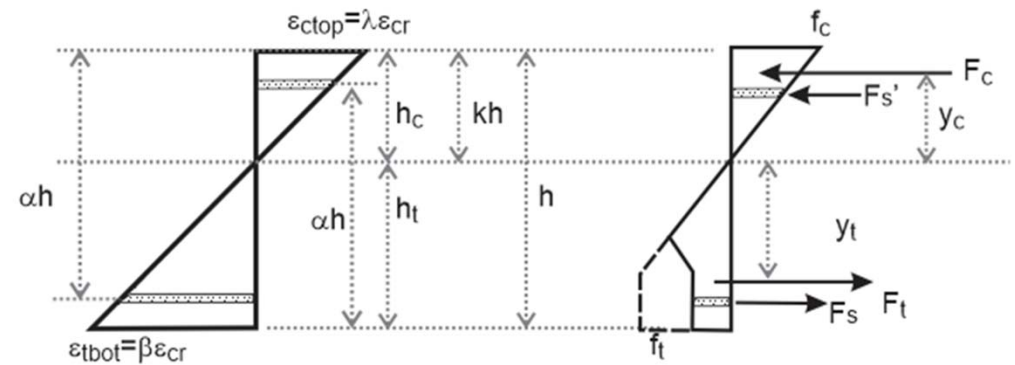
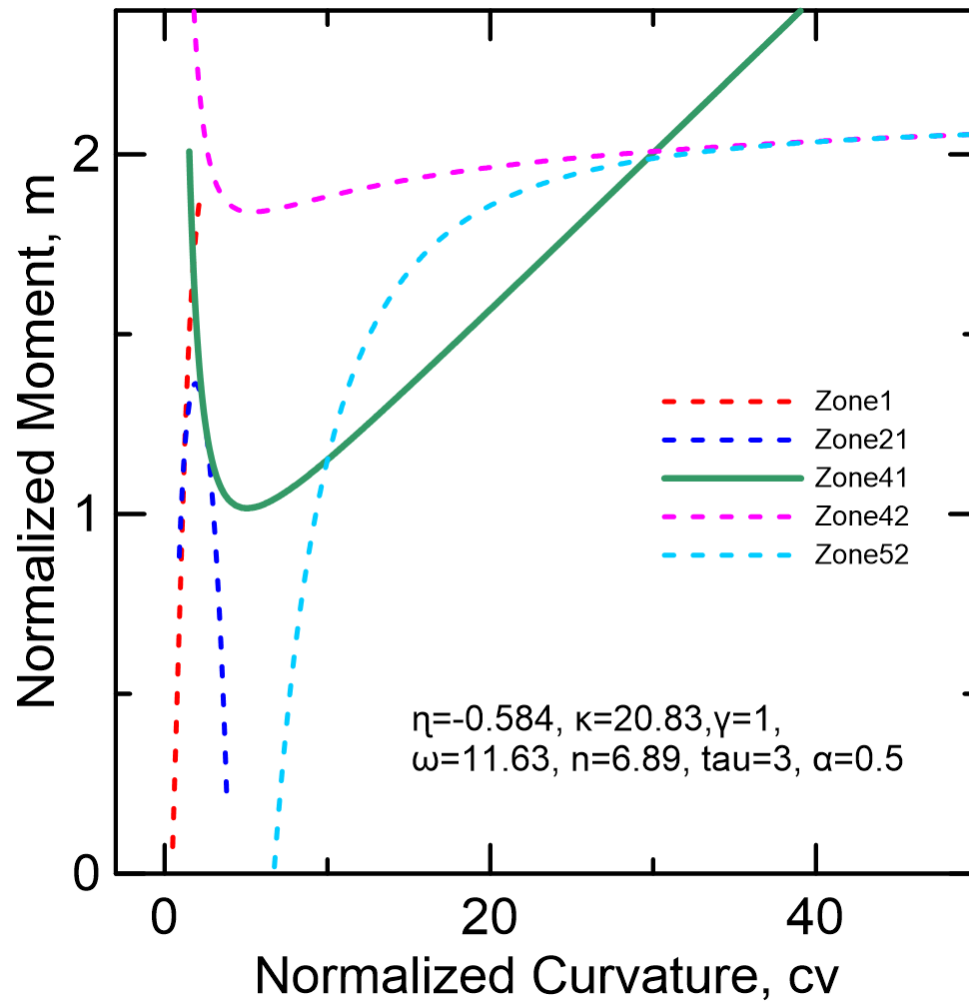
Zone2.1



Zone21

- compression is in elastic zone
- tension is second slope zone, first crack has formed
- Steel is elastic

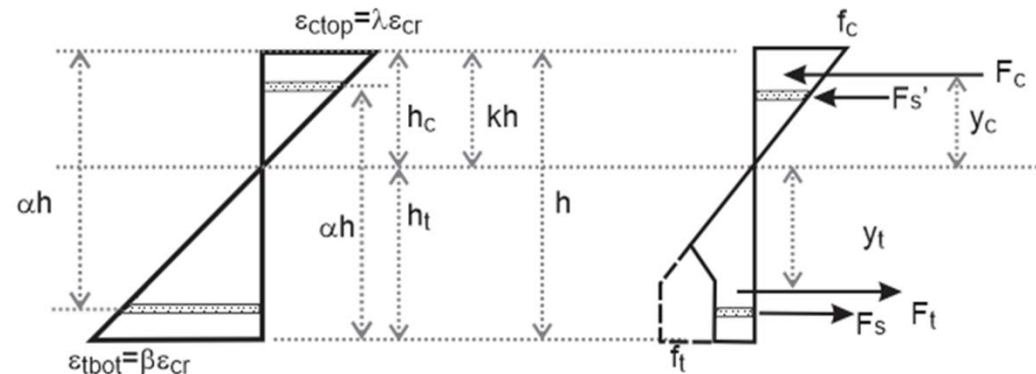
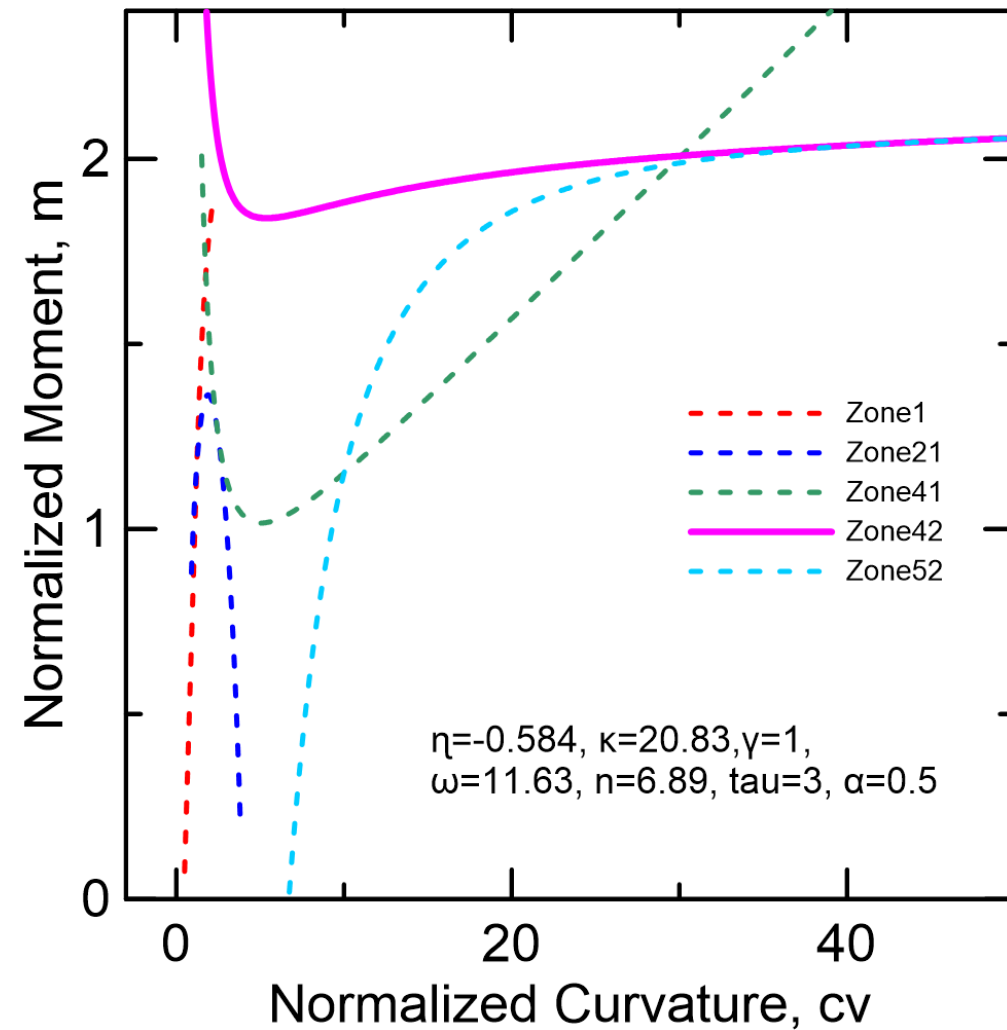
Zone4.1



Stage41 :

- compression is in elastic zone
- tension is in residual third range
- Steel is elastic

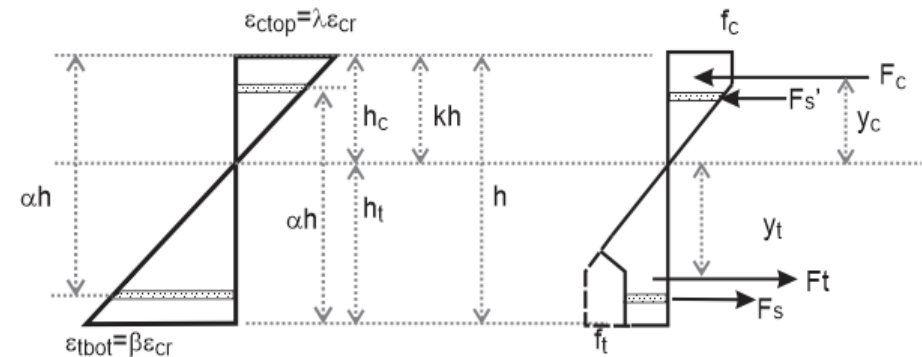
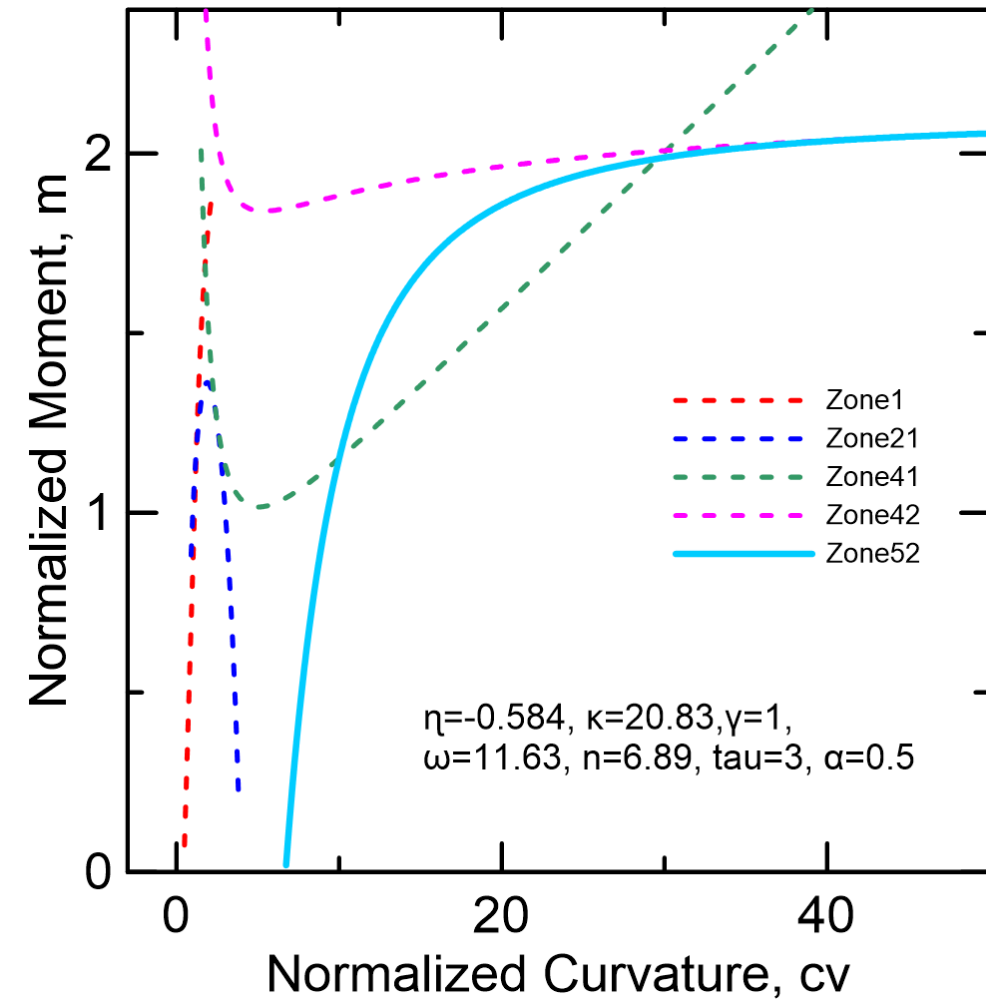
Zone4.2



Stage42 :

- compression is in elastic zone
- tension is in residual range
- Steel has yielded

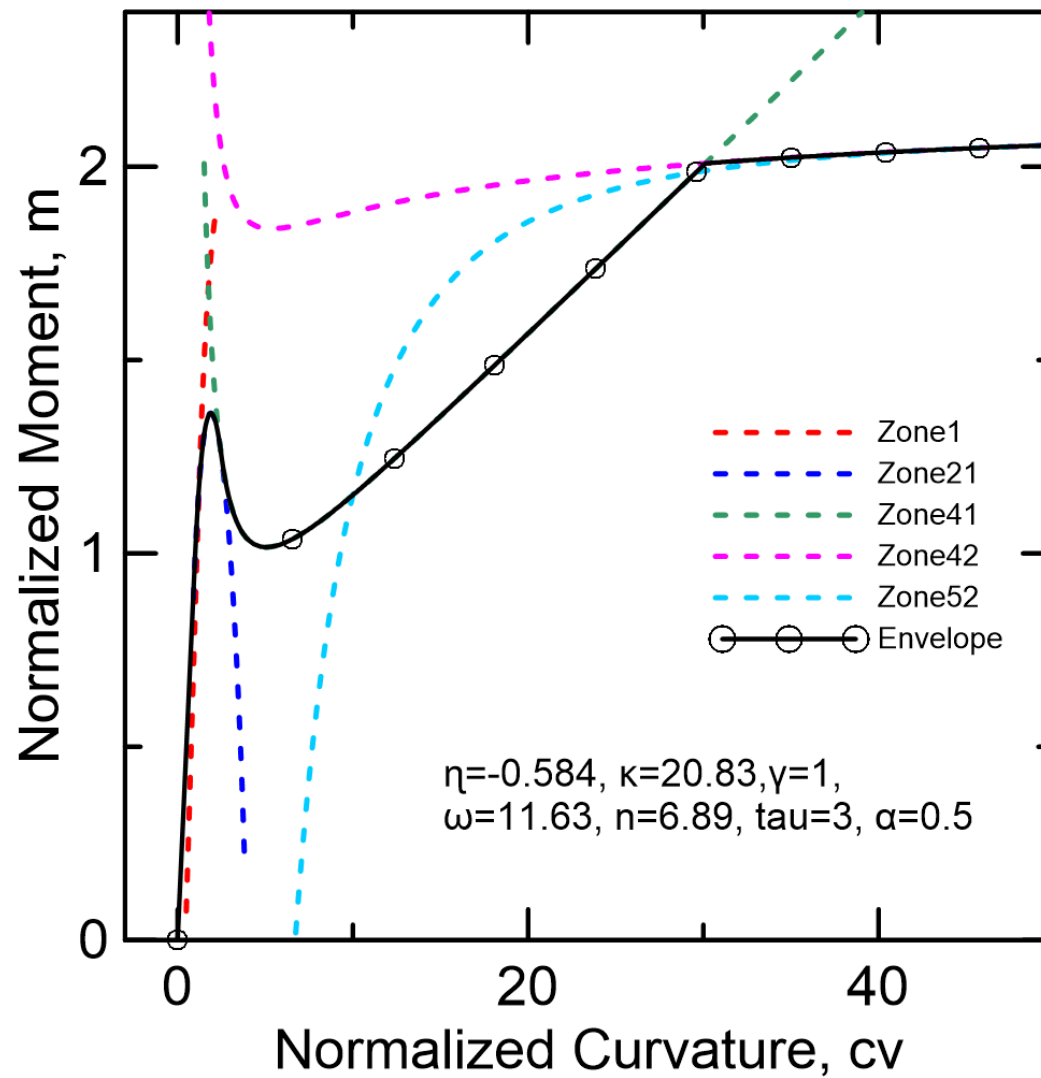
Zone5.2



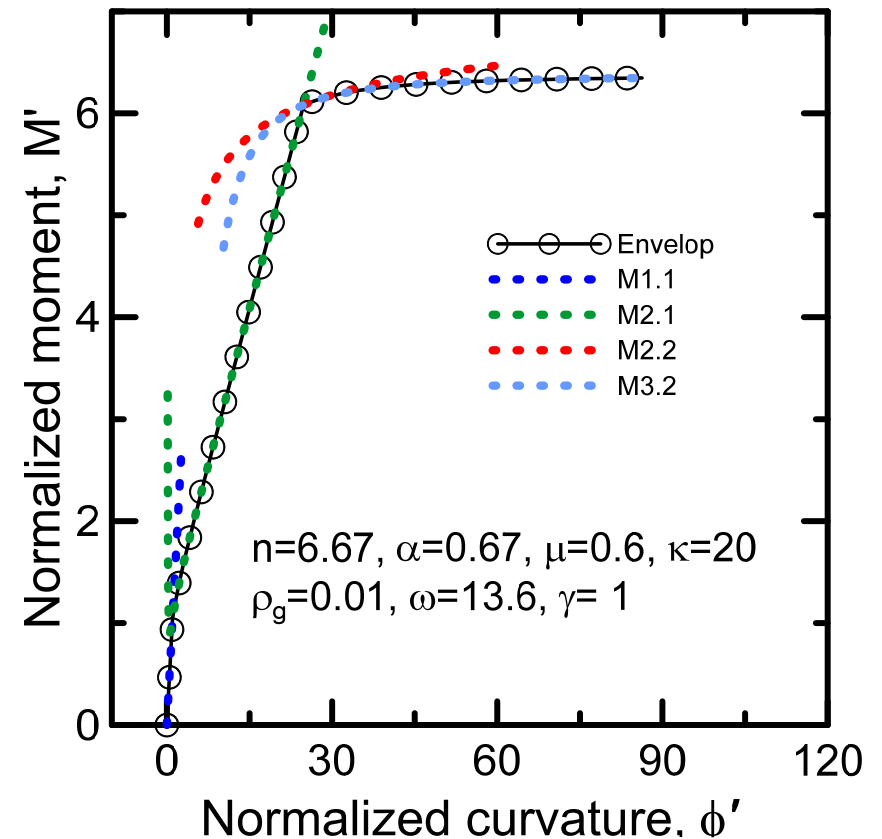
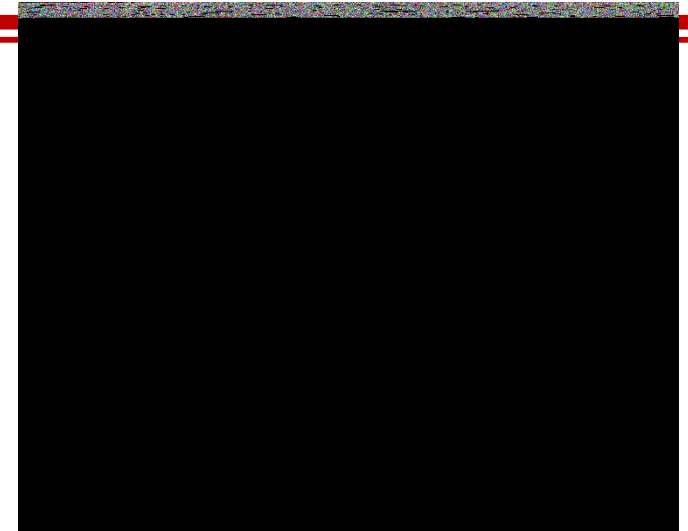
Stage52 :

- compression is in plastic zone
- tension is in residual range
- Steel has yielded

Envelope curve



Development of Envelope Moment-Curvature for Hybrid Response



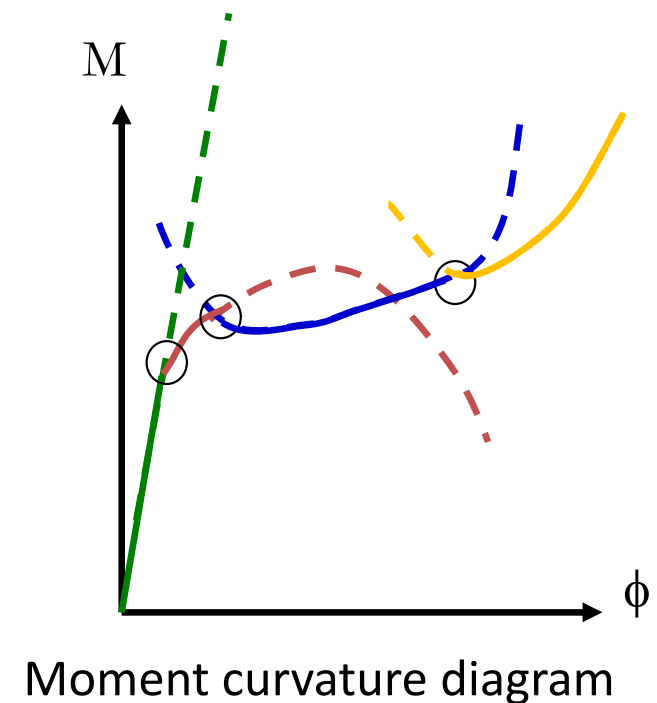
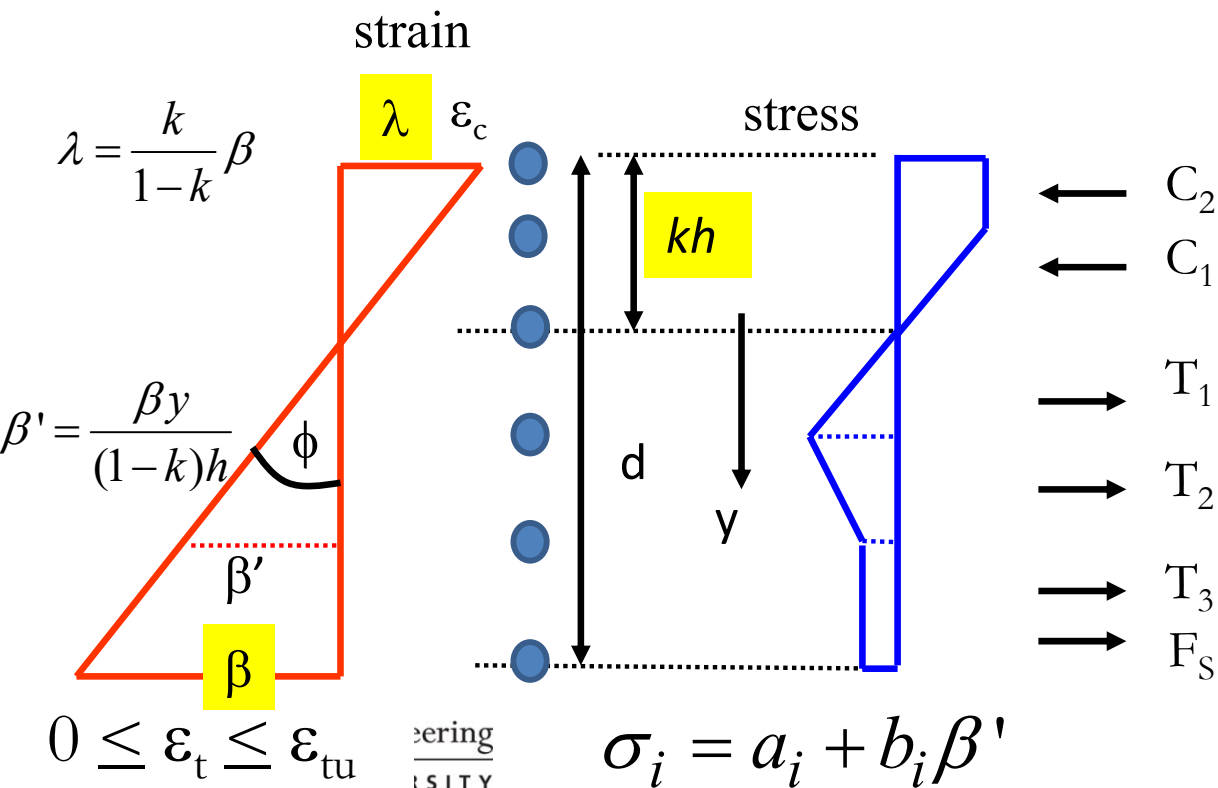
Change of variables, location of Neutral axis, $[\beta, \lambda]$ to $[\beta, k]$

- Incrementally impose $0 \leq \varepsilon_t \leq \varepsilon_{tu}$
- Strain Distribution, linear, β, β'
- Stress Distribution defined by linear segments
- $\Sigma F = 0$, determine k (Neutral axis)
- $M = \Sigma C_i y_{ci} + \Sigma T_i y_{ti}$ and $\phi = \varepsilon_c / k d$
- Normalization $M' = M / M_0$ and $\phi' = \phi / \phi_{cr}$

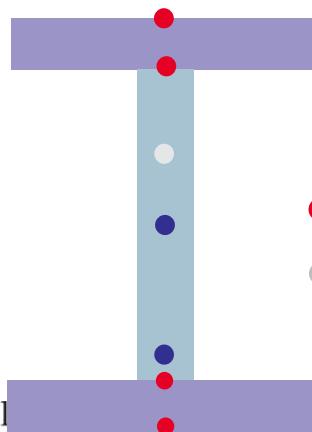
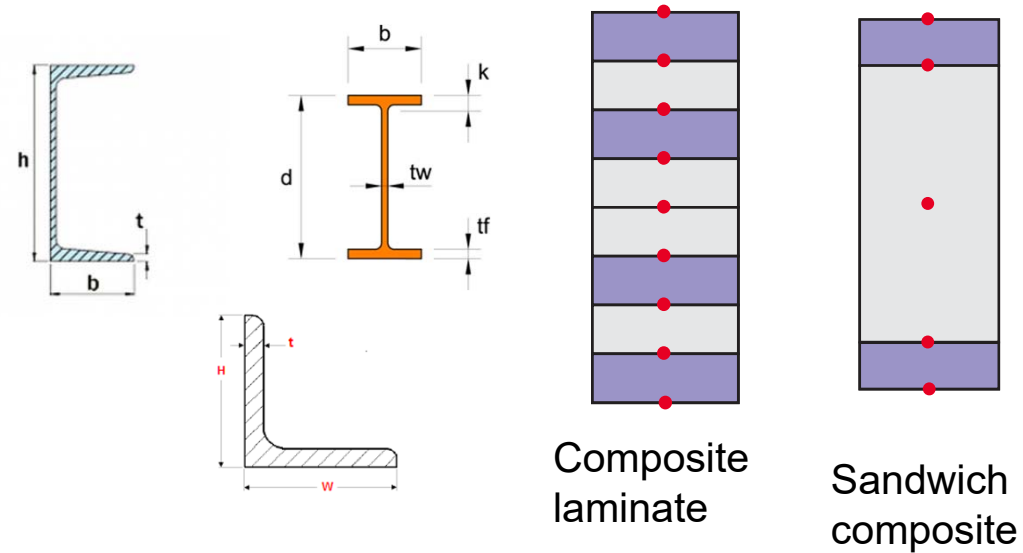
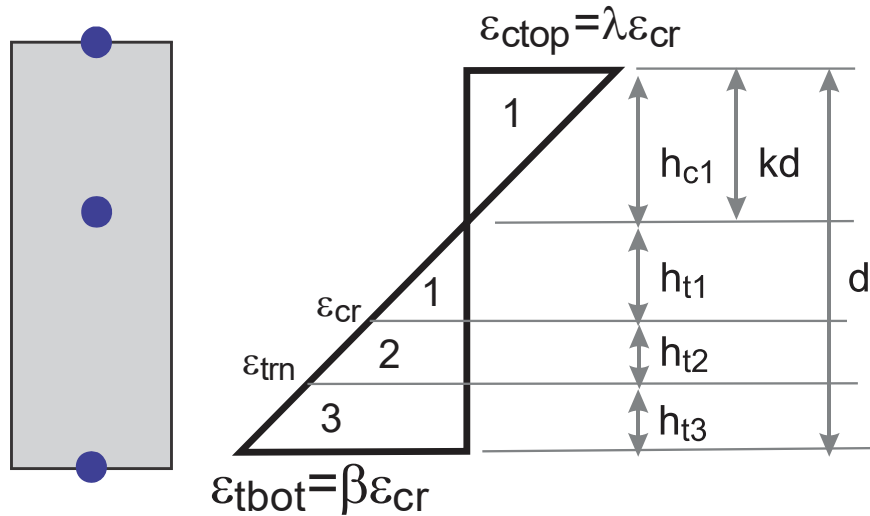
$$\sigma_i = a_i + b_i \beta'$$

$$F_{c1} = b \int_0^{kd} \sigma_i(y) dy$$

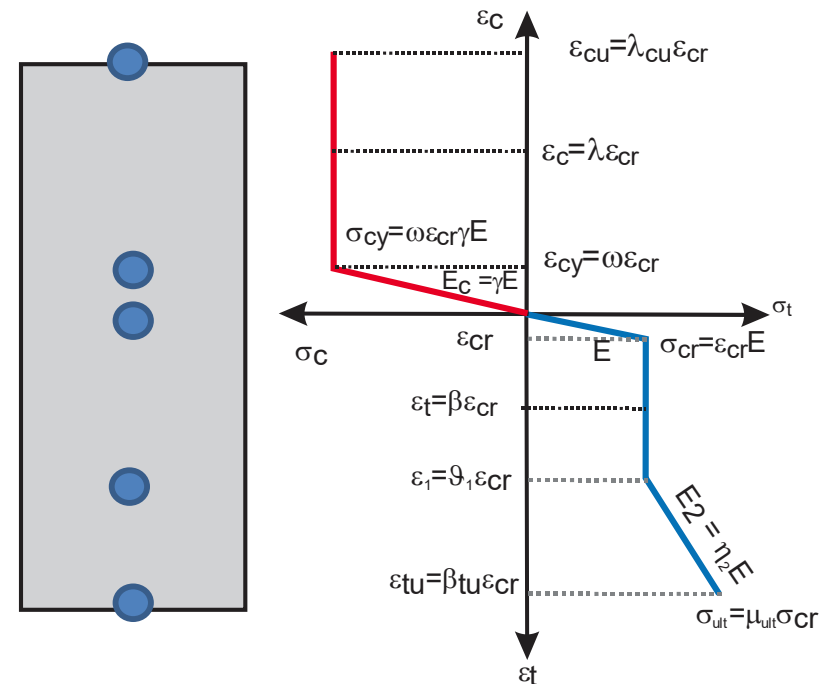
$$y_{c1} = \frac{b}{F_{c1}} \int_0^{kd} \sigma_i(y) y dy$$



Control Points, Geometrical and Material



- stress-strain control point
- geometrical control point
- neutral axis control point



Example for Control points for geometry and sections for a doubly Reinforced Concrete section

Define the strain distribution variable along the depth of the section

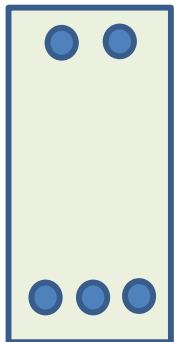
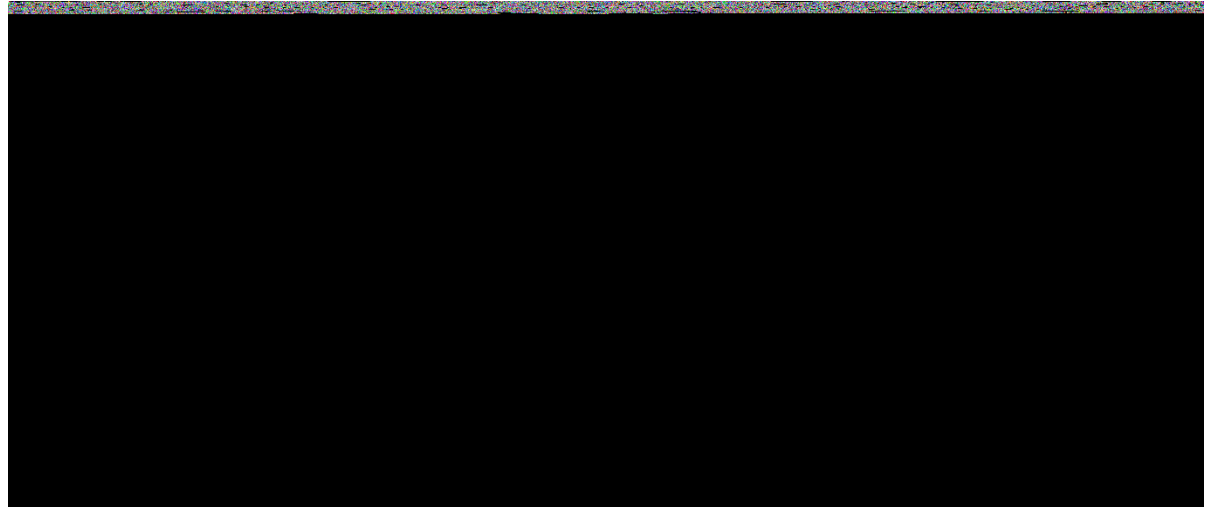
$$\beta' = \frac{\varepsilon(y)}{\varepsilon_{cr}} = \beta'(\beta, k, \frac{y}{h}) = \frac{\beta y}{(1-k)h}$$

λ = Compressive strain

β' = Tensile strain across depth

k = Neutral axis location

β = Max. Tensile strain, indep. variable



Geometrical control point	Depth, $\frac{y}{h}$	Strain Magnitude
Extreme compression fiber	$-kh$	$\lambda = \beta' = \frac{-\beta k}{1-k}$
Compression reinforcement	$-(\alpha + k - 1)$	$\chi' = \beta' = \frac{-(\alpha + k - 1)\beta}{1-k}$
Neutral axis	0	$\beta' = 0$
Tension reinforcement	$(\alpha - k)h$	$\chi = \beta' = \frac{(\alpha - k)\beta}{1-k}$
Extreme tensile fiber	$(1-k)h$	$\beta' = \beta$

Derivation of Moment-Curvature Relationship

Incrementally impose

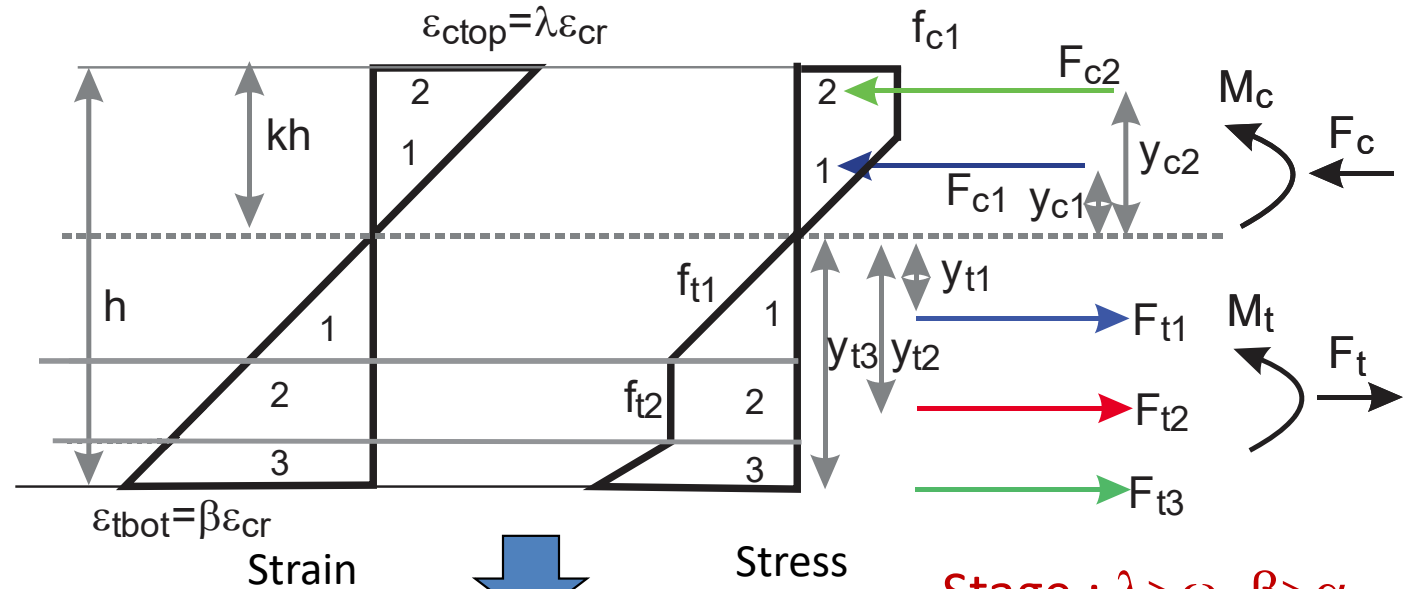
$$0 \leq \varepsilon_t \leq \varepsilon_{tu} \quad \beta$$

Strain Distribution
Stress Distribution

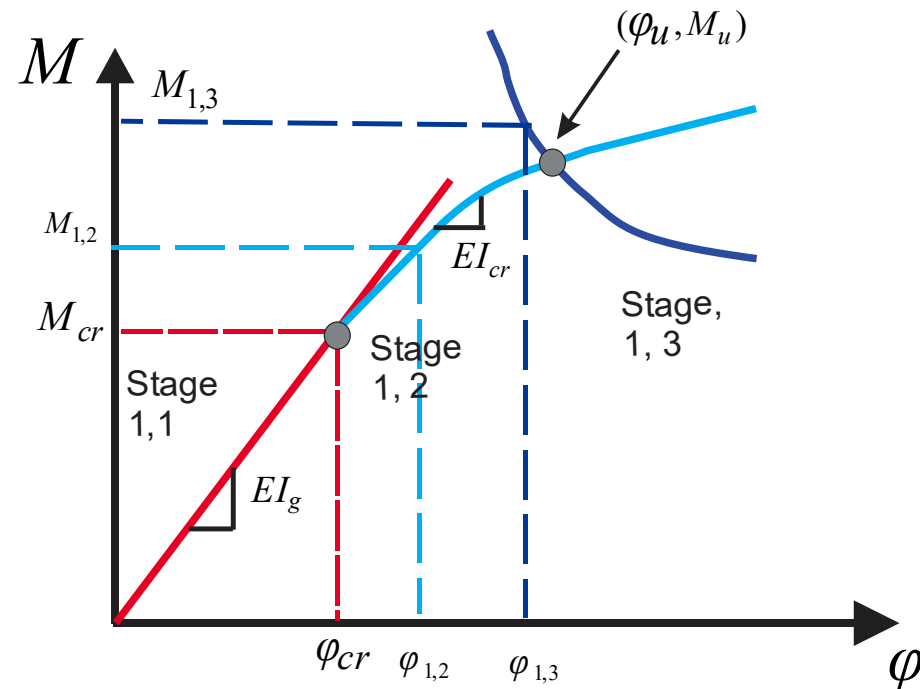
$$\Sigma F = 0, \text{ determine } k$$

$$\text{Moment: } M = \Sigma F_{ci} y_{ci} + \Sigma F_{ti} y_{ti}$$

$$\text{Curvature: } \phi = \varepsilon_c / kh$$



Stage : $\lambda > \omega, \beta > \alpha$



Evaluation of Moment-Curvature using Piecewise Functions

Stage	Tension	TRC tensile strain	Compression	TRC compressive strain
Zone 1.1	Elastic	$0 \leq \beta < 1$	Elastic	$0 \leq \lambda < \omega$
Zone 1.2	Distributed Cracking	$1 \leq \beta < \vartheta_1$	Elastic	$0 \leq \lambda < \omega$
Zone 1.3	Hardening	$\vartheta_1 \leq \beta < \beta_{tu}$	Elastic	$0 \leq \lambda < \omega$
Zone 2.2	Distributed Cracking	$1 \leq \beta < \vartheta_1$	Perfectly plastic	$\omega \leq \lambda < \lambda_{cu}$
Zone 2.3	Hardening	$\vartheta_1 \leq \beta < \beta_{tu}$	Perfectly plastic	$\omega \leq \lambda < \lambda_{cu}$

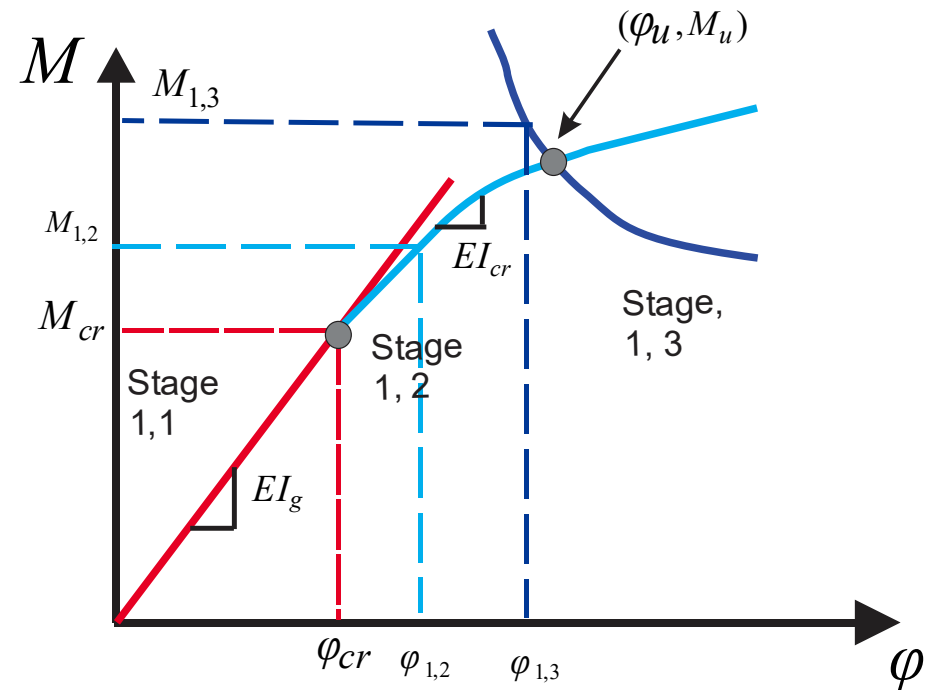
$$\phi = \frac{\beta \varepsilon_{cr}}{(1-k)h}$$

$$\phi_{cr} = \frac{2\varepsilon_{cr}}{h}$$

$$M_{cr} = \frac{1}{6}bd^2E\varepsilon_{cr}$$

$$\phi = \phi'_{i,j} \phi_{cr}$$

$$M_n = m_n M_{cr}$$



Design guide for strain hardening Systems

- 1st Crack tensile strain, (material property) $\varepsilon_{cr} = \frac{\sigma_{cr}}{E}$
- Allowable tensile strain, (tensile limit, β_{max}) $\beta_{max} = \frac{\varepsilon_t}{\varepsilon_{cr}} \leq \alpha = \frac{\varepsilon_{trn}}{\varepsilon_{cr}}$
- post crack tensile modulus $\eta = \frac{E_{cr}}{E} = \frac{\sigma_{peak} - \sigma_{cr}}{E} (\varepsilon_{trn} - \varepsilon_{cr})$
- compressive yield strain, Reduction factor $\omega = \frac{\phi f'_c}{\sigma_{cr}} \quad \phi=0.8$
- Neutral axis depth and Normalized moment and Curvature at any given strain:

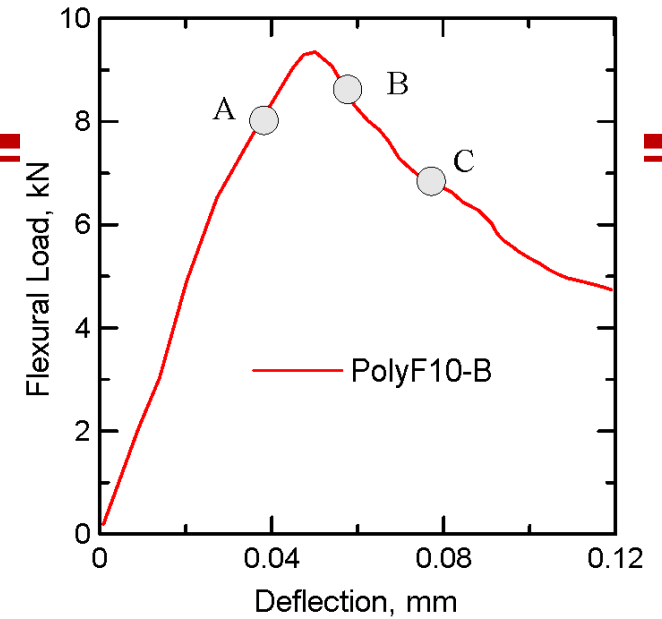
$$k = \frac{C_1 - \sqrt{\beta^2 C_1}}{C_1 - \beta^2} \quad M' = C_2 \frac{k^2 - 2k + 1}{\beta^2} + \frac{2\beta k^3}{1-k} \quad \phi' = \frac{\beta}{2(1-k)}$$

$$C_1 = \eta(\beta^2 - 2\beta + 1) + 2\beta - 1 \quad M_{cr} = \frac{\sigma_{cr} b h^2}{6} \quad \phi_{cr} = \frac{2\varepsilon_{cr}}{d} \quad \phi = \phi'(\beta, \eta) \phi_{cr}$$

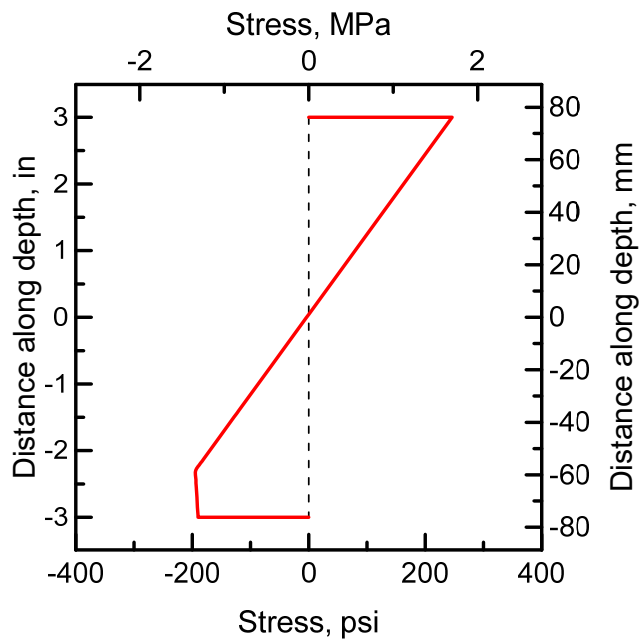
$$C_2 = C_1 + 2C_1\beta - \beta^2$$

$$M_n = M'(\beta, \eta) M_{cr} \quad \phi = \phi'(\beta, \eta) \phi_{cr}$$

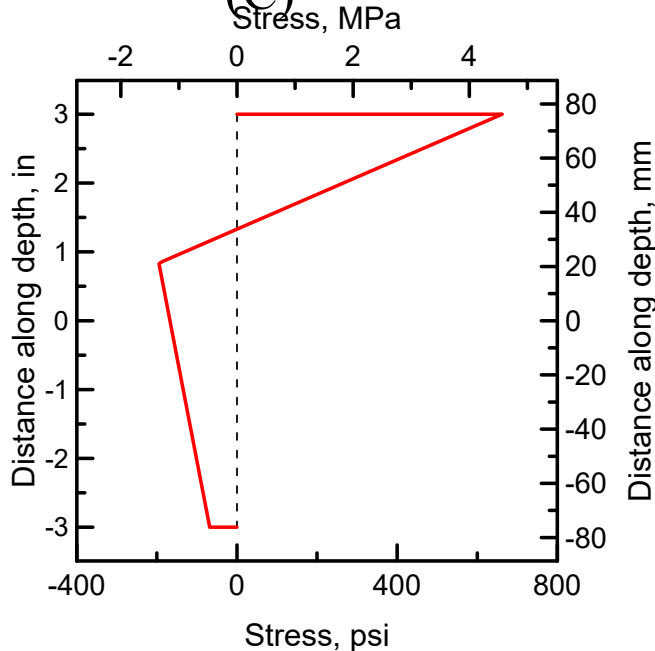
Evolution of Stress Distribution Profile



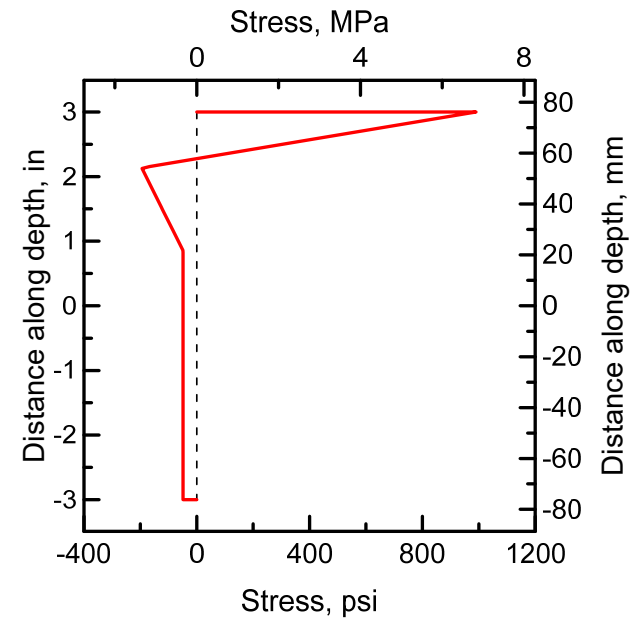
(A)



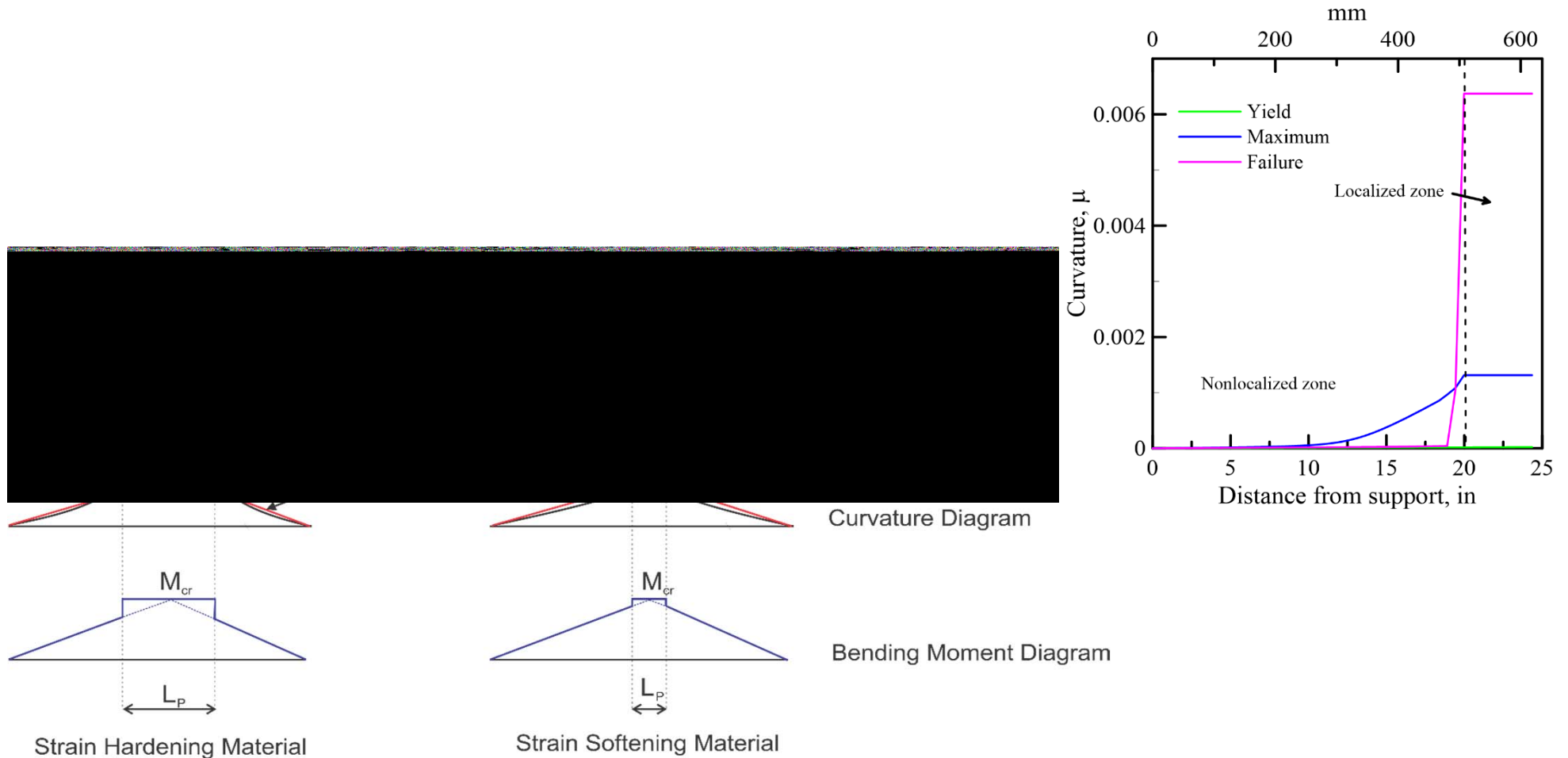
(C)



(B)



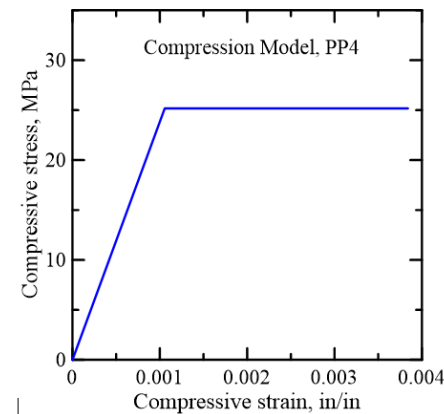
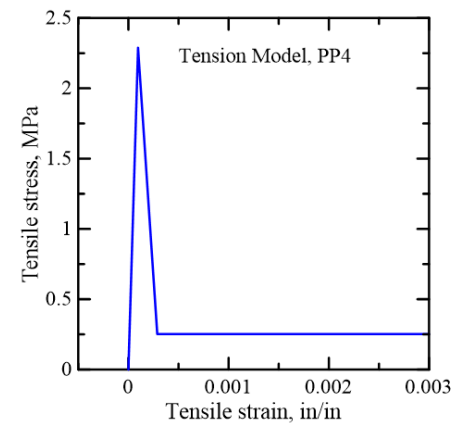
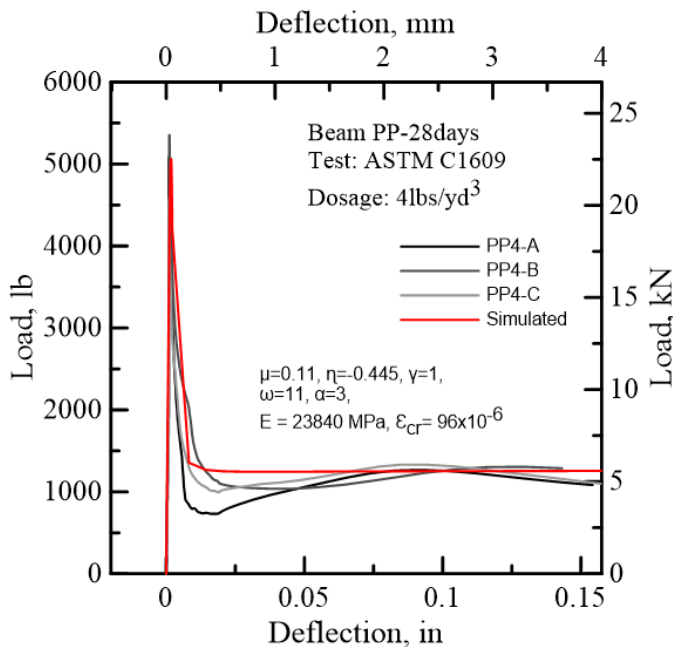
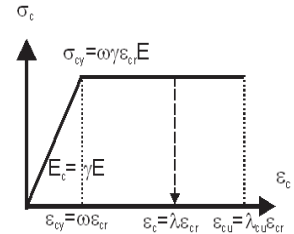
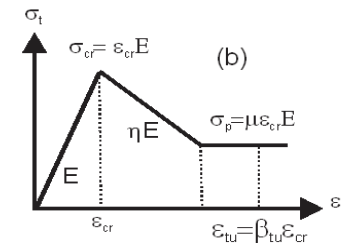
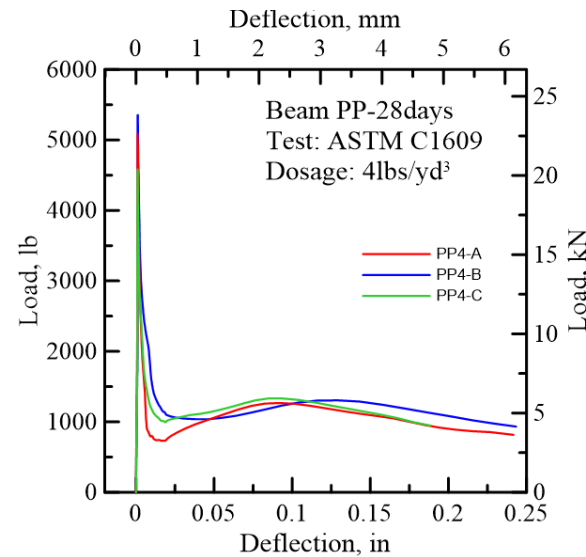
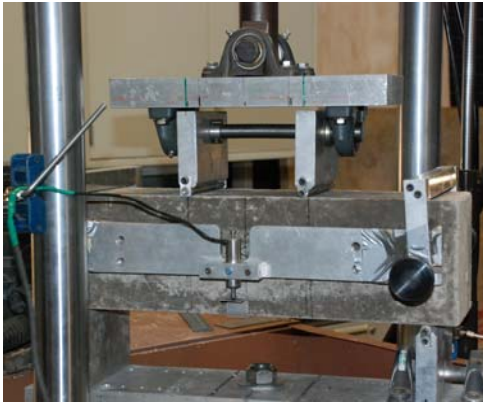
Modeling the localization effect and effect of residual tensile strength under flexure using deflection criteria



(a) schematic figure comparing in strain hardening and softening materials.

(b) curvature in different stages of loading

Standard Design Procedure using ACI 544.4R method (Parametric-based design)



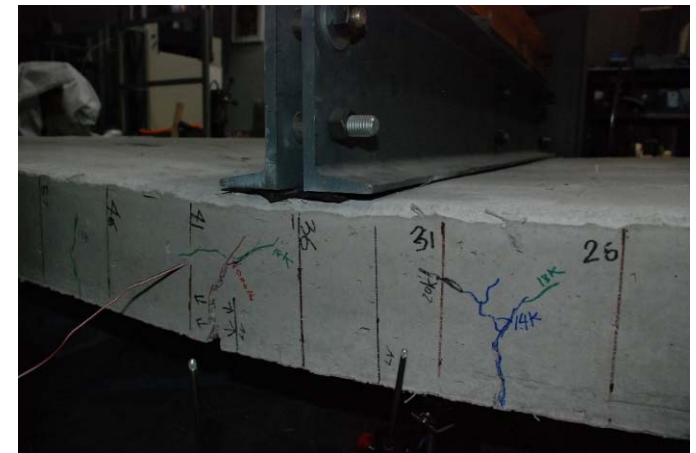
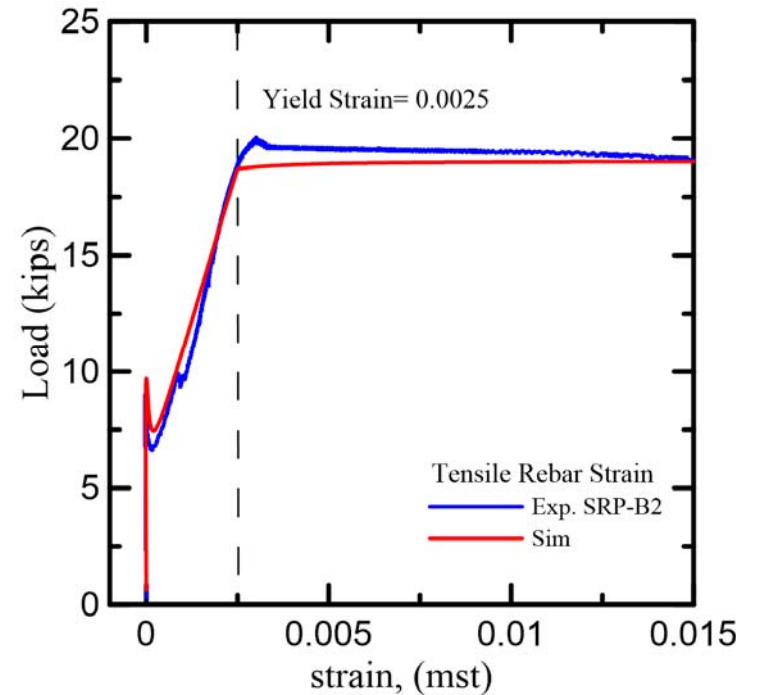
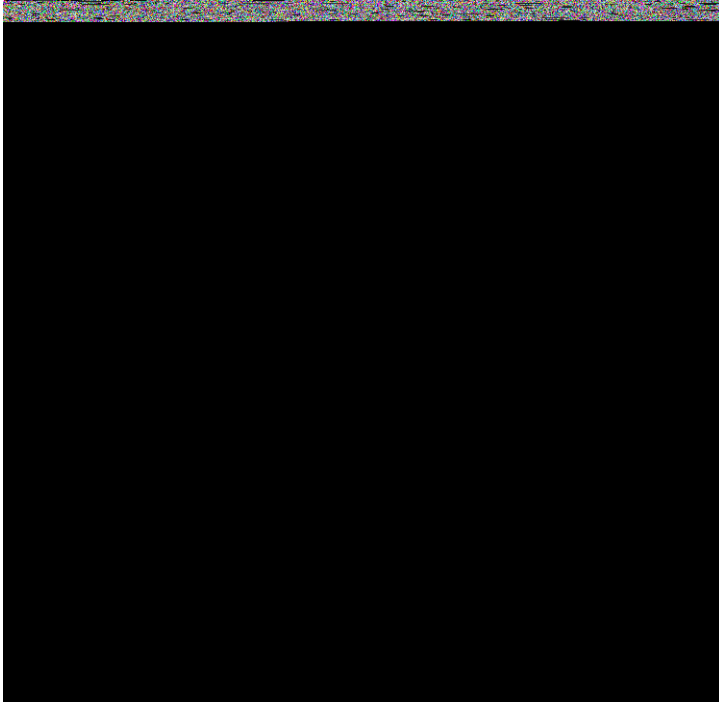
Using these parameters, the location of neutral axis, moment capacity, and curvature of the section can be obtained at any imposed strain

Hybrid Reinforced Concrete Design & Back-Calculation Software (Compiled Matlab GUI)

- Design parameters can be specified based on specifications on maximum allowed crack width, specific deflection, maximum curvature, or stiffness criteria.
- Output will produce 12 plots of various responses

Panel flexural tests Simulation

Variables Include: support Conditions, rebar configuration, and fiber content

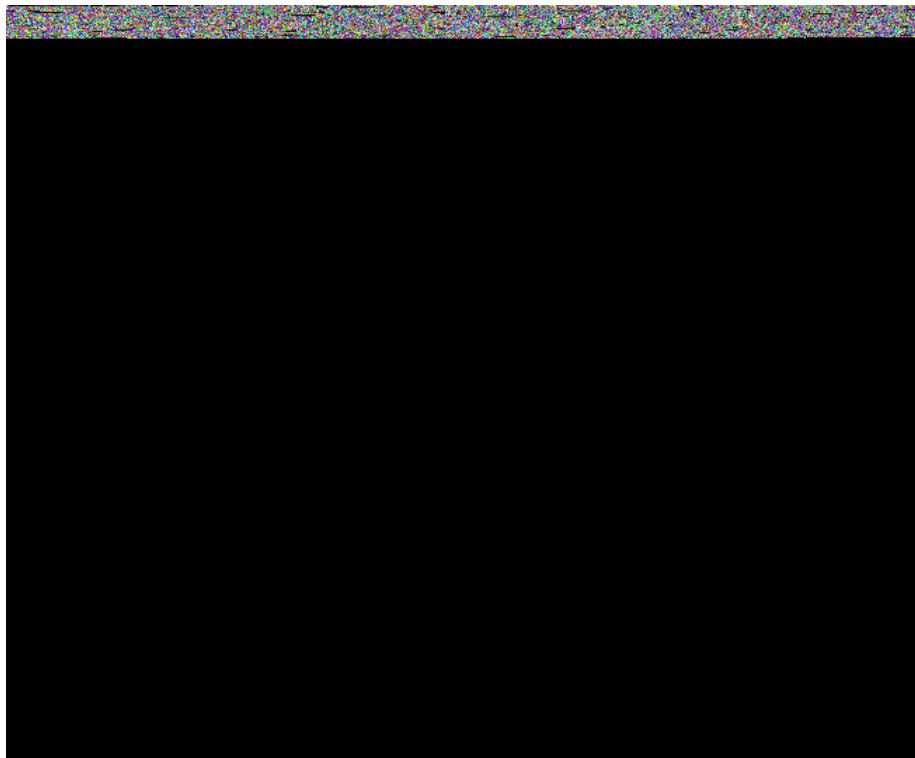


Force distribution in conventional RC and HRC



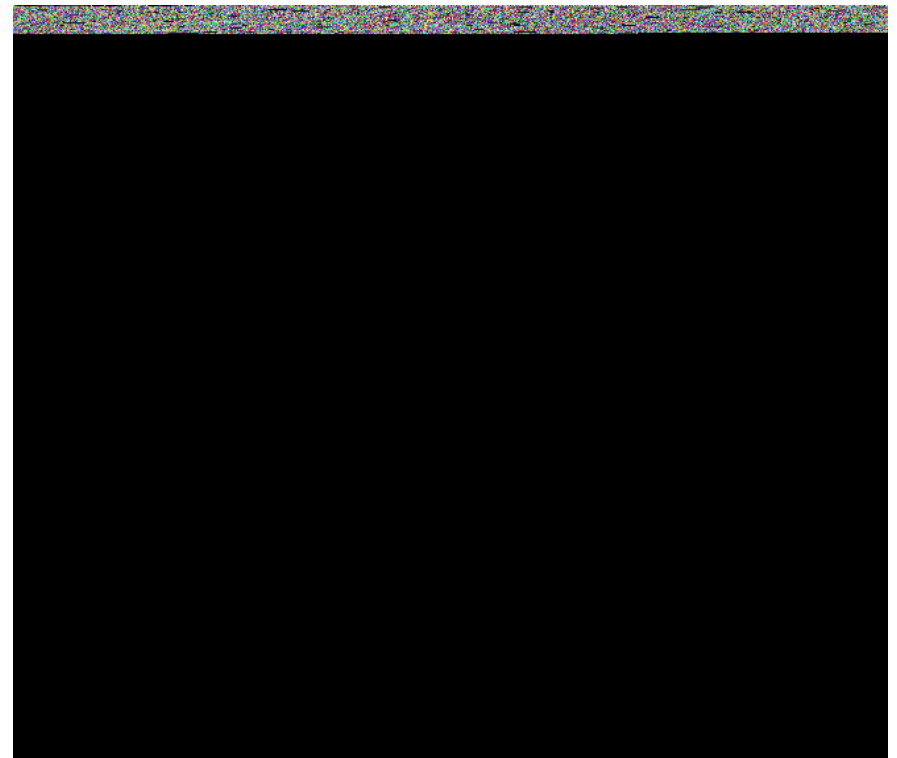
conventional RC

Sudden change as soon as cracking occurs with rebar tension taking over 100% of the tension load within the serviceability range



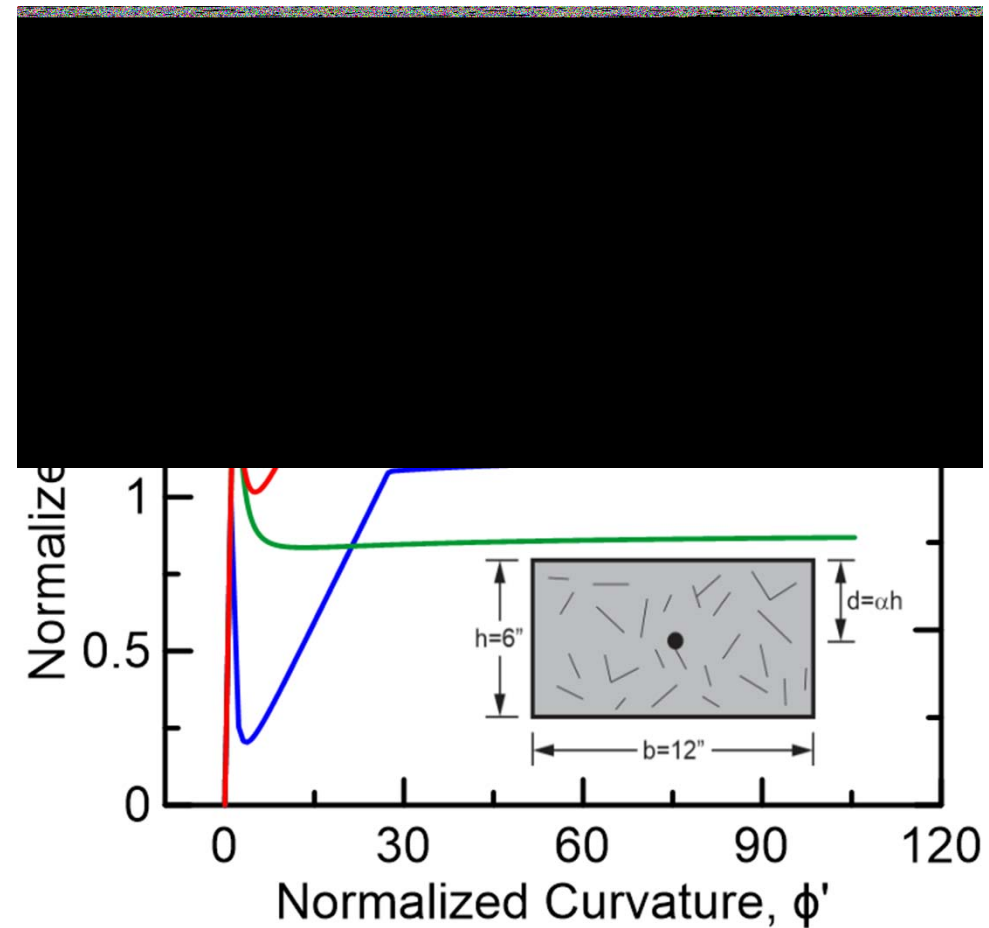
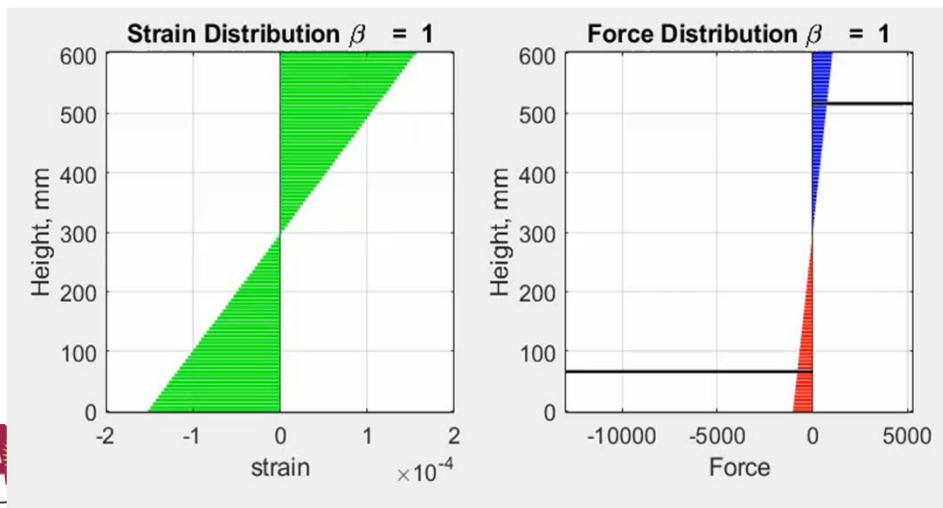
Hybrid RC

With 1% fiber, there is gradual change with cracking with rebar tension taking over only 20% of the tension load at limit state



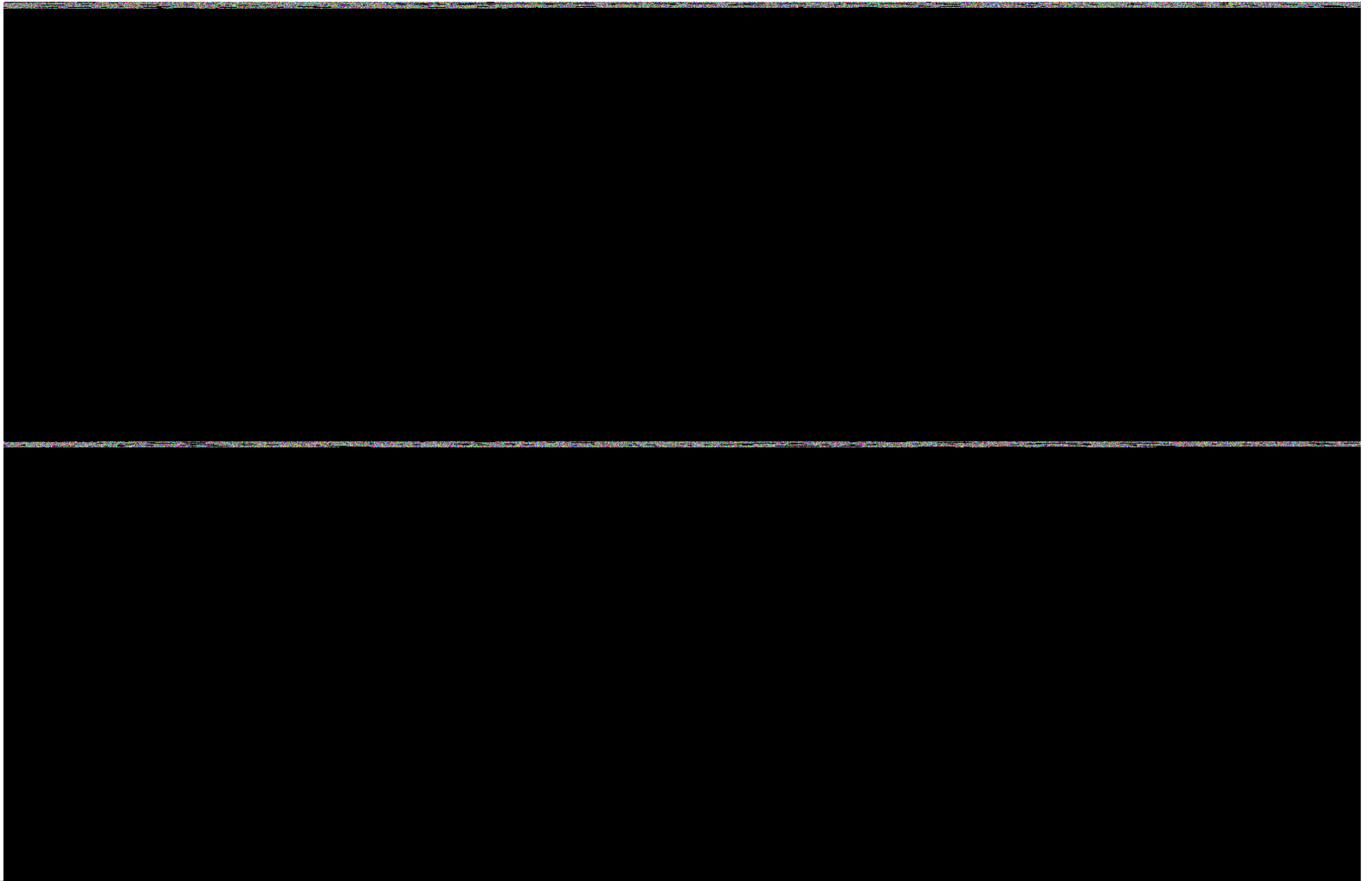
Three main contributions of the Model Representing the Synergy of fiber and rebar combination

1. Example Slab on Ground with a single rebar at the mid-height
2. Simulated Moment-Curvature Response comparing HRC with plain FRC and conventional rebar
3. Additive contribution of fiber and steel based on compatibility of strain analysis
4. Capability to use a serviceability-based limit strain or curvature analysis



Fracture and plasticity models

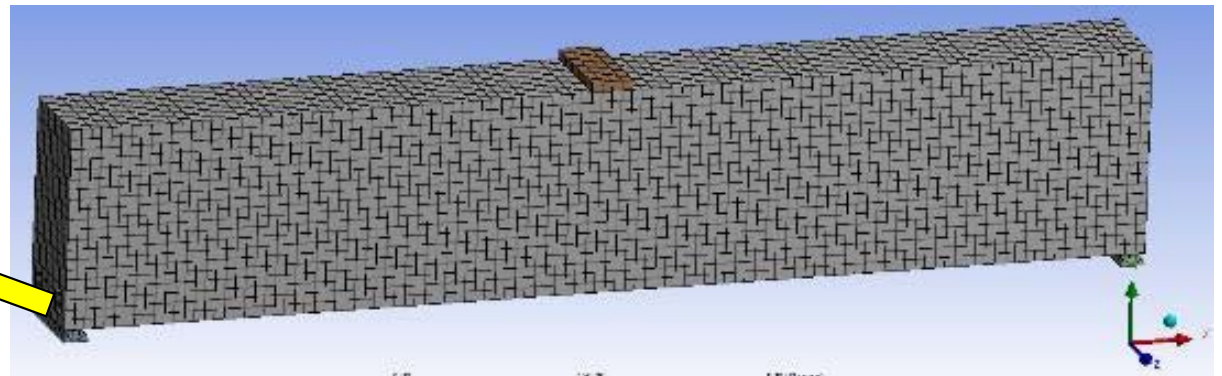
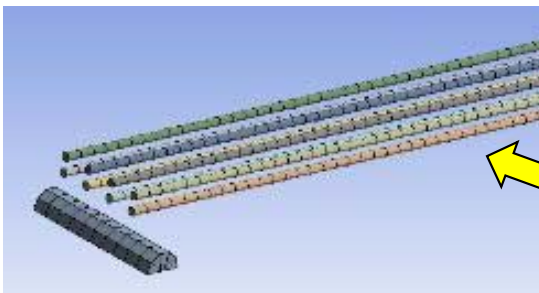
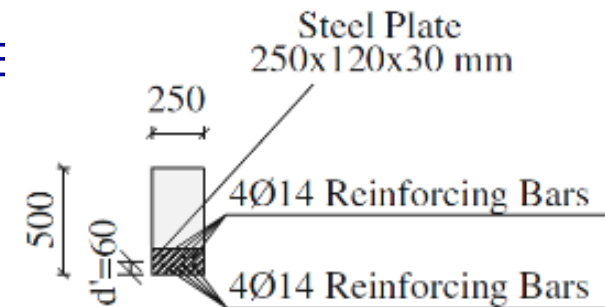
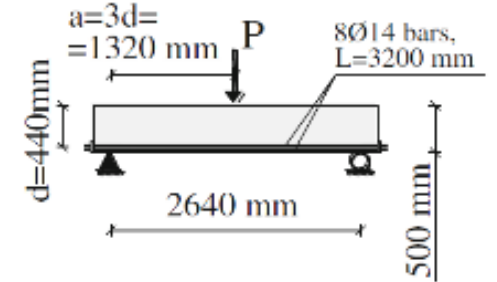
- The effect of material and Structural ductility



Finite Element Analysis on Shear Behavior

- Limitation of analytical based flexural model
 - Cross sectional analysis
 - Inclined growth
 - Damage
- Finite element analysis in LS-DYNA
 - Damage concrete model: MAT159 (*MAT_CSCM)
 - Piecewise model for rebar: MAT024 (*MAT_PIECEWISE_LINEAR)
 - Interface: *Lagrange_IN_SOLID

Beams H500



Stress Distribution

LS-DYNA user input

Time = 6.8497

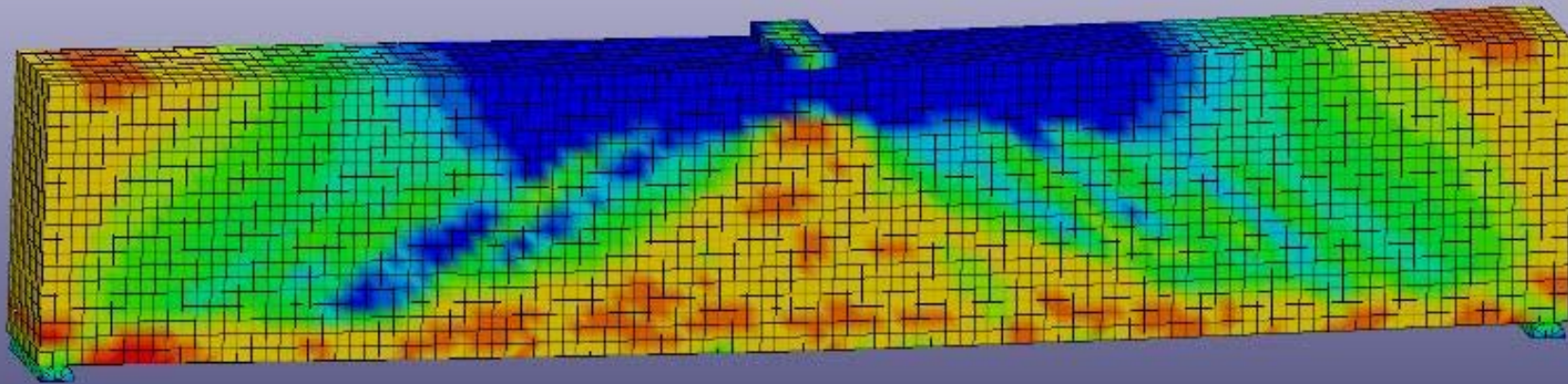
Contours of X-stress

min=-52.7734, at elem# 11761

max=2.41204, at elem# 22185

Fringe Levels

2.400e+00
1.260e+00
1.200e-01
-1.020e+00
-2.160e+00
-3.300e+00
-4.440e+00
-5.580e+00
-6.720e+00
-7.860e+00
-9.000e+00



LS-DYNA user input

Time = 9.084

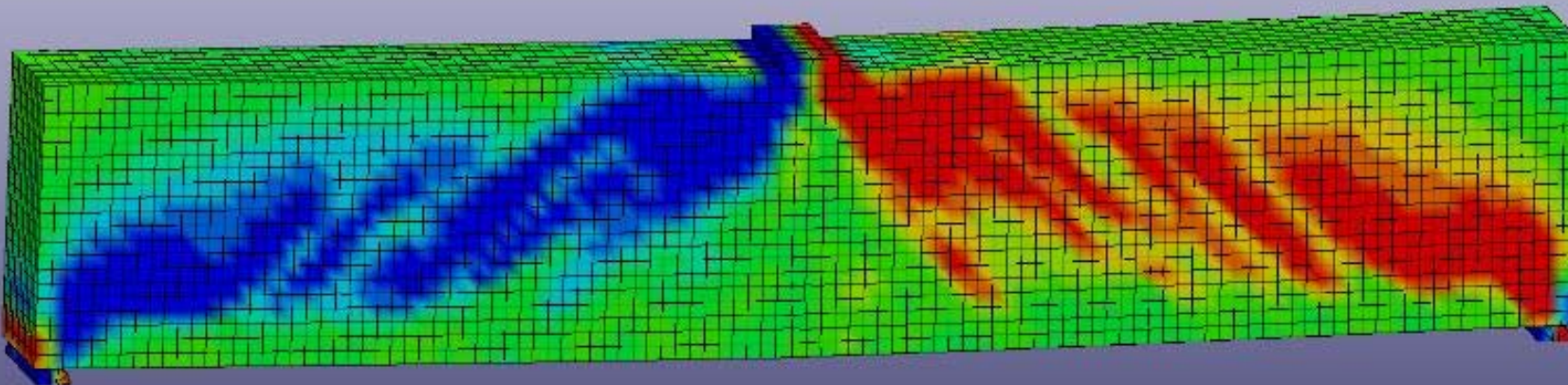
Contours of XY-stress

min=-12.8386, at elem# 12358

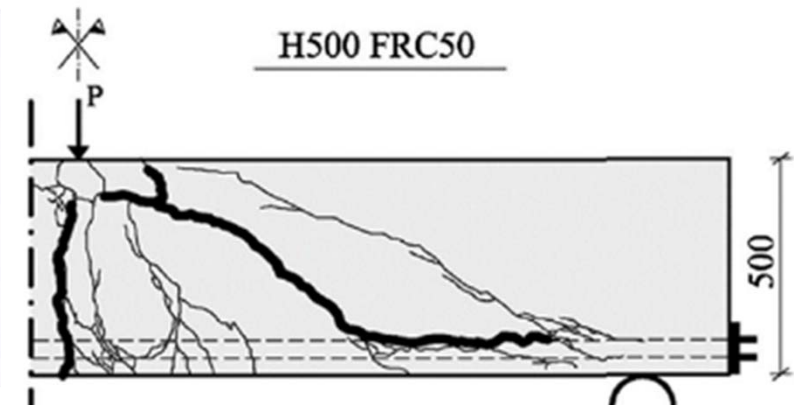
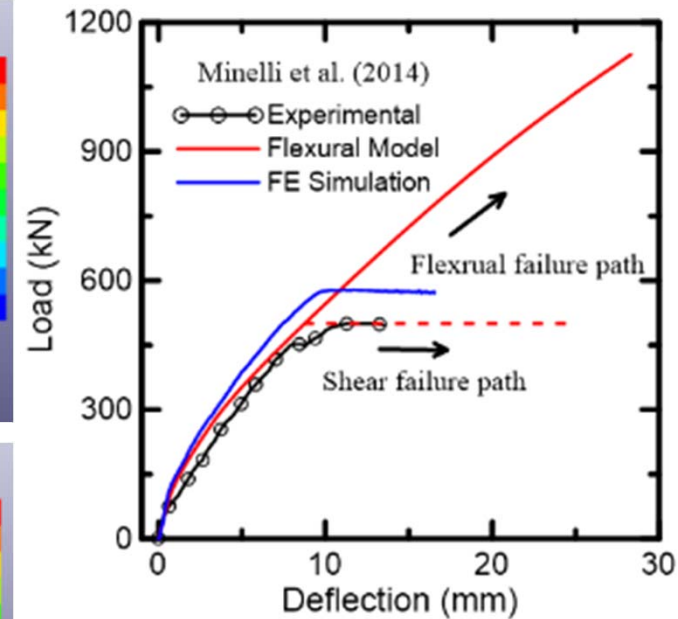
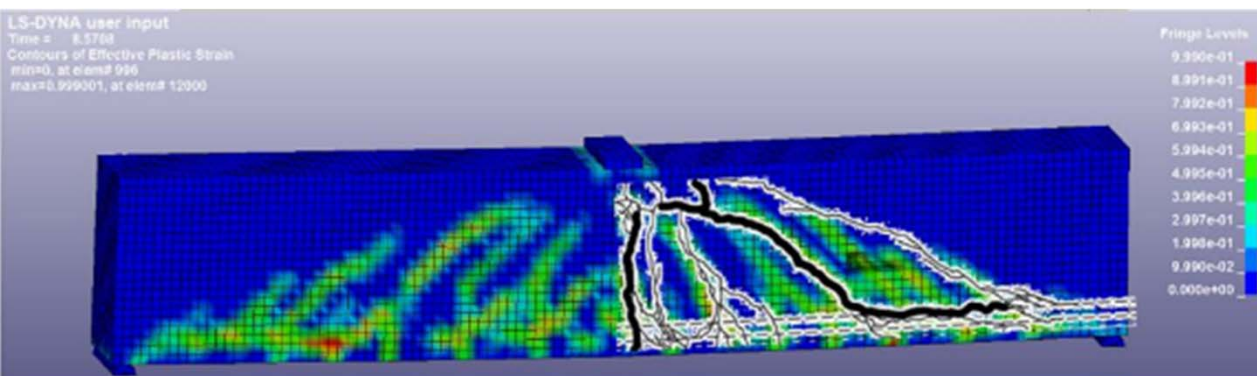
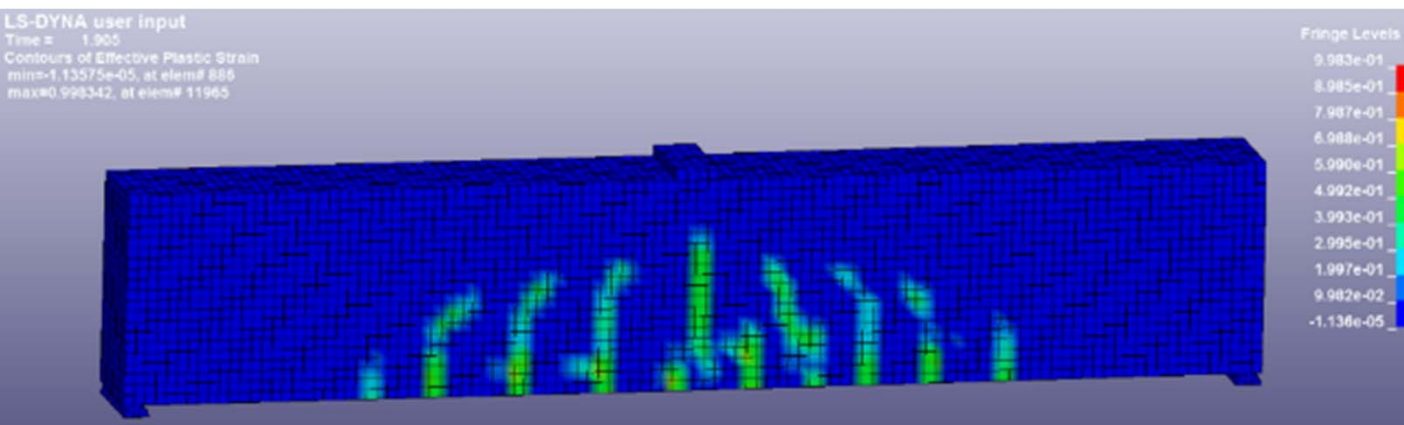
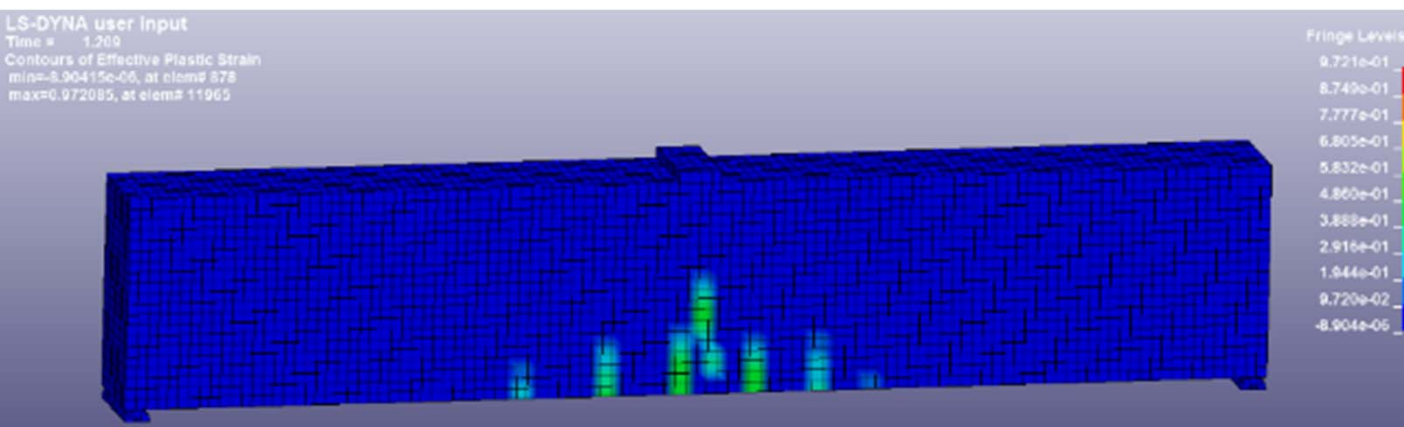
max=13.2817, at elem# 11135

Fringe Levels

3.000e+00
2.400e+00
1.800e+00
1.200e+00
6.000e-01
2.220e-16
-6.000e-01
-1.200e+00
-1.800e+00
-2.400e+00
-3.000e+00

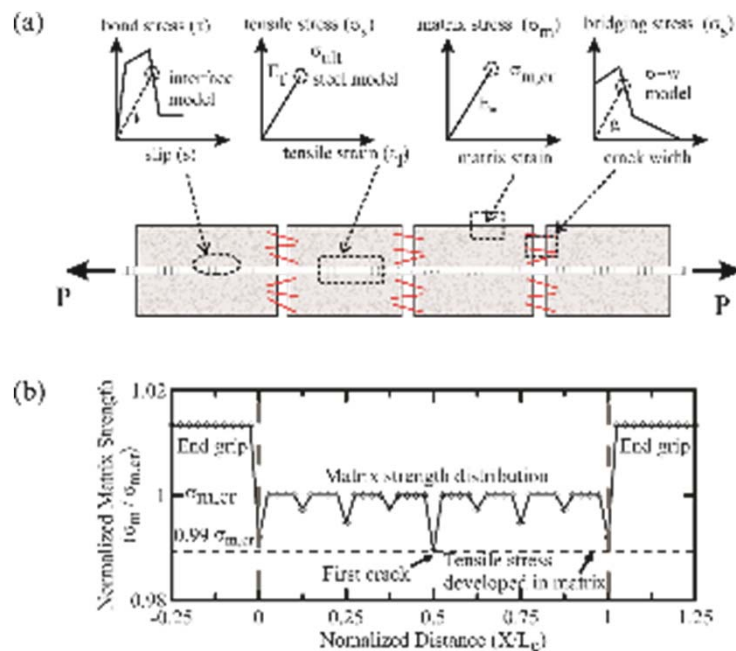


Distributed Damage

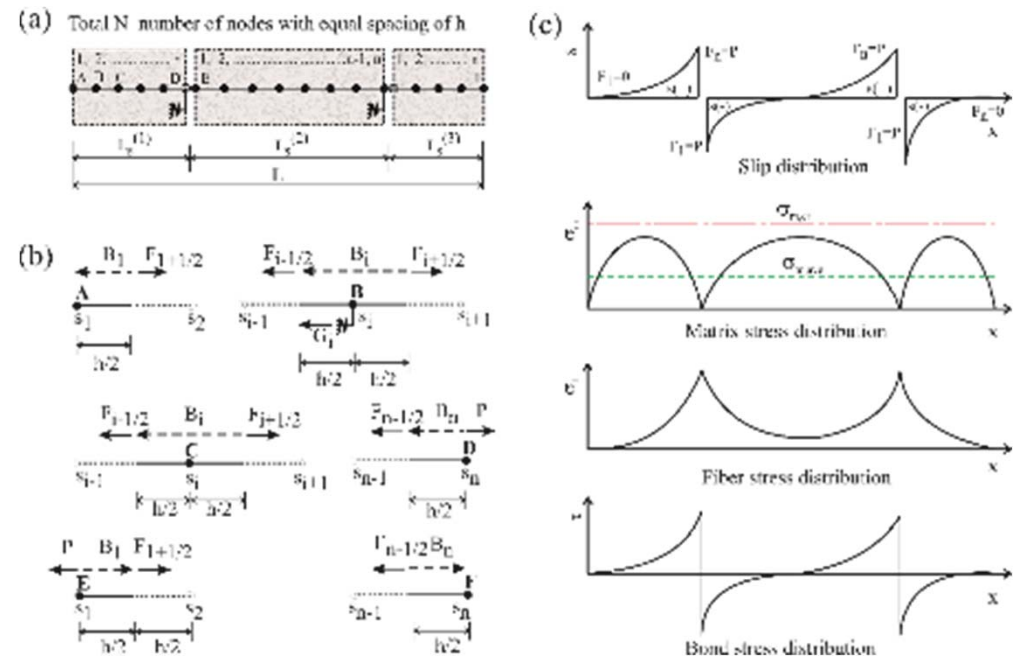


Tension Modeling and simulation studies

A cracked tension specimen is idealized as a series of 1-D pullout segments. The matrix is assumed to be brittle with no strain-softening response. Thus, the load at any crack is solely carried by the longitudinal yarns. The individual pullout segments are subjected to load applied at crack locations.

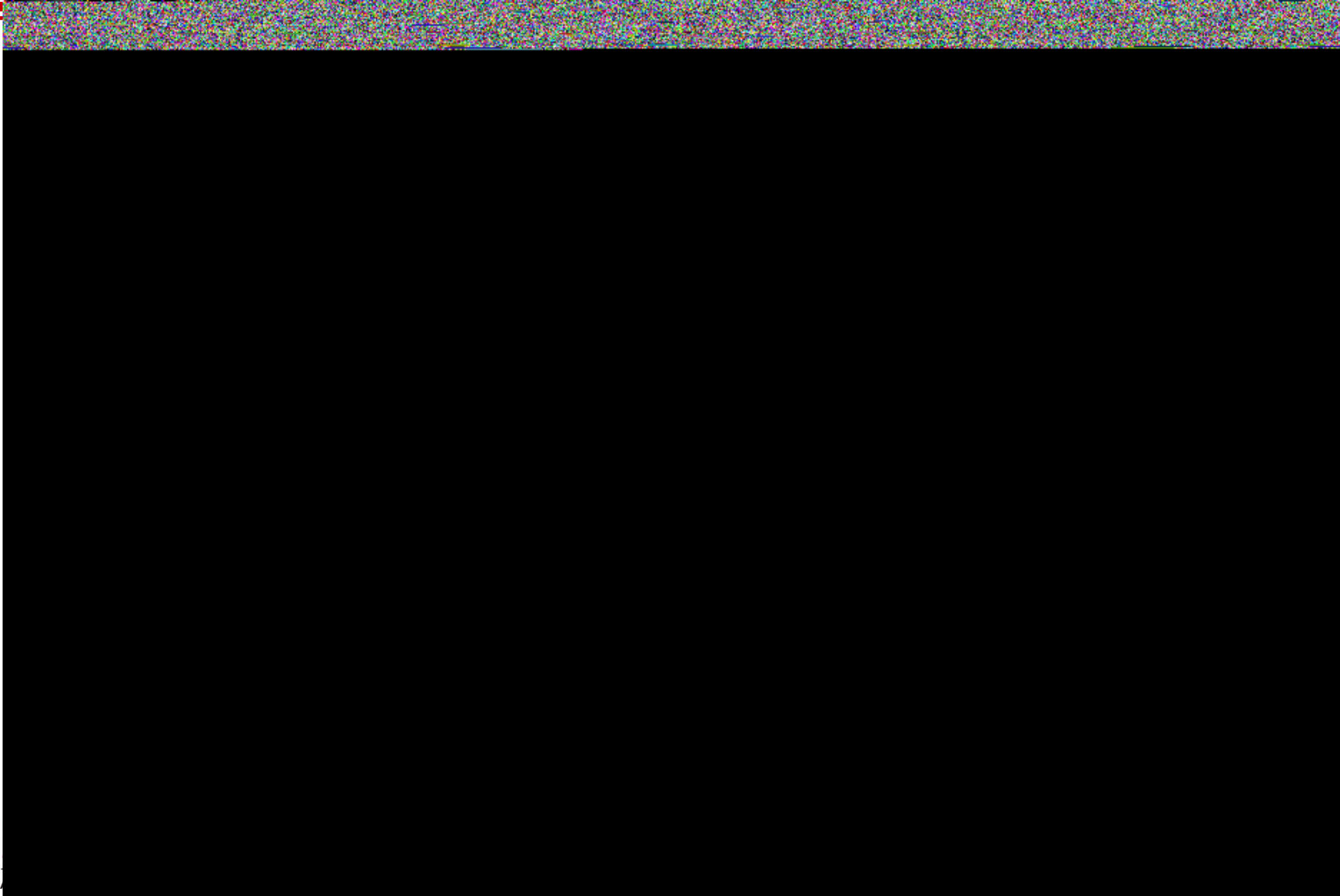


(a) Cracking mechanisms including matrix cracking, fiber bridging in terms of stress-crack width relation, interface bonding stress-slip model and tensile stress-strain relationship of longitudinal reinforcement; (b) stress distribution in matrix at cracked state.

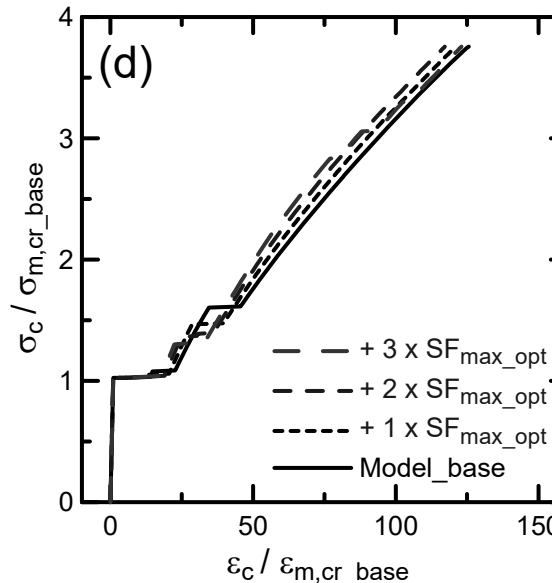
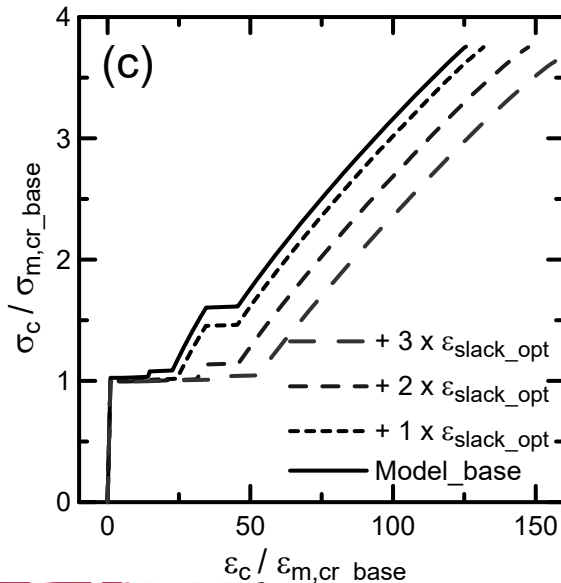
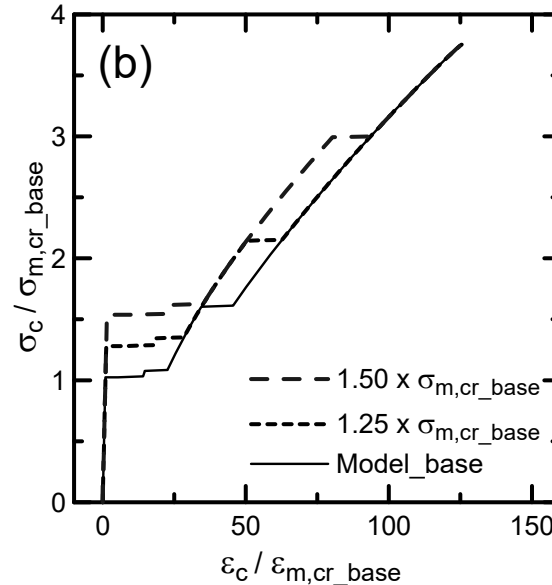
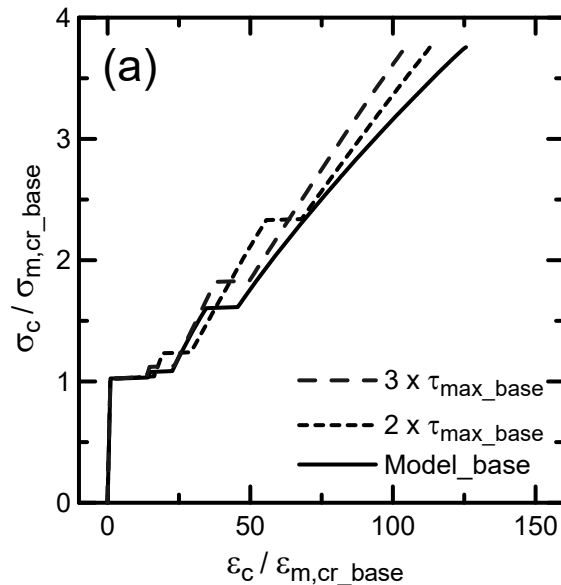


Finite difference model: (a) cracked concrete member, (b) free body diagram of six representative nodes labeled as "A"-"F", (c) distributions of slip, matrix stress, rebar stress and bond stress.

Tension Modeling and simulation studies



Effects of the four cracking criteria



Changes in normalized composite stress-strain responses due to the variations

a) Increase in bond strength marginally increases the post crack stiffness

b) Increase in matrix strength improves the stress at BOP

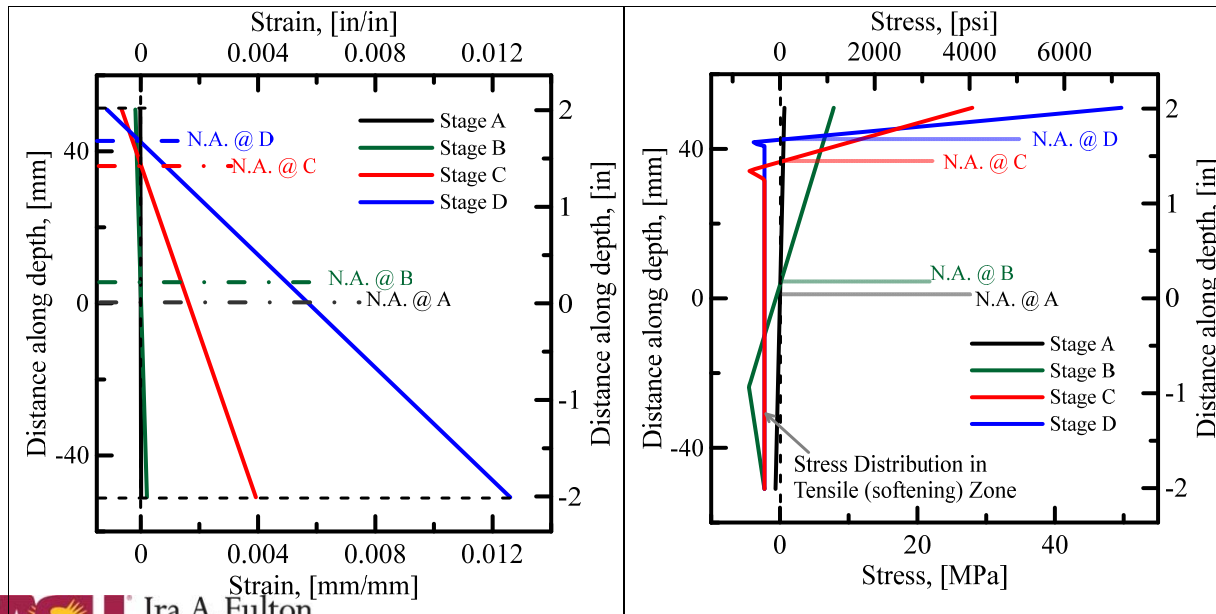
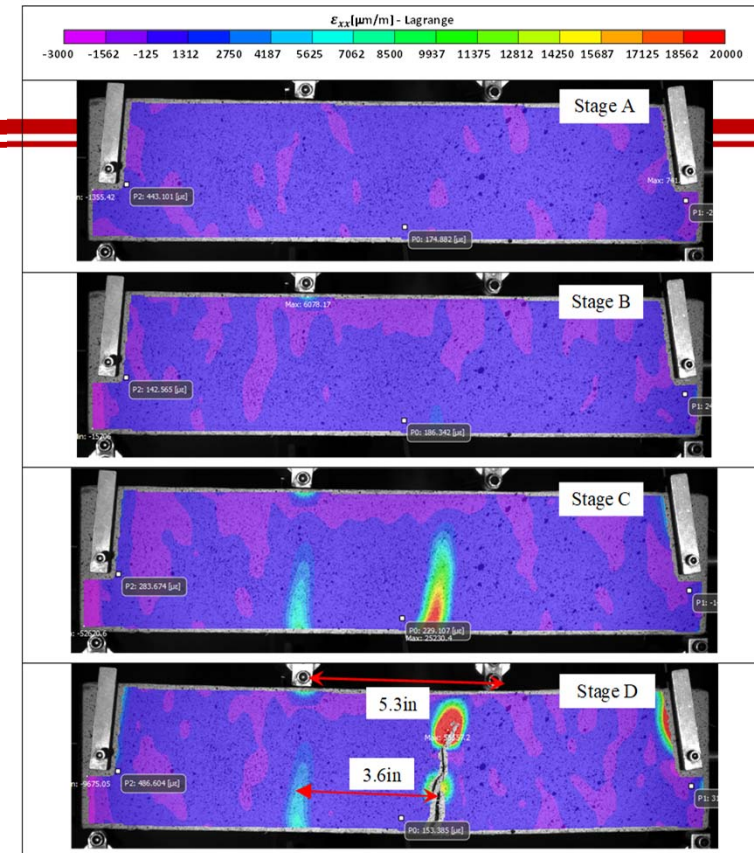
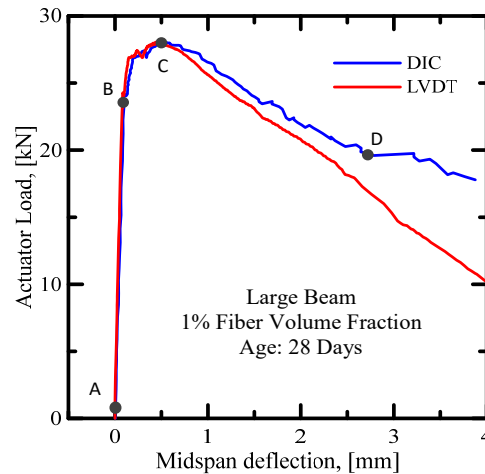
c) Effect of slack delay the action of longitudinal yarn to carry the load; more slack results in larger strain at early cracking stages

d) Increase in restraining levels of spring force has a positive effect in toughening post crack stiffness.

Applications of the UHPC Model

- Validation of the DIC Measurements in Flexural Crack Growth

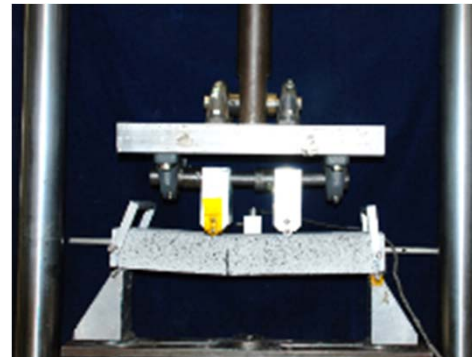
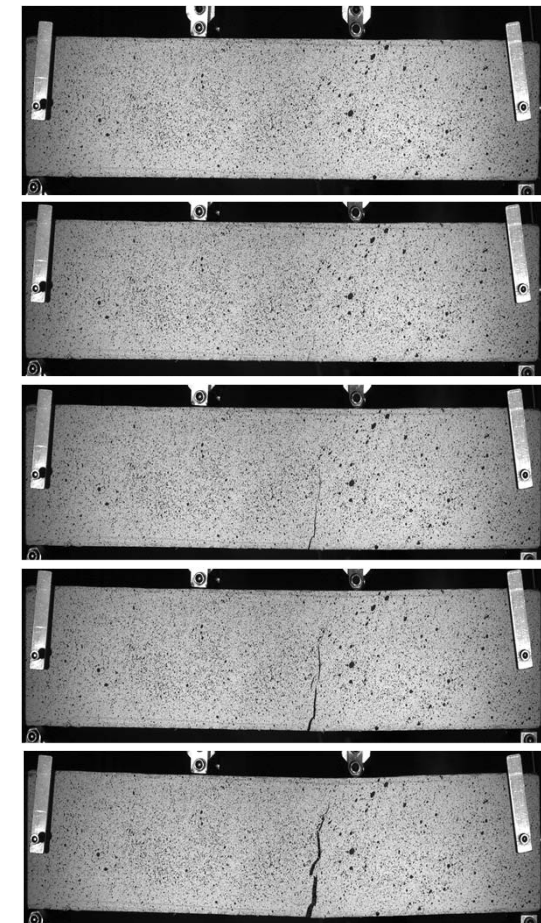
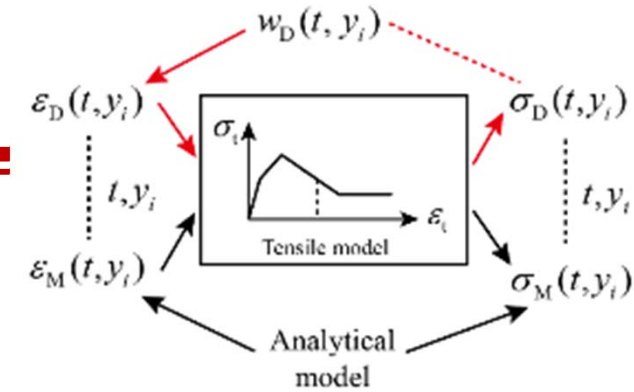
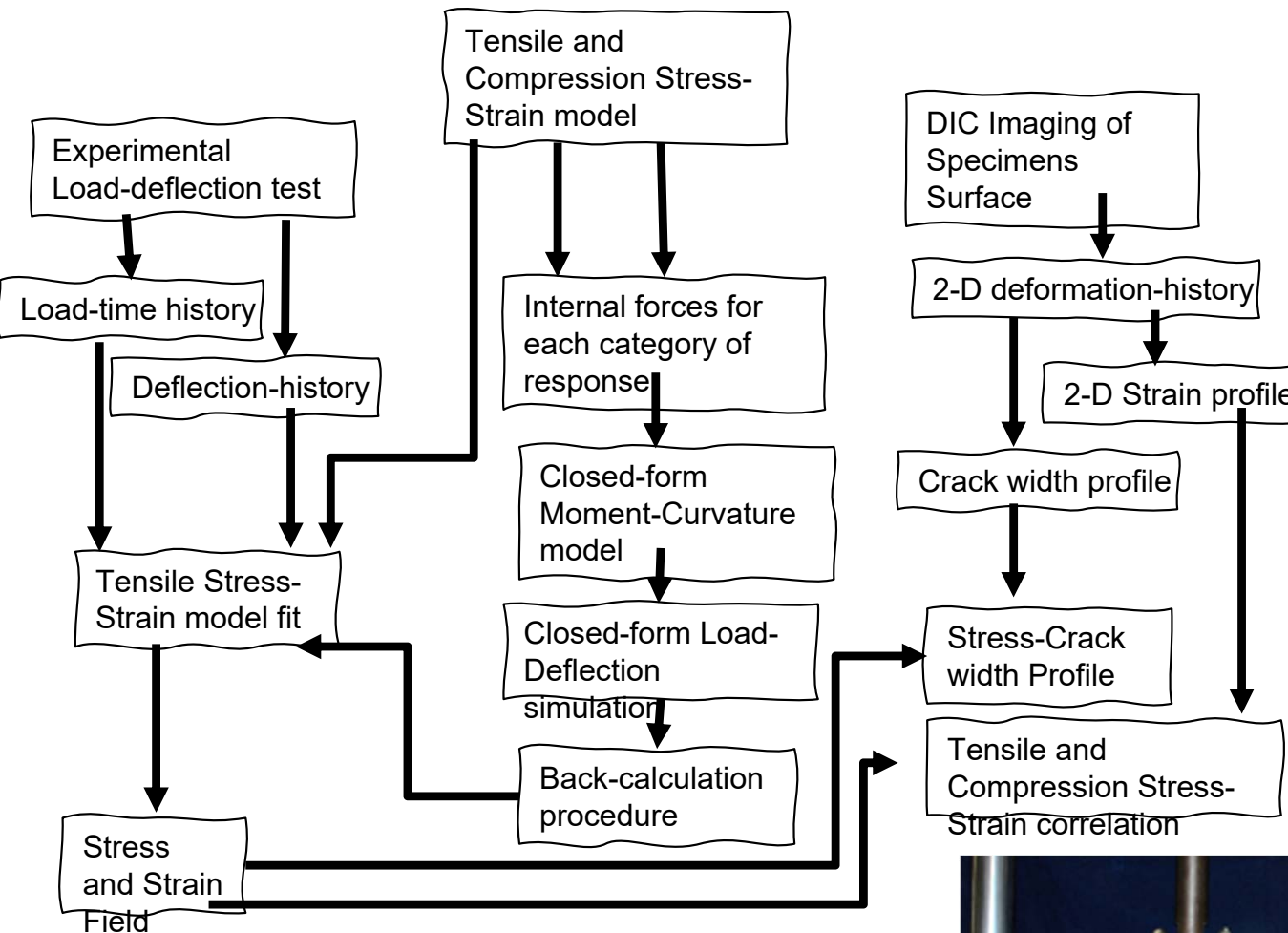
Stage	Characteristic
A	Initiation of the deformation
B	Initiation of the non-linear response
C	Response to the peak load
D	End of DIC data analysis (beam failure)



UHPC in Flexure

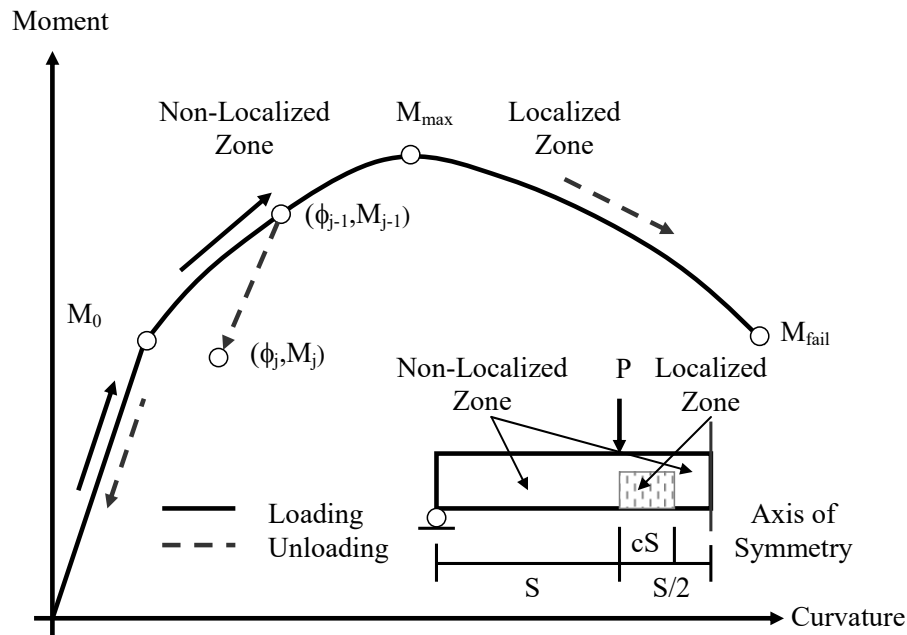
UHPC large Specimens, more opportunity for the bridging zone to develop

Correlation of tension and flexure tests is useful in the validation of results



Localization Rules

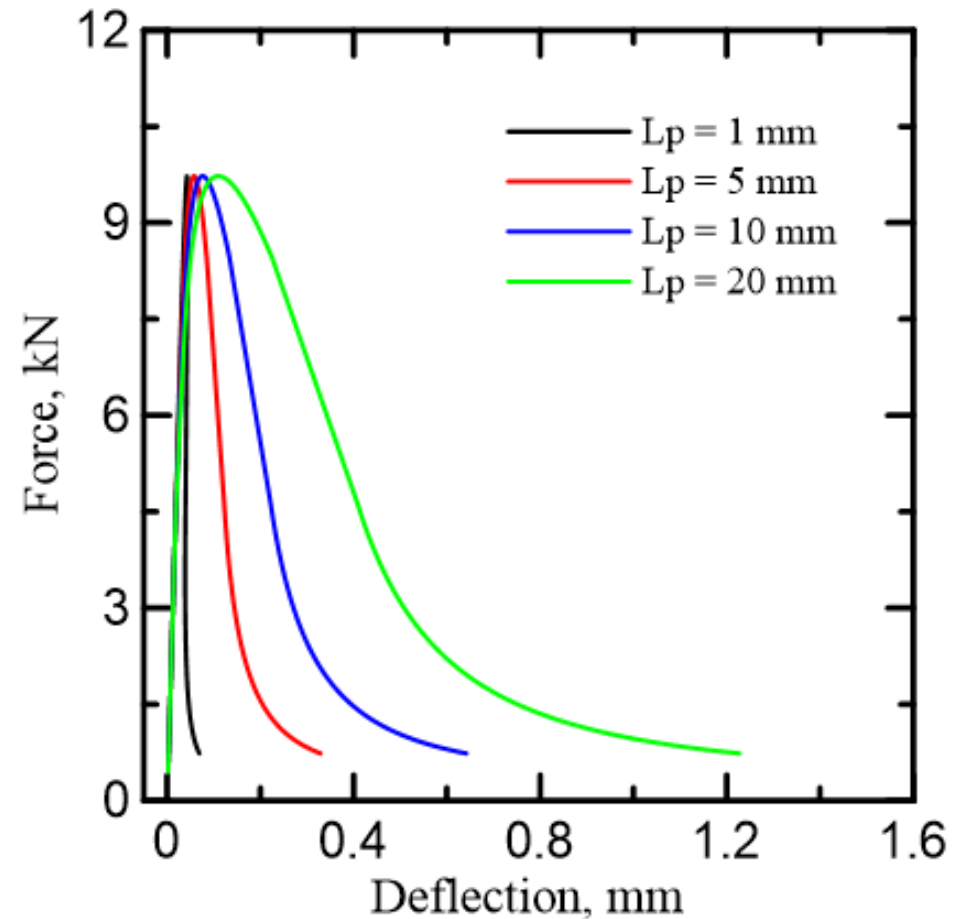
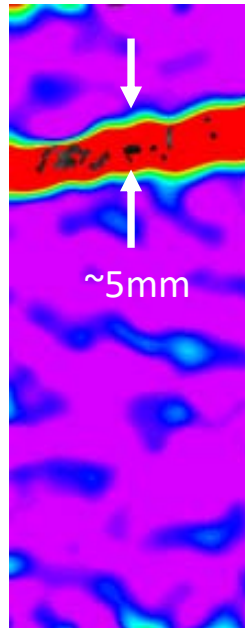
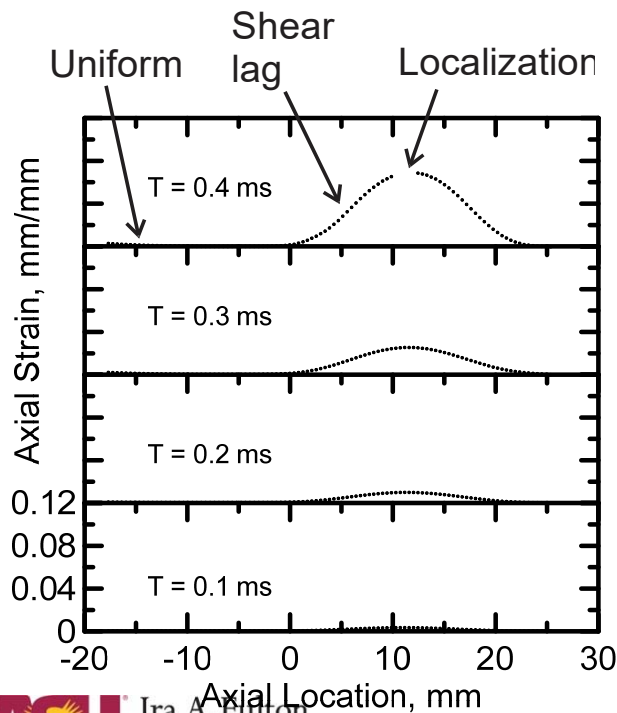
- Flexural specimen is loaded beyond the peak strength;
- Distinct zones develop as the deformation localizes in the cracking region an average response over the crack spacing region ;
- Results are used as a smeared crack in conjunction with the moment-curvature diagram
- Measurement from DIC



$$\phi_j = \phi_{j-1} - \xi \frac{(M_{j-1} - M_j)}{EI}$$

Effect of Localization Zone Size, L_p

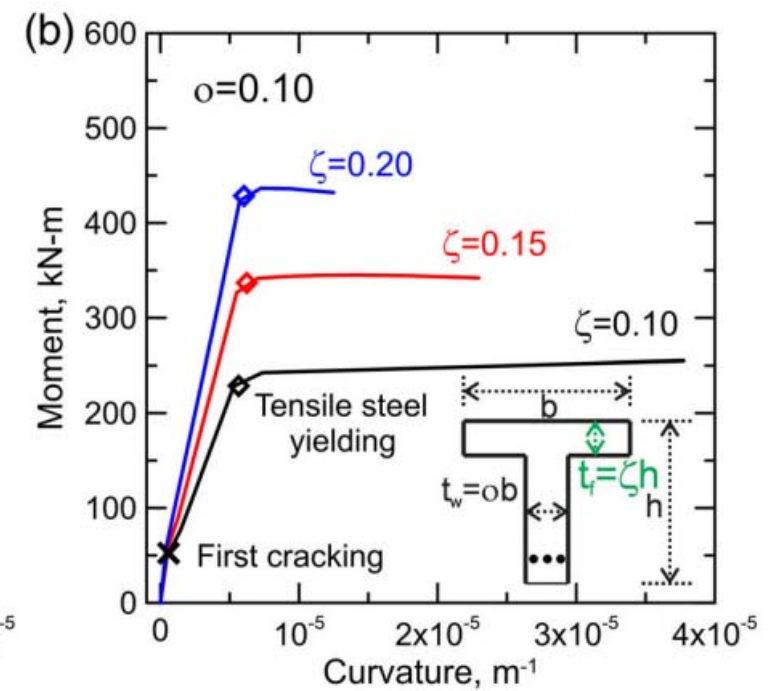
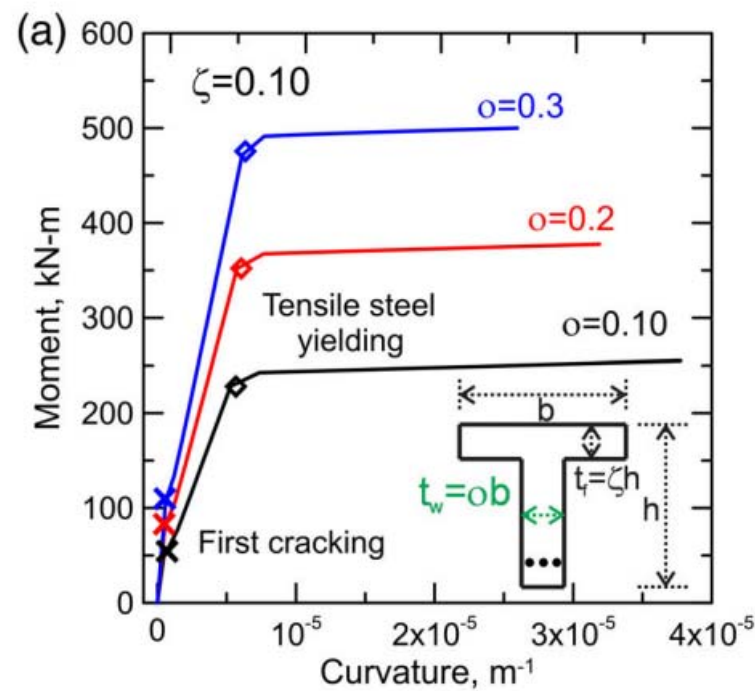
- Affects the general softening behavior in the post peak zone
- The simulated residual load capacity is not sensitive
- Model parameters:
- $b=40\text{mm}$, $h=40\text{mm}$, $L=125\text{mm}$
- $E=45\text{GPa}$, $\varepsilon_{cr}=280\mu\varepsilon$, $\alpha=90$, $\eta=-0.011$, $\mu=0.021$, $\beta_{tu}=300$, $\gamma=0.95$, $\omega=8.33$, $\lambda_{cu}=27$



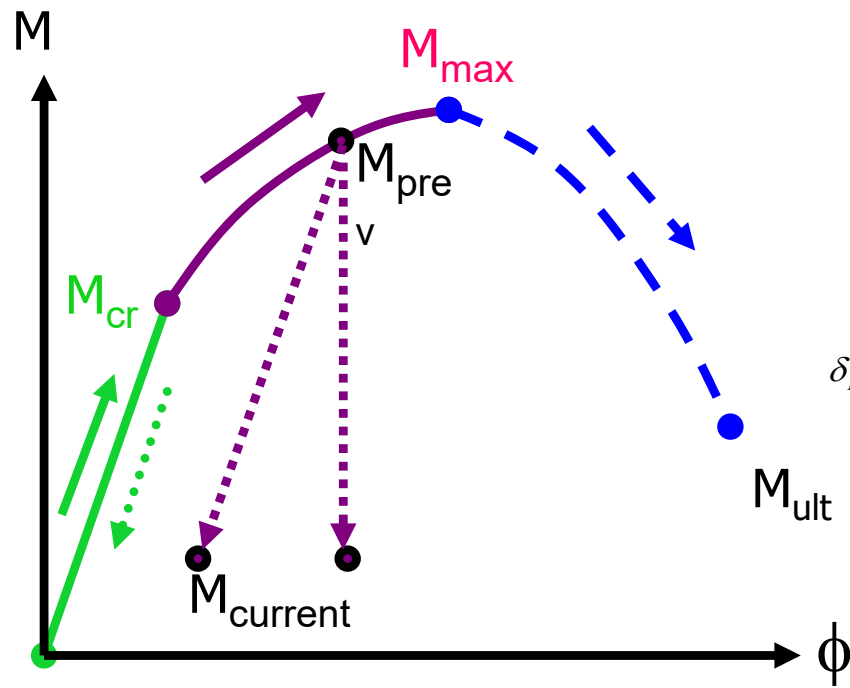
Moment–curvature equations for T-beams

- Yao Y, Mobasher B, Wang J, Xu Q. Analytical approach for the design of flexural elements made of reinforced ultra-high-performance concrete. *Structural Concrete*. 2020; 1–20. <https://doi.org/10.1002/suco.201900404>

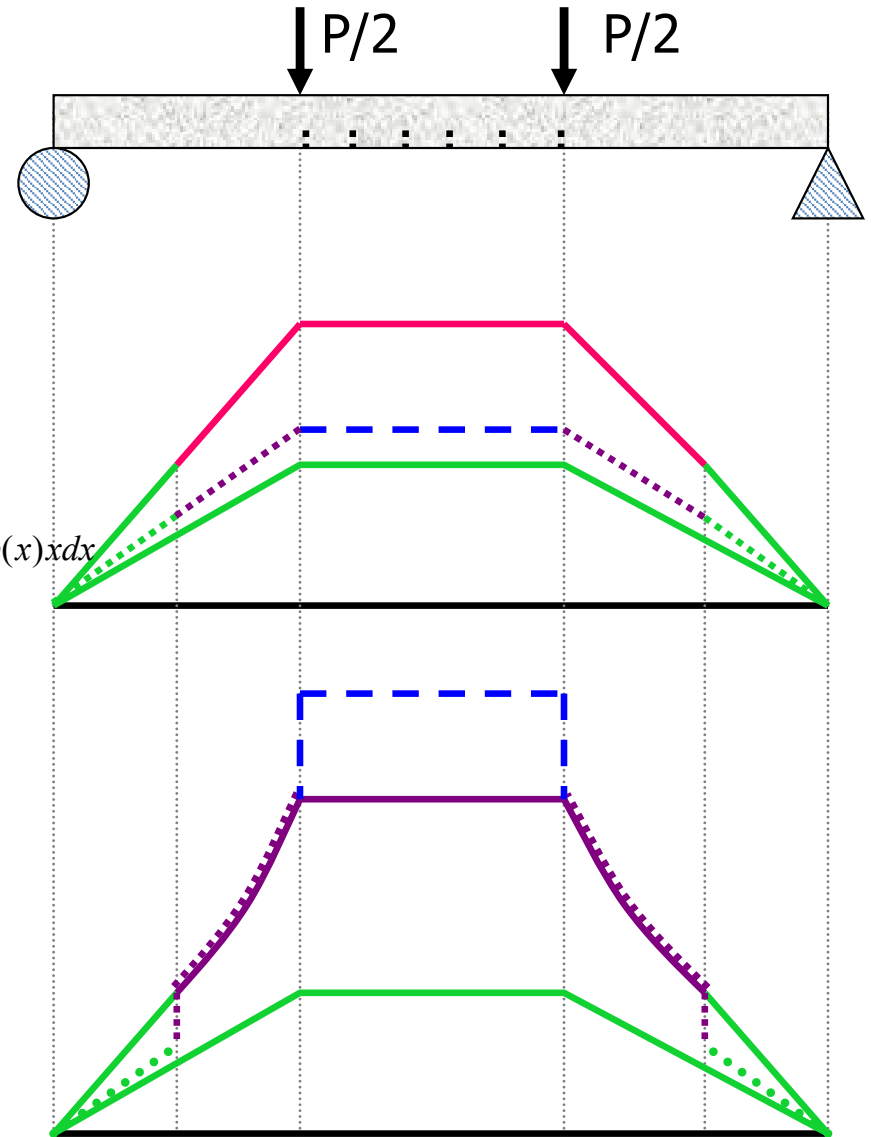
Moment–curvature equations for T-beams



Load Deflection by Moment Area Method

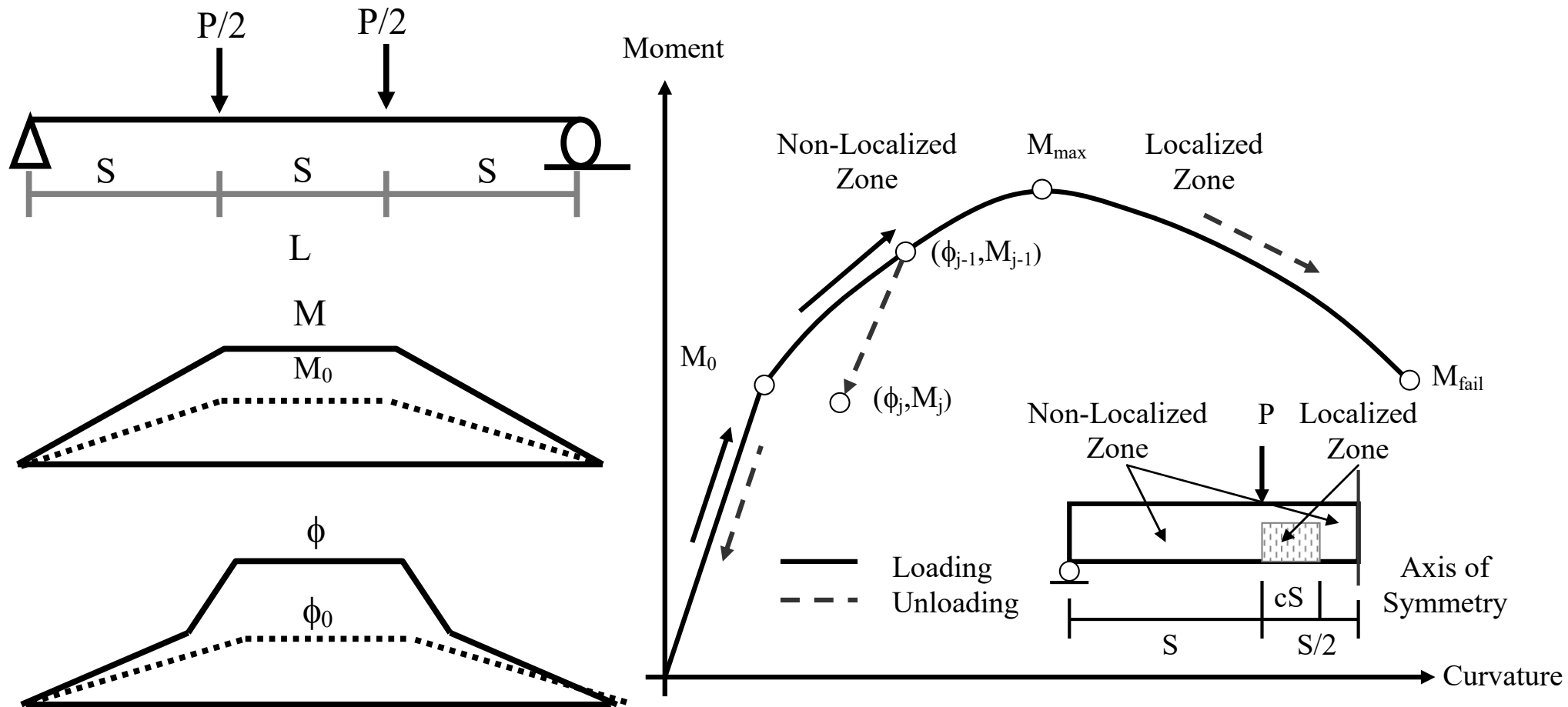


$$\delta_{L/2} = \int_0^{L/2} \phi(x) x dx$$



Load Deflection Response

- Moment area method
- Crack localization rules



Bilinear Moment-Curvature Model

$$M_{cr} = \frac{1}{6} b d^2 E \varepsilon_{cr} \quad \varphi_{cr} = \frac{2 \varepsilon_{cr}}{d}$$

$$\begin{aligned} M(\varphi) &= EI_g \varphi & 0 < M \leq M_{cr} & \quad 0 < \varphi \leq \varphi_{cr} \\ M(\varphi) &= \frac{M_p - M_{cr}}{\varphi_p - \varphi_{cr}} (\varphi - \varphi_{cr}) + M_{cr} & M_{cr} \leq M \leq M_p & \quad \varphi_{cr} \leq \varphi \leq \varphi_p \end{aligned}$$

Normalized moment-curvature

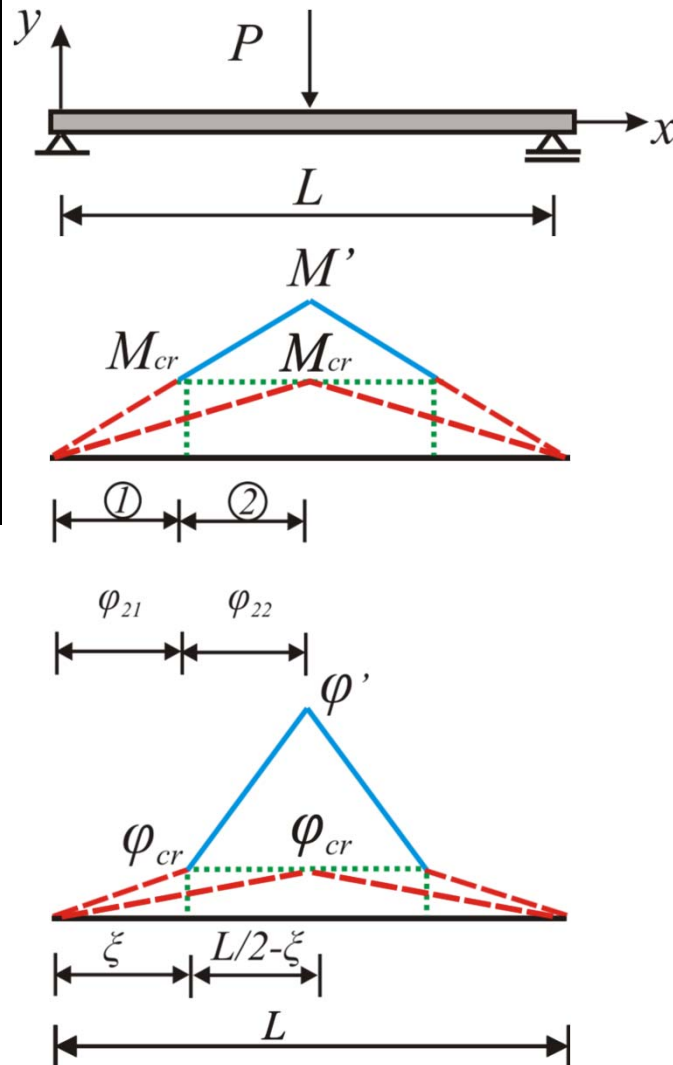
$$m(\kappa) = \kappa$$

$$m(\kappa) = \frac{m_p - 1}{\kappa_p - 1} (\kappa - 1) + 1$$

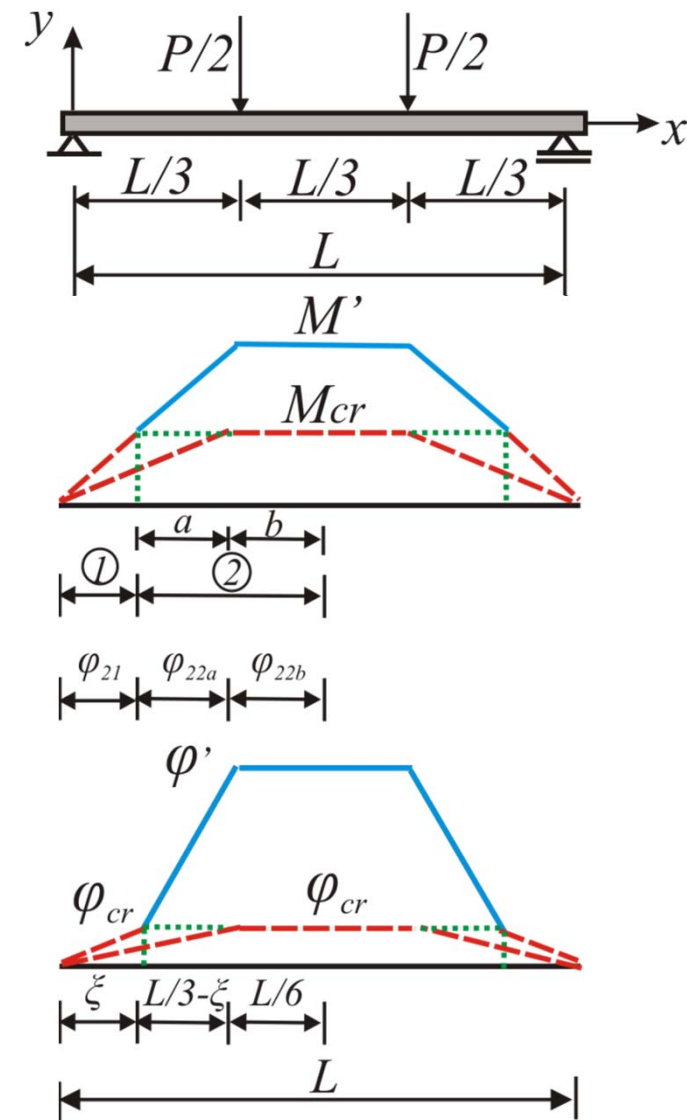
$$m(\kappa) = \frac{M(\varphi)}{M_{cr}} \quad \kappa = \frac{\varphi}{\varphi_{cr}} \quad m = \frac{M_u}{M_{cr}} \quad \kappa_p = \frac{\varphi_p}{\varphi_{cr}} \quad \eta = \frac{EI_{cr}}{EI_g}$$

1-D Member: Moment-Curvature Distributions

3PB



4PB



■ Use static equilibrium to get moment distributions

■ Moment distributions

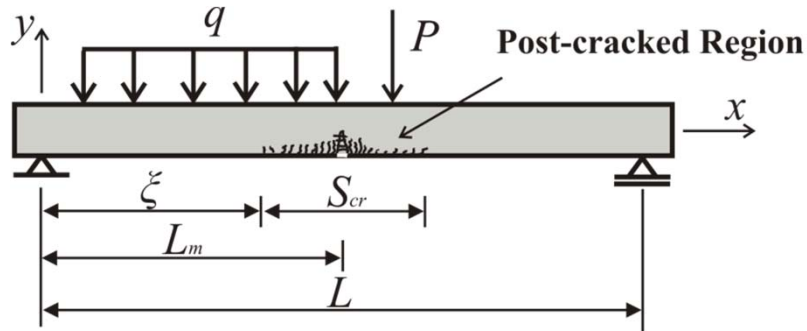


Bilinear moment-curvature



Curvature distributions

Analytical Deflection Solutions



➤ Rotation

$$\text{Region ① : } \theta_{21} = \int_0^x \varphi_{21}(x) dx + C_1 \quad 0 \leq x \leq \xi$$

$$\text{Region ② : } \theta_{22} = \int_0^{\xi} \varphi_{21}(x) dx + \int_{\xi}^x \varphi_{21}(x) dx + C_2 \quad \xi \leq x \leq L/2$$

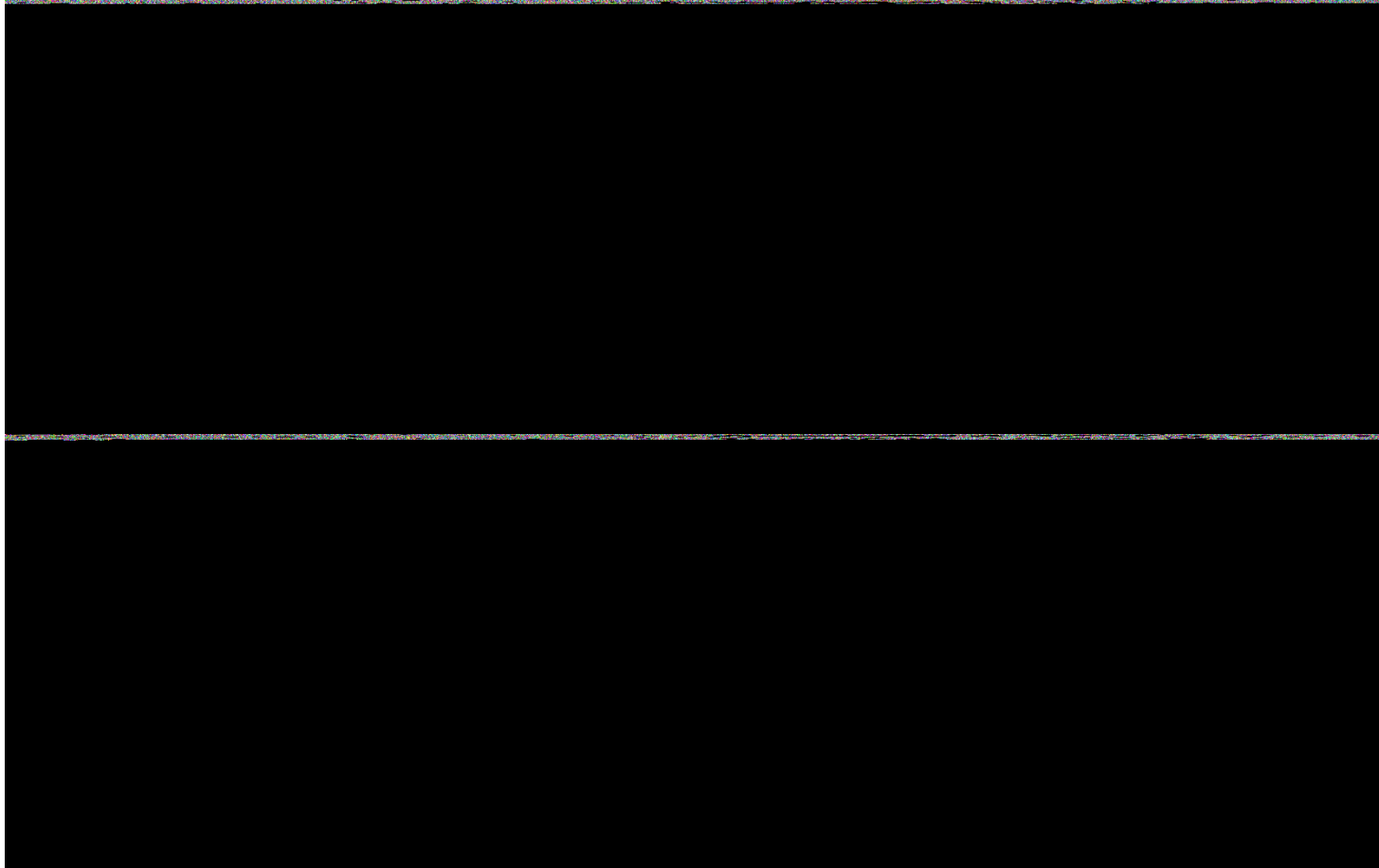
➤ Deflection

$$\text{Region ① : } \delta_{21} = \int_0^x \theta_{21}(x) dx + C_3 \quad 0 \leq x \leq \xi$$

$$\text{Region ② : } \delta_{22} = \int_0^{\xi} \theta_{21}(x) dx + \int_{\xi}^x \theta_{22}(x) dx + C_4 \quad \xi \leq x \leq L/2$$

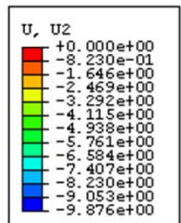
The constants of integration defined in terms of C_i are numerical values which are determined based on the boundary conditions.

Solutions are available for other loading types

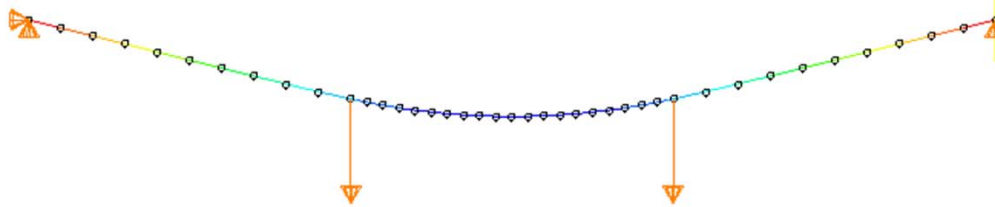
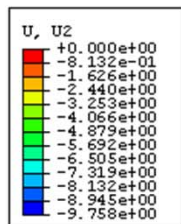


- Different equilibrium equations affecting Moment distribution
- Simply supported beam and cantilever beam.
- 8 different loading types

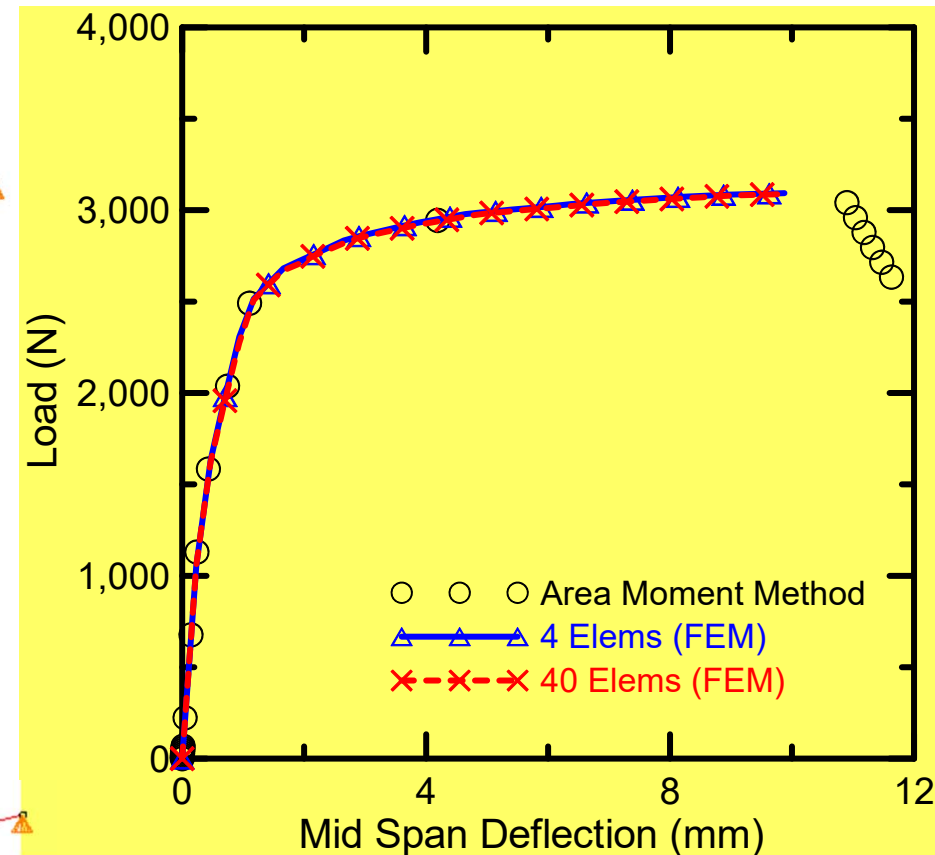
Use of Moment-Curvature response as material property of 2 node, 3- DOF per node beam elements



Ferrocement, $V_f = 1.51\%$, 4- B23 bending elements,

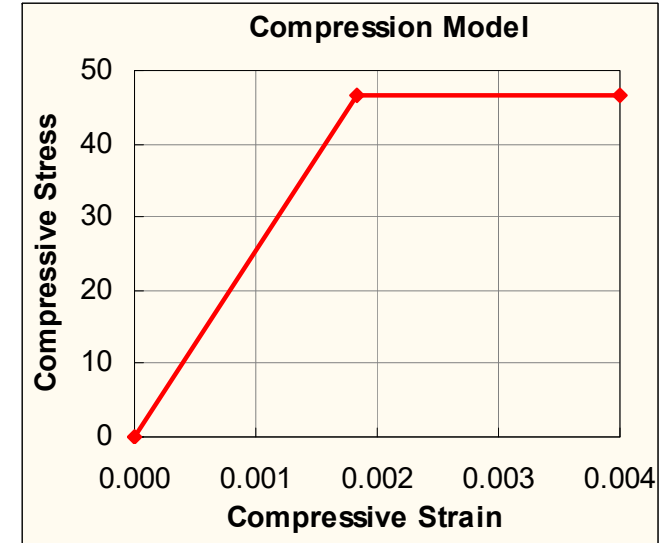
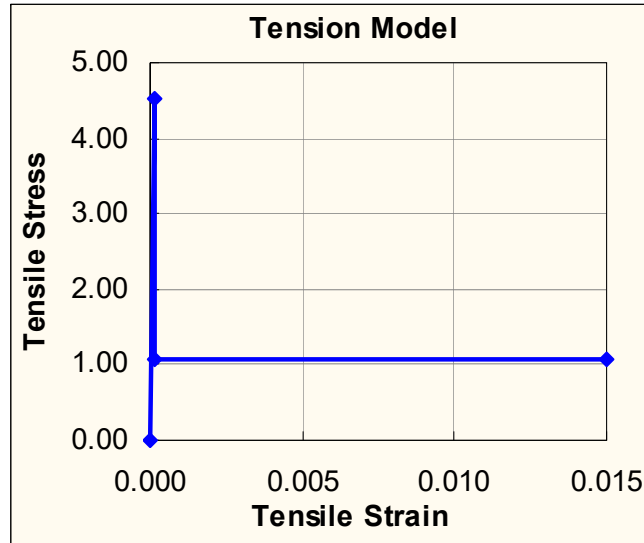
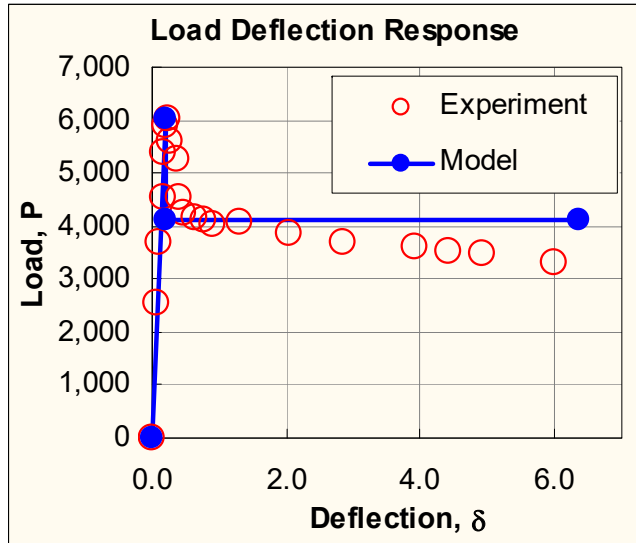


Ferrocement, $V_f = 1.51\%$, 40- B23 bending elements,



Soranakom, C., and Mobasher, B., "Correlation Of Tensile And Flexural Behavior Of Fiber Reinforced Cement Composites," Ferro8, proceedings of the 8th International Ferrocement and thin reinforced cement composites Conference, Bangkok, Thailand, 2006.

Back Calculation of Material Properties



Beam Size

test method **4** point bending
Don't need L_p **0**

$b =$ **100**
 $d =$ **100**

$L =$ **750**
 $S =$ **250**

Material Model

$E =$ **25400**
 $\epsilon_{cr} =$ **0.0001781**

tension

$\mu =$ **0.24**
 $\beta_{tu} =$ **84.24**

strain

0.0000000
 $\epsilon_{cr} =$ 0.0001781
 $\epsilon_{cr} =$ 0.0001781
 $\epsilon_{tu} =$ 0.0150000

stress

0.00
 $\sigma_{cr} =$ 4.52
 $\sigma_p =$ 1.09
 $\sigma_{tu} =$ 1.09

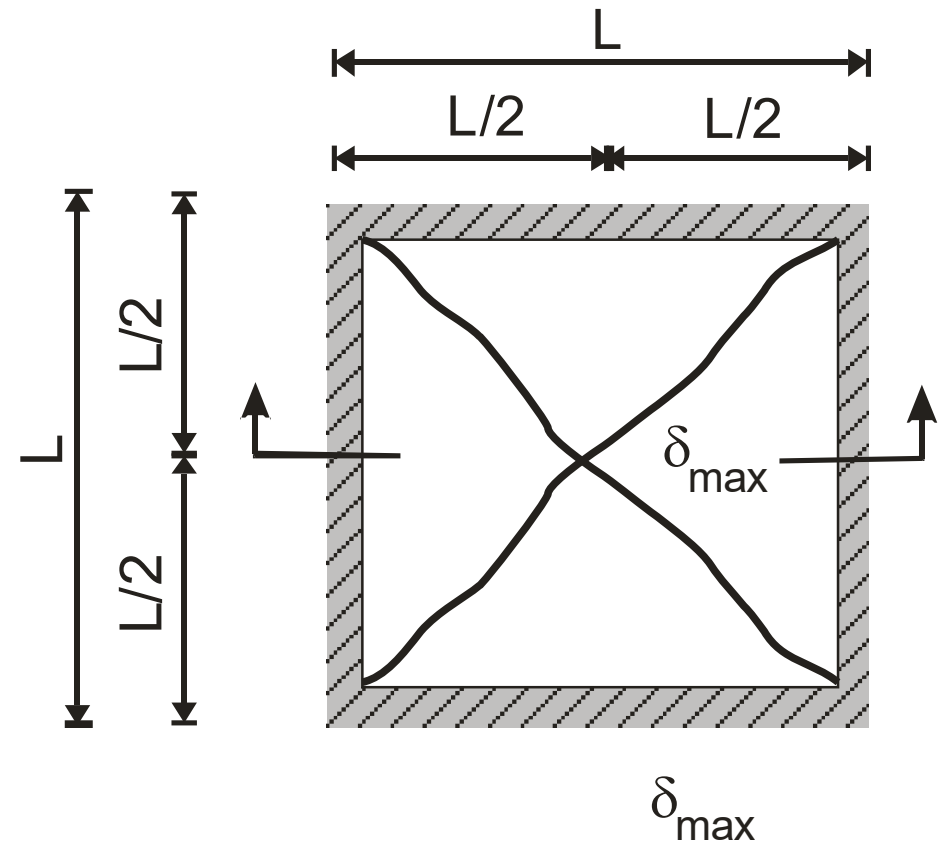
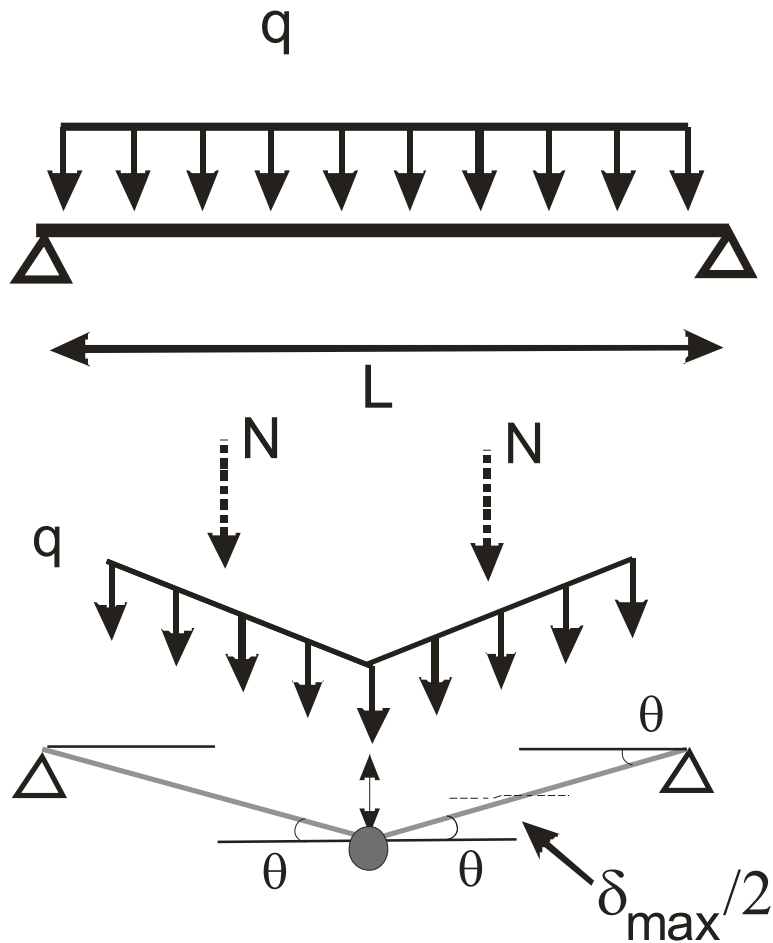
compression

$\omega =$ **10.30**
 $\lambda_{cu} =$ **22.46**

0.0000000
 $\epsilon_{cy} =$ 0.0018341
 $\epsilon_{cu} =$ 0.0040000

0.00
 $\sigma_{cy} =$ 46.59
 $\sigma_{cu} =$ 46.59

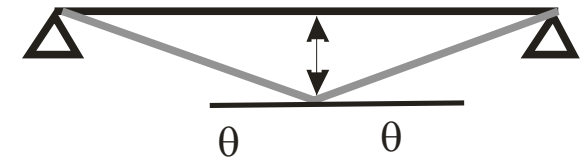
Plastic Analysis Beam and Panel Flexure



$$W_{int} = W_{ext}$$

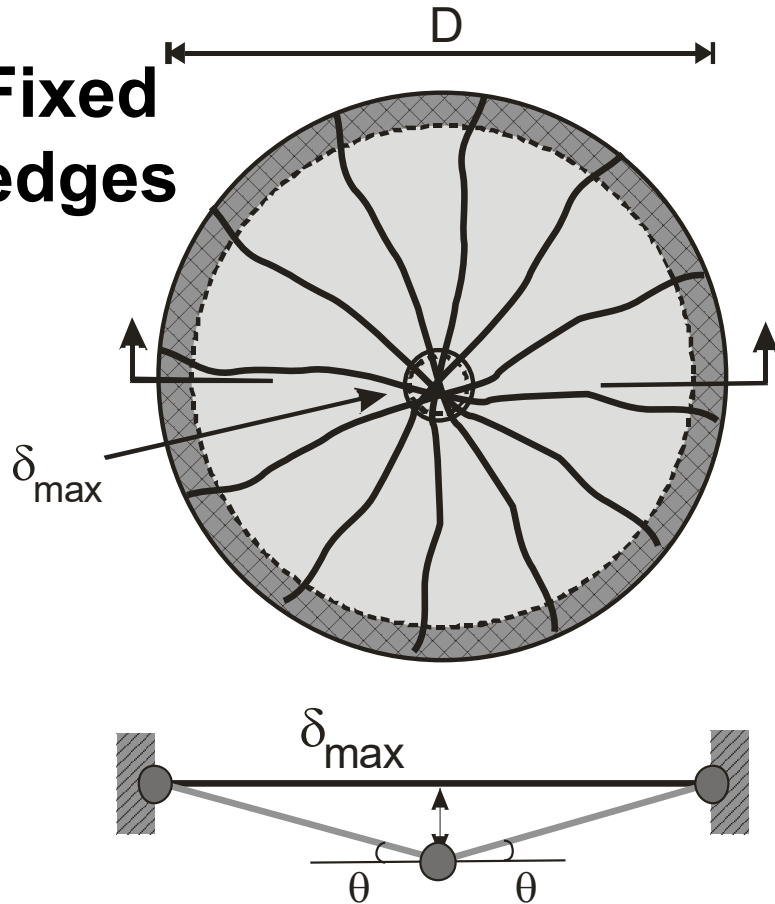
$$2M_P\theta = 2M_P\left(\frac{2\delta}{L}\right) = 2\left[q\left(\frac{L}{2}\right)\left(\frac{\delta}{2}\right)\right]$$

$$M_P = \frac{q_{ult}L^2}{8}$$



Plastic analysis, Round Panel- Fixed or Free

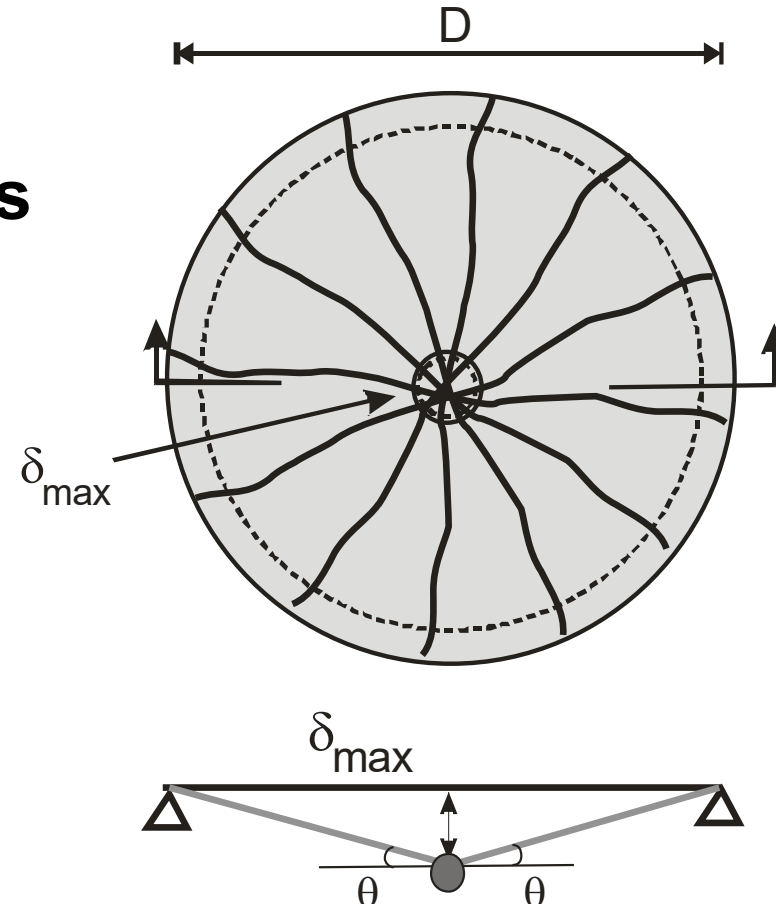
Fixed edges



$$W_{int} = W_{ext} \quad P_{ult} = 4\pi M_P$$

$$4\pi M_P = \frac{2\pi R}{2} \frac{q_{ult} R}{3}$$

Free edges

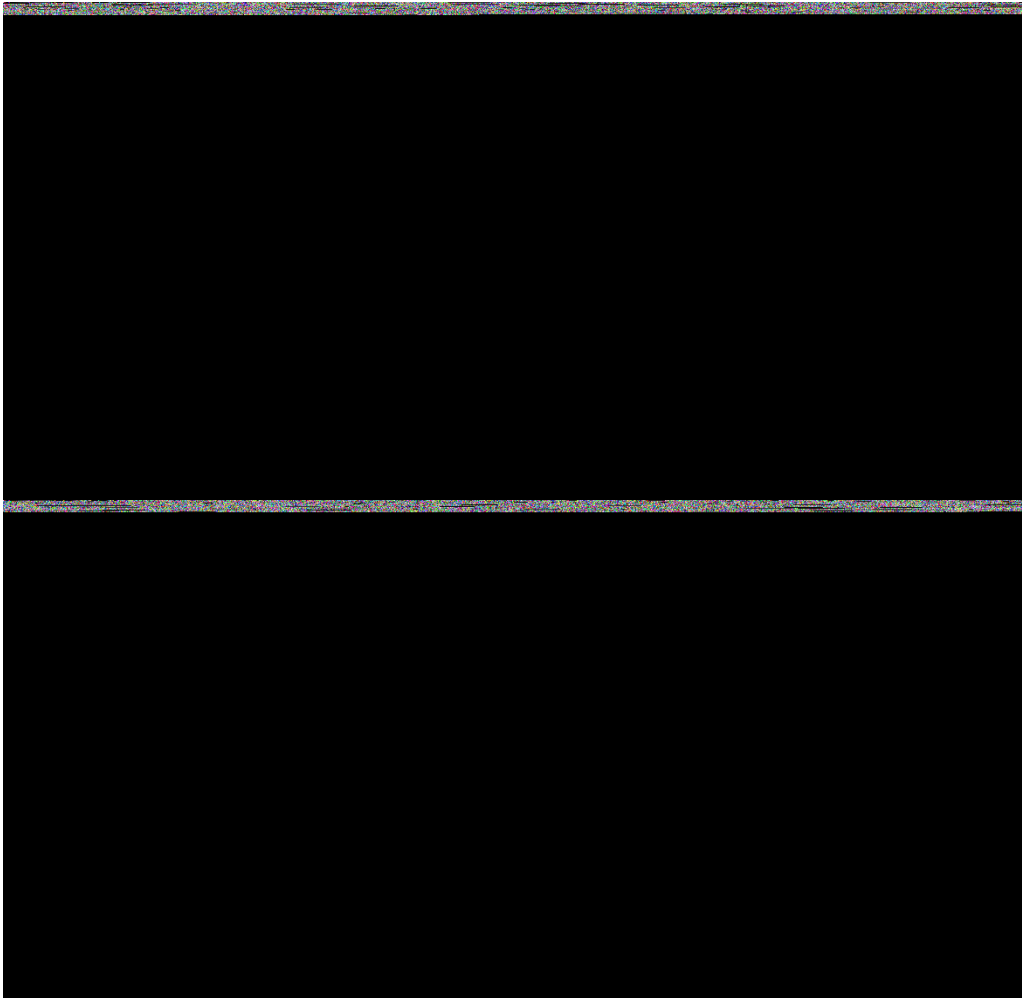


Concentrated Load $P_{ult} = 2\pi M_P$

Distributed Load $M_P = \frac{q_{ult} R^2}{12}$

2-D Member: Slab and Panel

Model Approach



Type of Members

Round Panel



Rectangle Panel



Square Panel

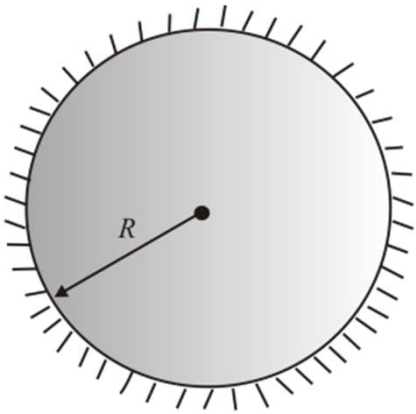


Boundary conditions {
Simply supported
Clamped

- Stage I: Elastic Solution
- Stage II : Yield Line Approach

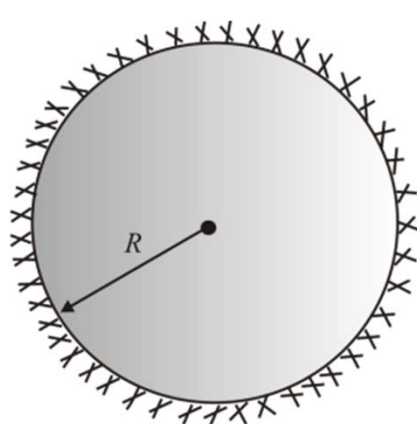
Elastic Solution for Panels

Case 1.1



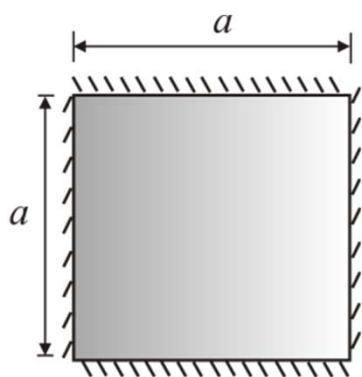
$$\frac{PR^2}{16\pi D} \left(\frac{3+\nu}{1+\nu} \right)$$

Case 1.2



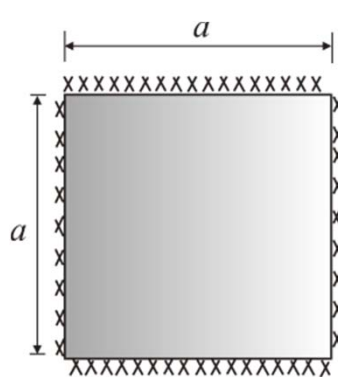
$$\frac{PR^2}{16\pi D}$$

Case 3.1

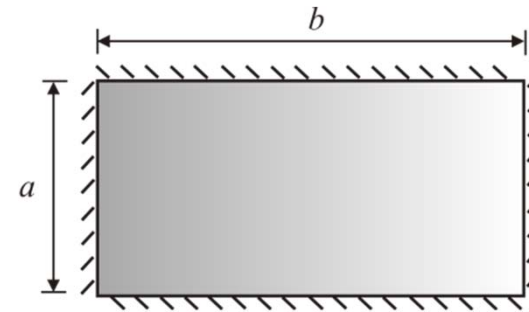


$$0.00406 \frac{qa^4}{D}$$

Case 3.2

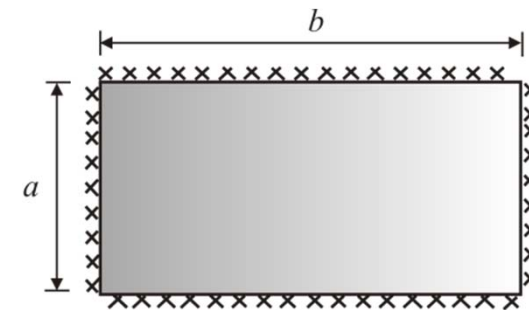


$$\frac{0.032}{2} (1-\nu^2) \left(\frac{qa^4}{Et^3} \right)$$



Case 2.1

$$\propto \frac{qa^4}{D}$$



Case 2.2

$$\frac{0.032(1-\nu^2)}{1 + \left(\frac{a}{b} \right)^4} \left(\frac{qa^4}{Et^3} \right)$$

Round Panel – Point load (P) at center
Rectangular/Square panel— Uniform load (q)



Clamped support



Simply supported

Moment Rotation



Plastic Solution: Yield Line Moment– Applied Load

$$W_{\text{ext}} = W_{\text{int}}$$

$$\Sigma(N \delta) = \Sigma(m l \theta)$$

Work done in moving loads = Work done in rotating yield lines

Assumptions:

- Failure takes place according to the assumed pattern
- Yield lines are straight and end at slab boundary
- Yield lines at vertexes are at 45° to the edges in case of square and rectangular slabs
- Hogging moment about the yield lines and sagging moment about the supports are equal

N – Load, δ – Defl., m – moment abt. YL, l – length of YL, θ - Rotation

Yield Line Moment – Applied Load Case study

Square panel with simply supported edges

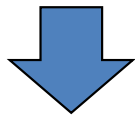
$$W_{\text{ext}} = W_{\text{int}}$$

$$\Sigma(N \delta) = \Sigma(m l \theta)$$

$$W_{\text{ext}} = \Sigma(N \delta) = 4 \times \left(q \frac{a^2}{4} \times \frac{1}{3} \right)$$

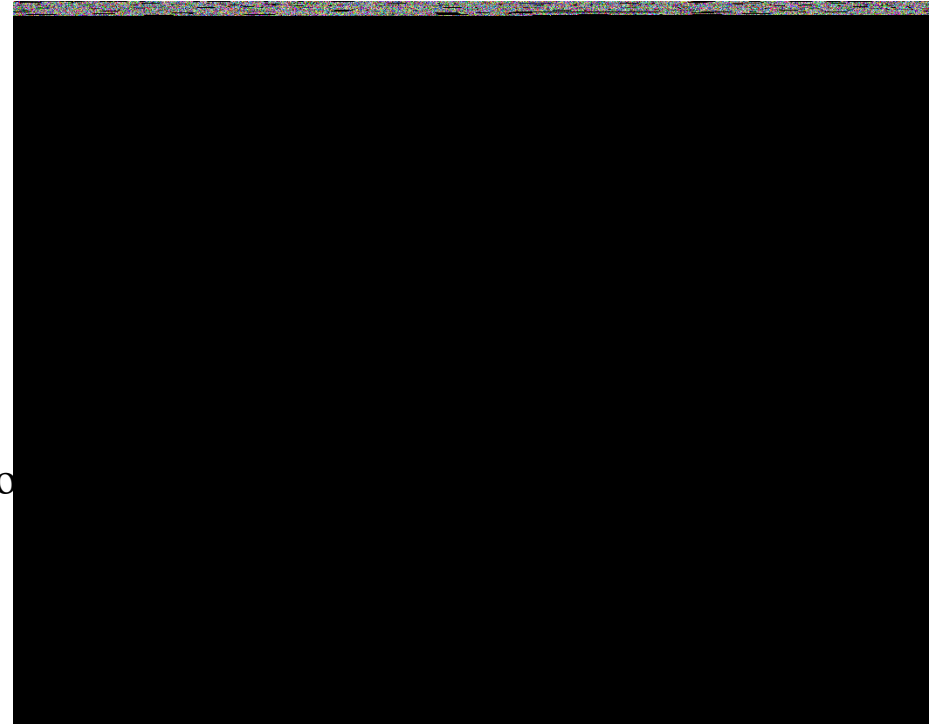
Equivalent point load Deflection of the centre

$$W_{\text{int}} = \Sigma(m l \theta) = 4 \times \left(m a \frac{1}{0.5a} \right)$$



$$m = \frac{qa^2}{24}$$

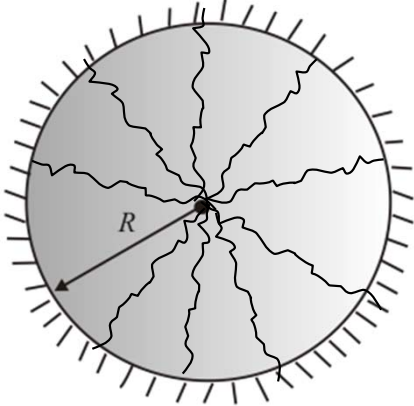
Uniform load: q



δ_{max} is unit

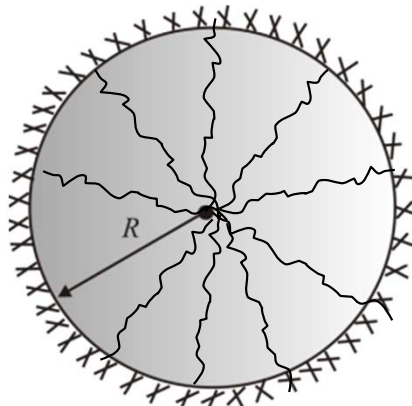
Yield Line Moment – Applied Load

Case 1.1

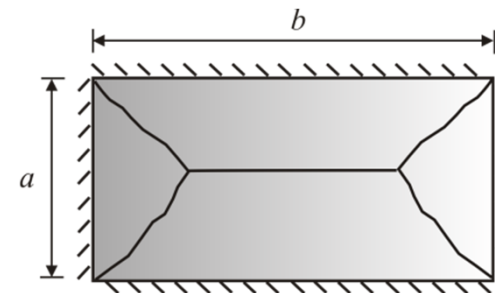


$$m = \frac{P}{2\pi}$$

Case 1.2

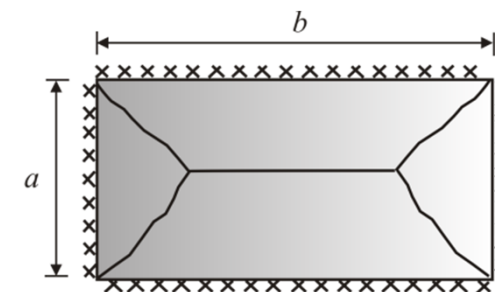


$$m = \frac{P}{4\pi}$$



Case 2.1

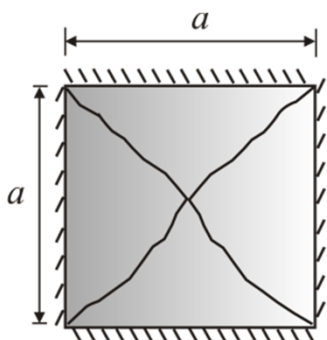
$$m = \frac{qb^2(3b-a)}{12(2b+2a)}$$



Case 2.2

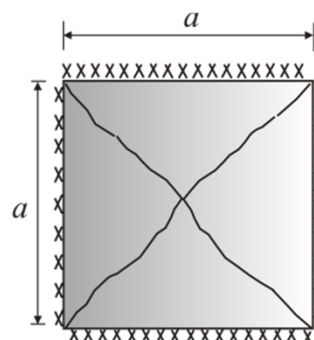
$$m = \frac{qb^2(3b-a)}{12(4b+4a)}$$

Case 3.1



$$m = \frac{qa^2}{24}$$

Case 3.2



$$m = \frac{qa^2}{48}$$

Round panel – Point load (P) at center
Rectangular/Square panel— Uniform load(q)



Clamped support



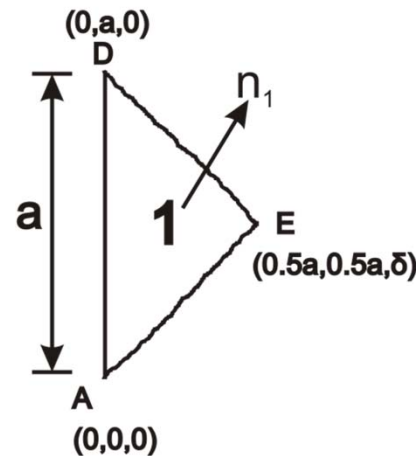
Simply supported

Moment Rotation



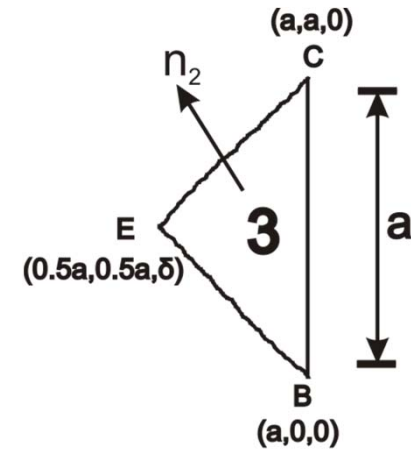
Curvature-Deflection Relationship for Square panels

Square panel



Equation of plane # 1

$$(a\delta)x - 0.5a^2z = 0$$



Equation of plane # 3

$$(a\delta)x + 0.5a^2z - a^2\delta = 0$$

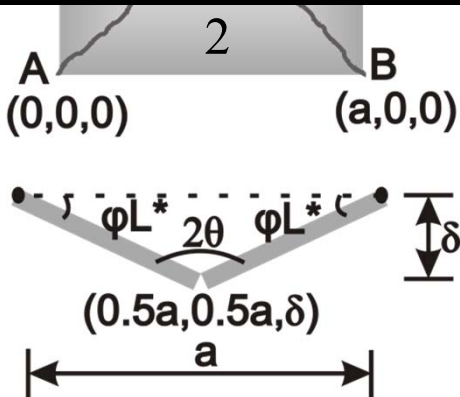
Angle between plane # 1 and #3

$$\cos 2\theta = \cos(\pi - 2\phi L^*) = \frac{a^2\delta^2 - 0.25a^4}{\sqrt{(a^2\delta^2 + 0.25a^4)}\sqrt{(a^2\delta^2 + 0.25a^4)}} = \frac{4\delta^2 - a^2}{4\delta^2 + a^2}$$

Simplifying

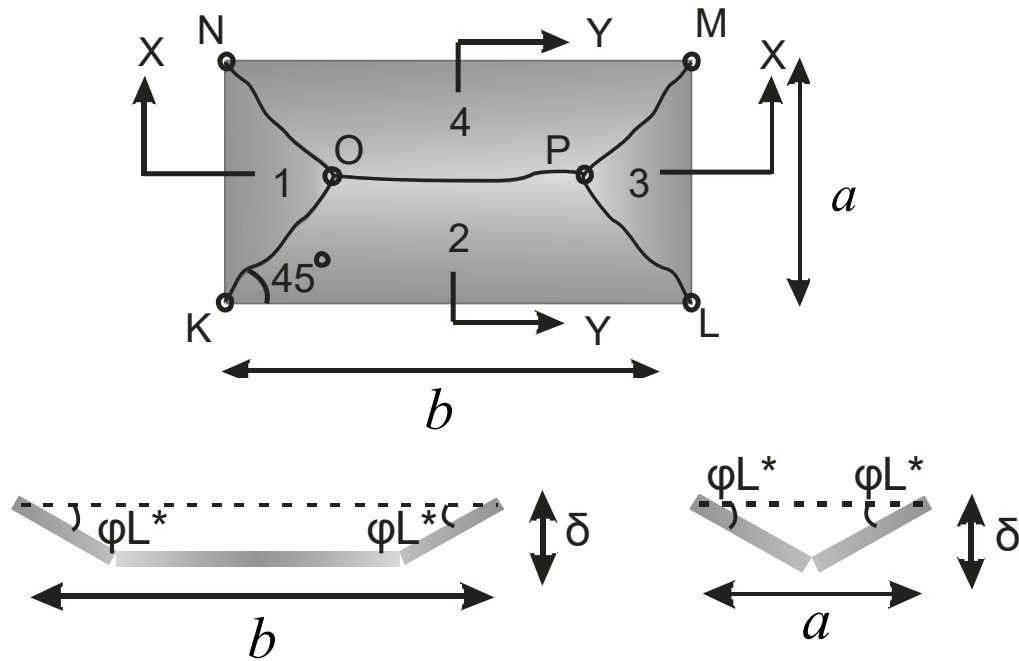
$$\delta = \frac{a}{2} \sqrt{\frac{1 - \cos 2\phi L^*}{1 + \cos 2\phi L^*}}$$

L^* is the hinge length



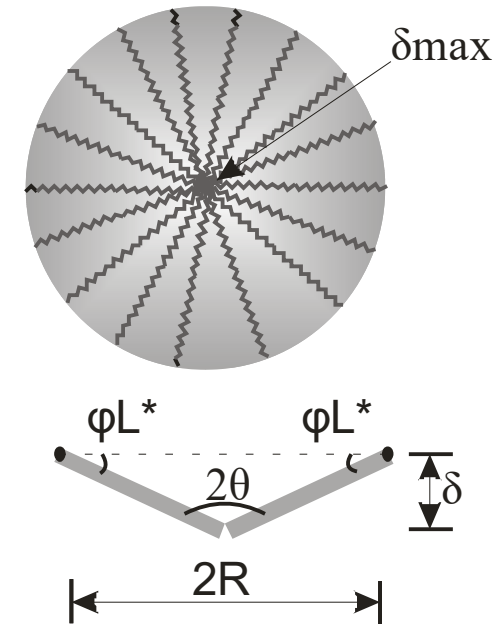
Curvature-deflection relationship

Rectangular panel



$$\delta = \frac{a}{2} \sqrt{\frac{1 - \cos 2(\phi_{14} L^*)}{\cos 2(\phi_{14} L^*)}}$$

Round panel



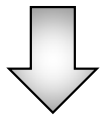
$$\delta = R \sqrt{\frac{2 - 2 \cos(\phi L^*)}{1 + 2 \cos(\phi L^*)}}$$

ϕL^* - Rotation, L^* - Hinge length, L - Length, ϕ - Curvature, δ - Deflection

Experimental Verification

Input

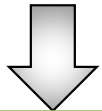
Dimension: b, d, L
Material properties: E, ϵ_{cr}



Calculate

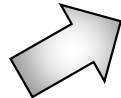
$$M_{cr} = bd^2 E \epsilon_{cr} / 6$$

$$\phi_{cr} = 2\epsilon_{cr} / d$$

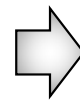


Select position x

Assume maximum q' and η
Loop from $q' = 0$ to q'_{max}



If $q' > 1$



Stage II
Calculate
 $M(i), P(i), m', \xi(i)$



If Region ①
 $0 < x \leq \xi(i)$
 $\delta(i) = \delta_{21}$



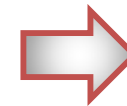
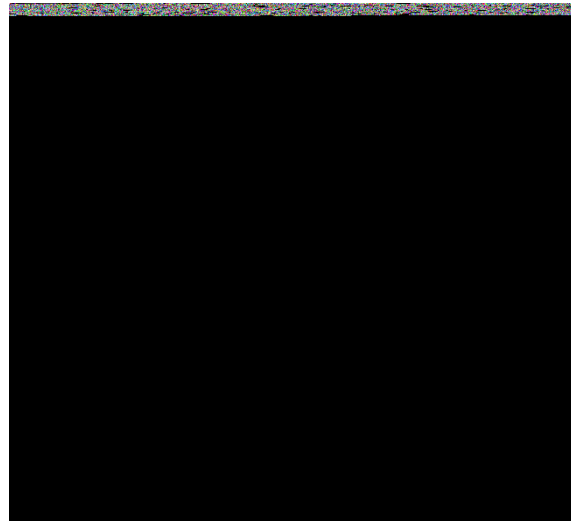
If Region ②
 $\xi(i) < x < L/2$
 $\delta(i) = \delta_{22}$

If $q' \leq 1$

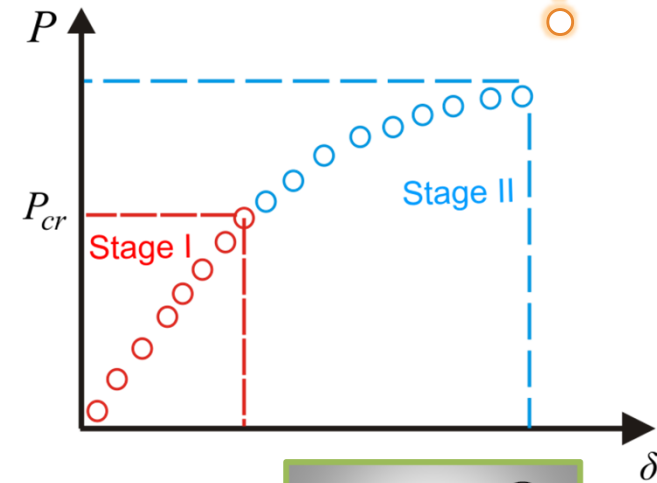


Stage I
Calculate
 $M(i), P(i), \delta(i) = \delta_1$

Moment-curvature



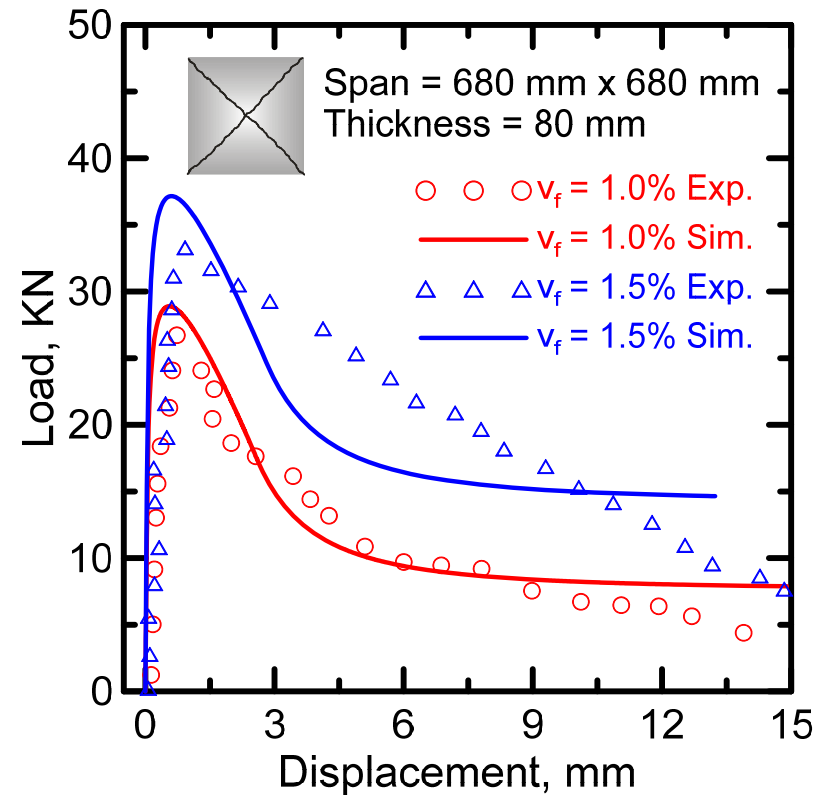
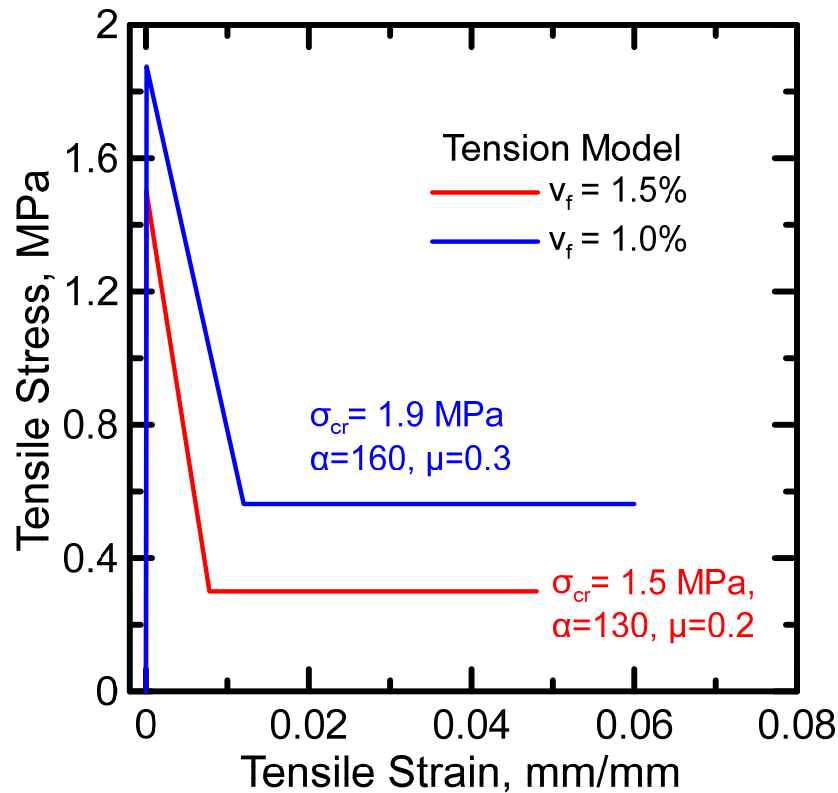
Load - deflection



Experimental Verification

Data Set 2 – Square Slab

Square slab with point load at center
680 mm x 680 mm, thickness is 80 mm
Steel fiber $v_f = 1.0\%$ and 1.5%



Simplified Design Approach

- **Step 1:** Determine the full range moment-curvature relationship using the closed-form equations
- **Step 2:** Obtain simplified relations using polynomial curve fit

$$\text{Stage 1: } m = \kappa$$

$$\text{Stage 2.1: } m = 2.805 - \frac{1.882}{\sqrt{\kappa}} \quad (\text{ascending})$$

$$m = (10^{-4})\kappa^2 - 7.966(10^{-3})\kappa + 2.45 \quad (\text{descending})$$

$$\text{Stage 3.1: } m = 0.592 + \frac{3949}{\kappa^2}$$

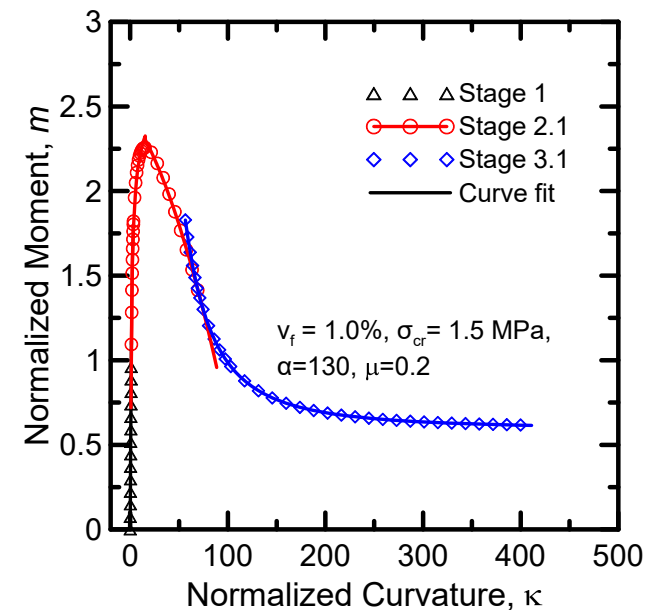
For clear span of 680 mm, thickness 80 mm, cracking stress as 1.5 MPa, moment–curvature relationship of Stage 2.1 can be shown as

$$M = \left(2.806 - \frac{1.882}{\sqrt{\phi / (1.5 \cdot 10^{-6})}} \right) \frac{1.5 \cdot 1 \cdot 80^2}{6} = \left(4489.6 - \frac{3.570}{\sqrt{\phi}} \right) N \cdot mm \quad (\text{ascending})$$

$$M = \left[-7.111(10^7)\phi^2 - 8497.1\phi + 3.920 \right] (10^3) N \cdot mm \quad (\text{descending})$$

Similar expression can also be calculated for Stage 3.1:

$$M = \left[946.72 + \frac{1.422(10^{-5})}{\phi^2} \right] N \cdot mm$$



Curve fit of the moment-curvature relationship

Cont'd

- **Step 3:** The boundary conditions are simply supported on all sides. The moment-load relationship for a point load for square slab is as follows:

$$P = 8m_0$$

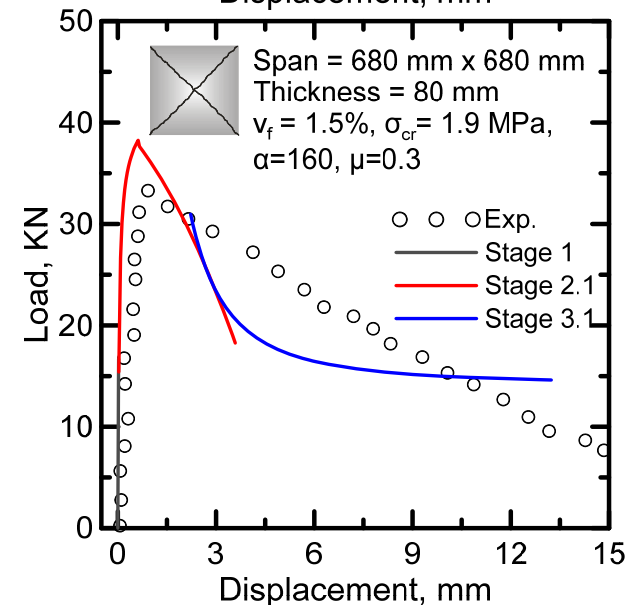
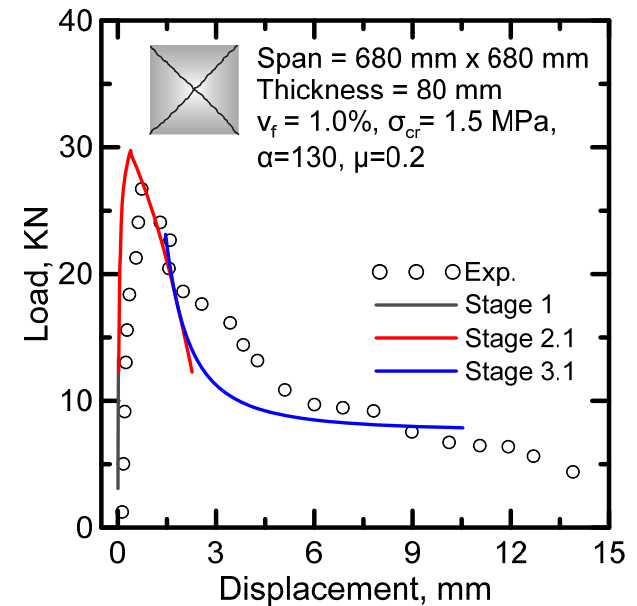
- **Step 4:** The curvature-deflection relationship for a square slab in cracked stages (2.1 and 3.1) are derived as

follows:

$$\text{Elastic stage: } \delta = \frac{0.00406PL^2}{D}, \quad D = \frac{Et^3}{12(1-\nu^2)}$$

$$\text{Cracked stage: } \delta = \frac{L}{2} \frac{\sqrt{\cos \phi L^* (1 - \cos \phi L^*)}}{\cos \phi L^*}$$

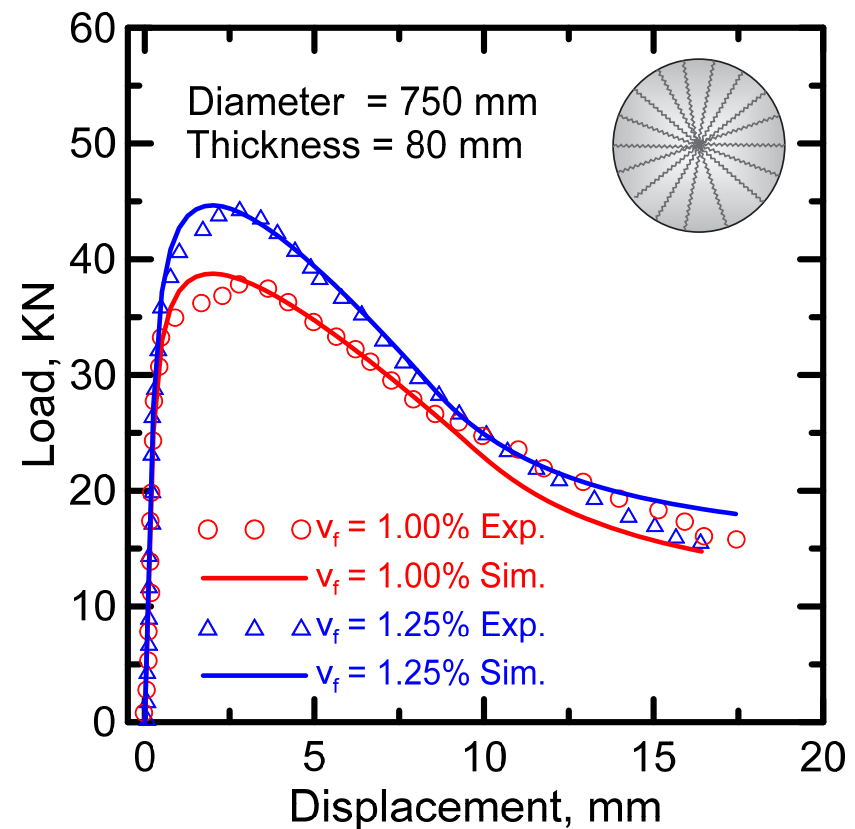
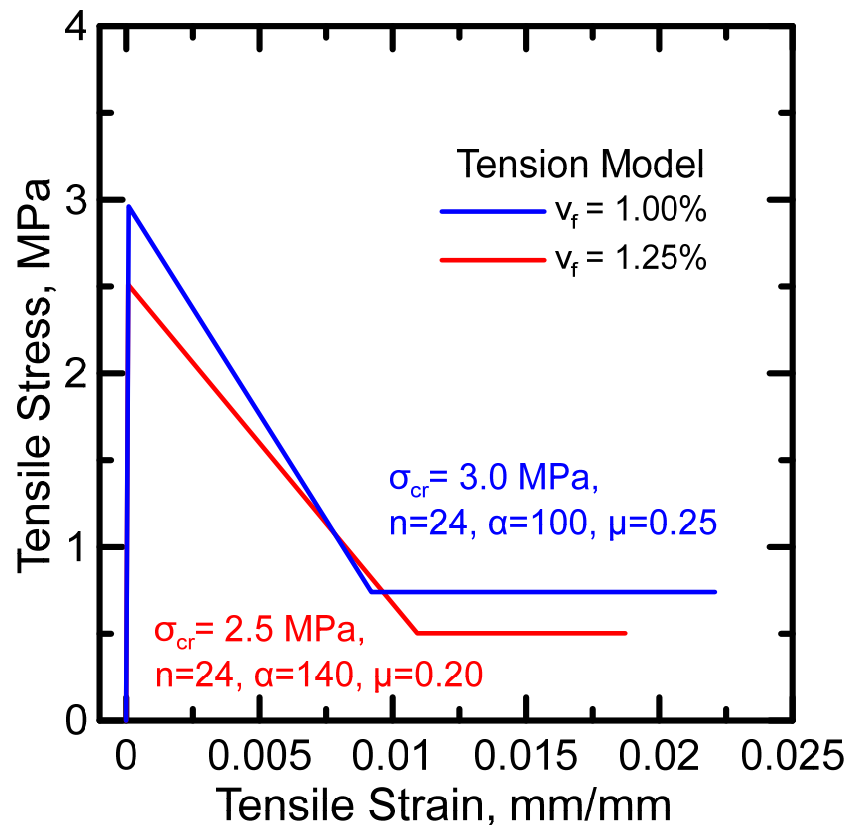
- **Step 5:** Calculate the load-deflection result using the results from Steps 3 and 4.



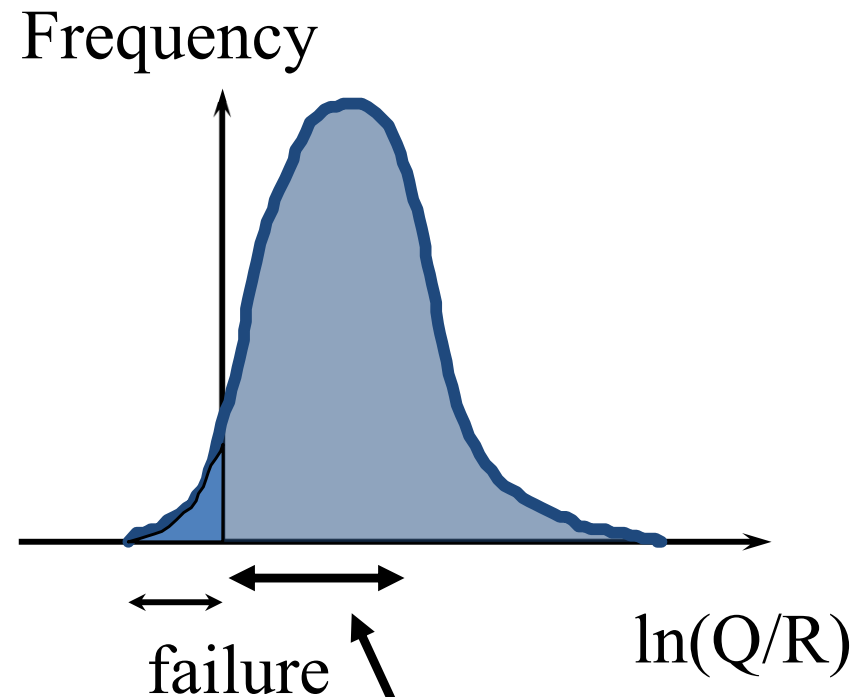
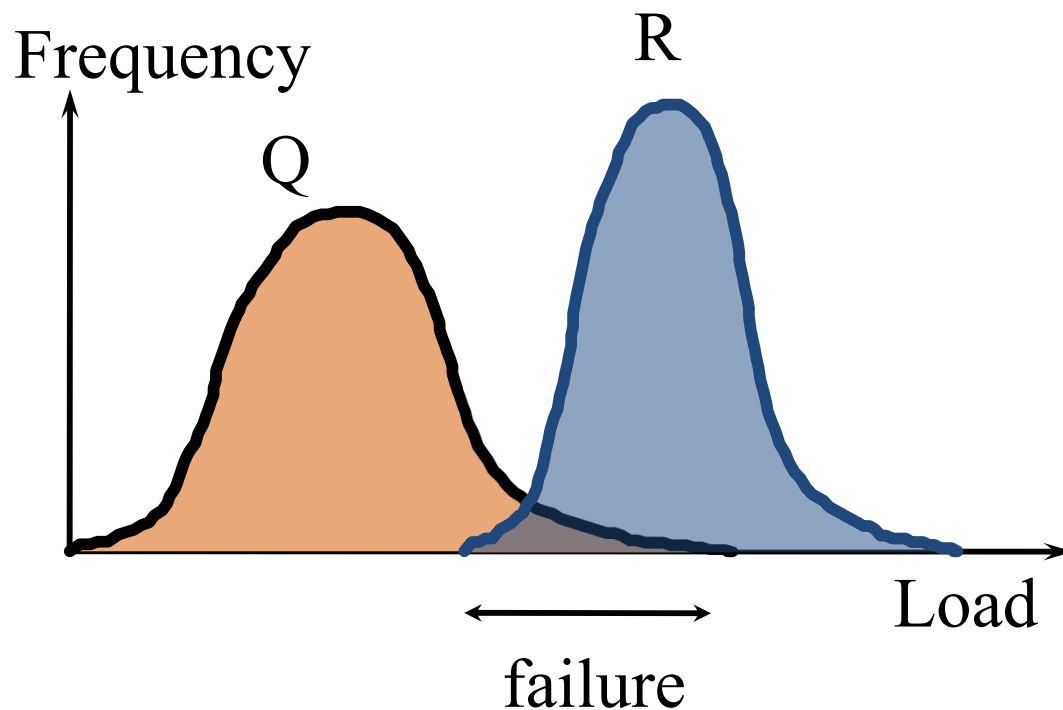
Experimental Verification

Data Set 3 – Round Panel

Round panel with point load at center
Diameter is 750 mm, thickness is 80 mm
Steel fiber $v_f = 1.0\%$ and 1.25%



Probability based model



V_R = Coefficient of variation of R
 V_Q = Coefficient of variation of Q

$$\beta = \frac{\ln(R/Q)}{\sqrt{V_R^2 + V_Q^2}}$$

β , number of standard deviations

Conclusions

- Moment curvature response of fiber reinforced concrete composites can be derived explicitly
- The material model can be used to predict the flexural behavior of:
 - strain hardening materials
 - strain softening materials
 - deflection hardening
 - deflection softening
- Direct application of uniaxial tensile stress strain response under-predicts the flexural response for material that is brittle.
- Ratio of Flexural strength to tensile strength can be predicted using the present model.

Conclusions

- Closed form solutions of the moment curvature response provide a direct and universal method to design and analyze FRC materials and structures.
- The higher the brittleness of the material the higher differences are observed in correlating the tensile and flexural responses.
- When used in conjunction with the moment-area method and crack localization rules, moment- curvature responses can be used to predict load deflection response of various loading geometries.
- There are direct applications for design and analysis of FRC structures.

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- 2. Marikunte, S., and Aldea, C., and Shah, S.P., “Durability of Glass Fiber Reinforced Cement Composites: Effect of Silica Fume and Metakaolin,” *Adv Cement Based Materials*, V.5, No.3-4, Apr.-May. 1997, pp. 100-108.
- 3. Lim, T.Y., Paramasivam, P., and Lee, S.L. (1987). “Analytical Model for Tensile Behavior of Steel-Fiber Concrete.” *ACI Material Journal*, 84(4), 286-551.
- 4. Lim, T.Y., Paramasivam, P., and Lee, S.L. (1987). “Bending Behavior of Steel-Fiber Concrete Beams.” *ACI Structural Journal*, 84(6), 524-536.
- 5. Dupont, D., “Modeling and Experimental Validation of The Constitutive Law (s-e) and Cracking Behavior of Steel Fiber Reinforced Concrete”, PhD Dissertation, Catholic University of Leuven, 2003
- 6. Soranakom, C., and Mobasher, B., “Closed Form Solutions for Flexural Response of Fiber Reinforced Concrete Beams,” *Journal of Engineering Mechanics*, August, 2007.